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(54) MEDICAL MANIPULATOR AND MEDICAL MANIPULATOR

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(73) Assignee: OLYMPUS CORPORATION, Tokyo (JP)

(21) Appl. No.: 19/074,725

(22) Filed: Mar. 10, 2025

Related U.S. Application Data

(60) Provisional application No. 63/562,855, filed on Mar. 8, 2024, provisional application No. 63/635,021, filed on Apr. 17, 2024, provisional application No. 63/648,937, filed on May 17, 2024, provisional application No. 63/664,879, filed on Jun. 27, 2024, provisional application No. 63/669,306, filed on Jul. 10, 2024, provisional application No. 63/688,972, filed on Aug. 30, 2024, provisional application No. 63/691,009, filed on Sep. 5, 2024, provisional application No. 63/695,602, filed on Sep. 17, 2024, provisional application No. 63/733,398, filed on Dec. 12, 2024, provisional application No. 63/737,339, filed on Dec. 20, 2024.

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A61B 34/00 (2016.01)

(52) U.S. Cl.
CPC *A61B 34/71* (2016.02)

(57) **ABSTRACT**

A medical manipulator includes a manipulator flexible portion extending in a longitudinal direction; and two arms arranged at a distal end of the manipulator flexible portion, and the two arms are arranged in a left-right direction perpendicular to the longitudinal direction, each of the two arms includes a second bending portion, and distal ends of second bending portions of the two arms are spaced apart in the left-right direction by bending the second bending portions.

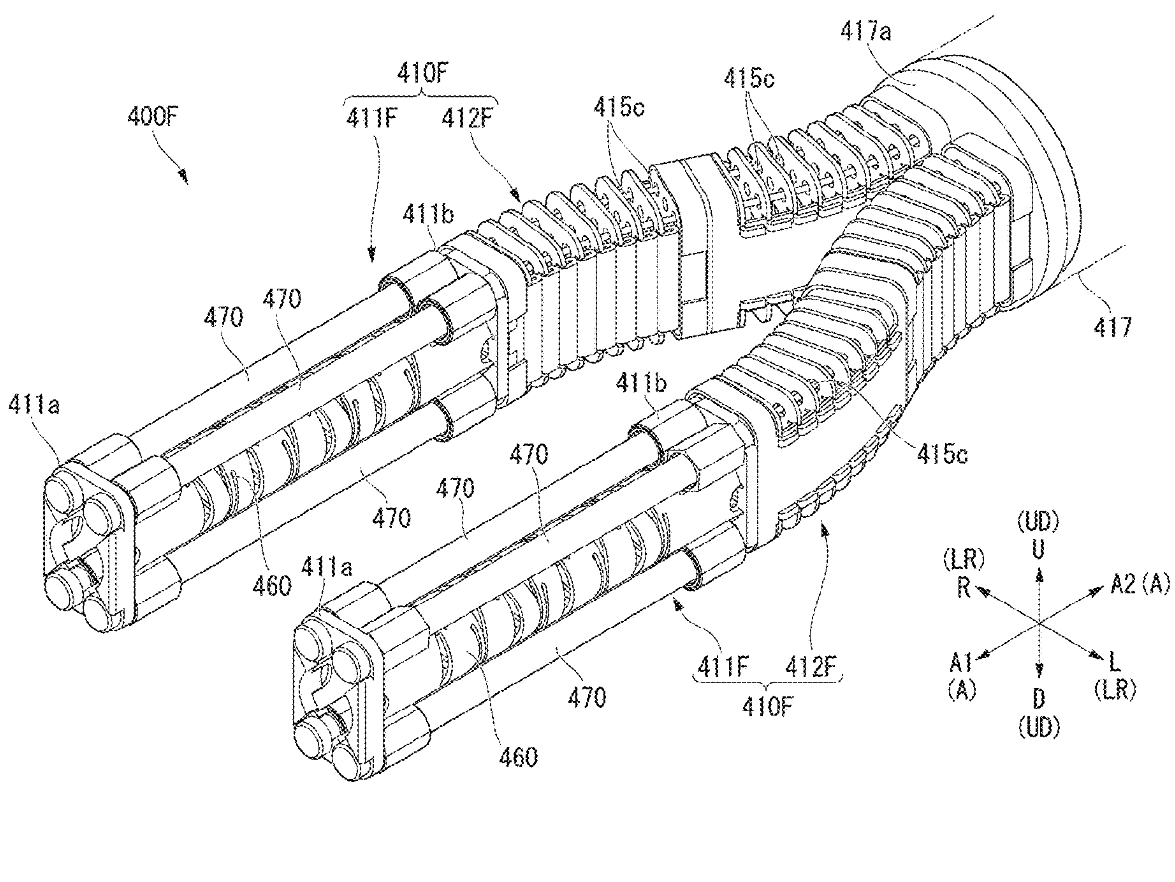
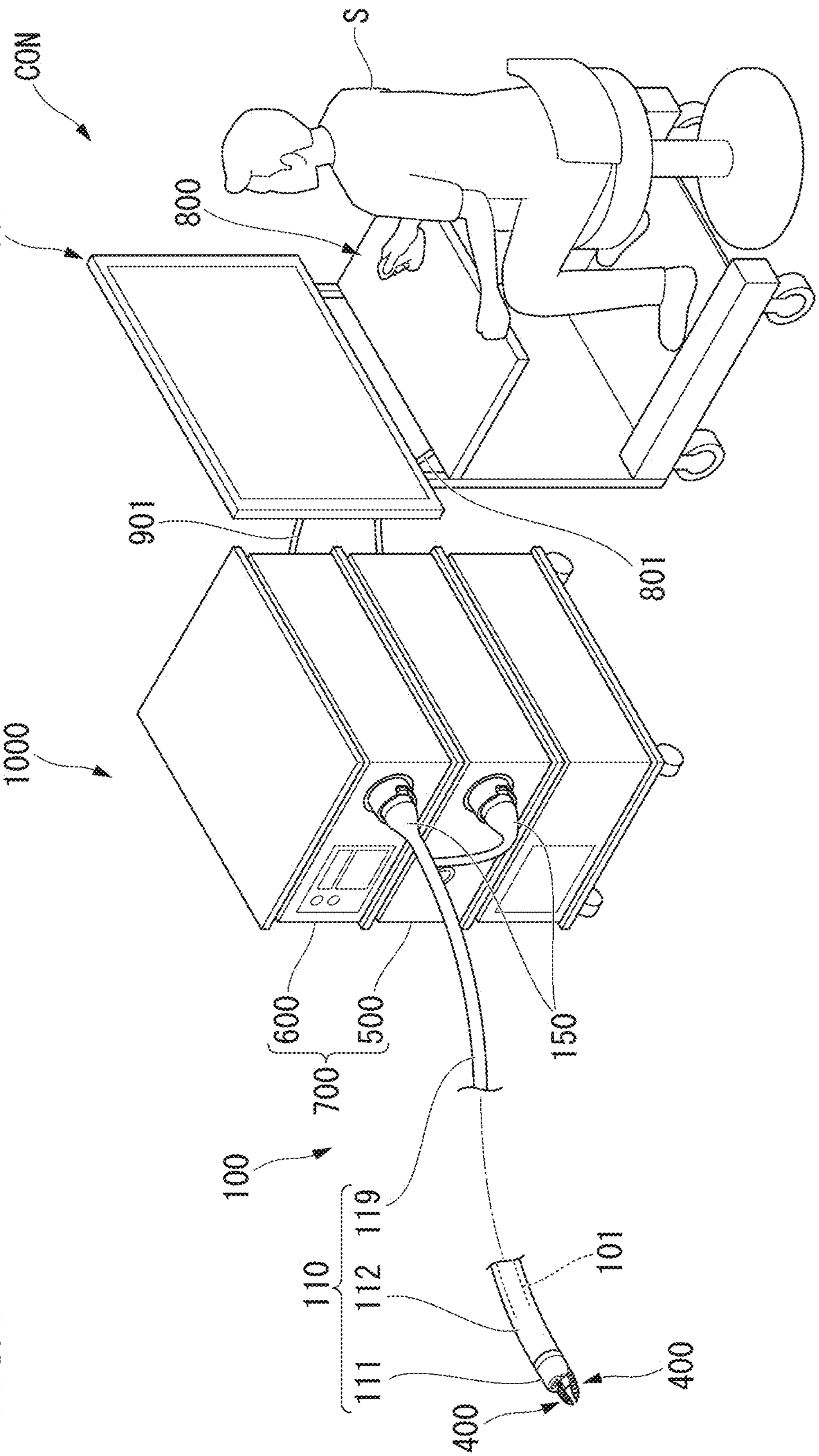


FIG. 1


CON
S
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110
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112
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101
400
400

FIG. 2

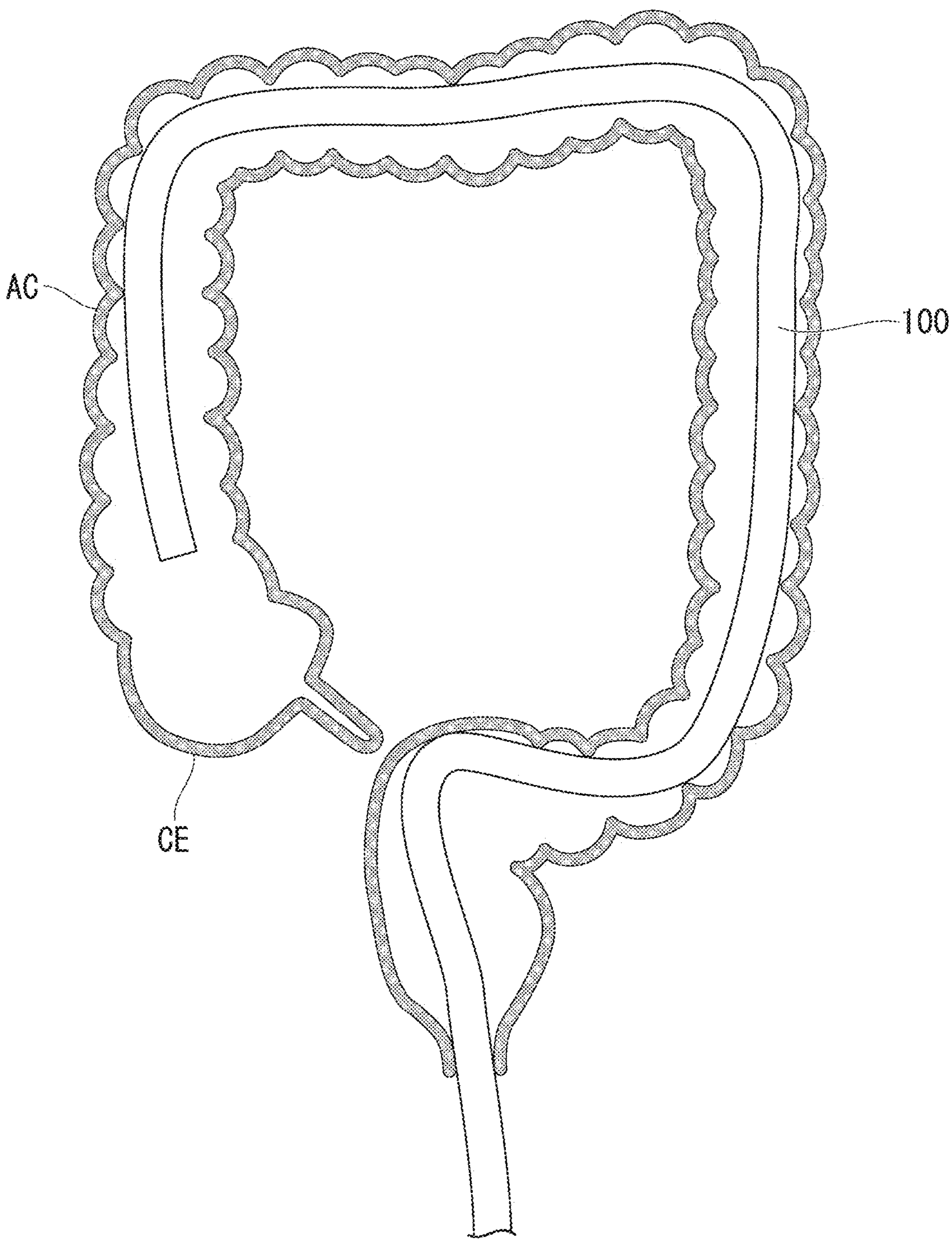


FIG. 3

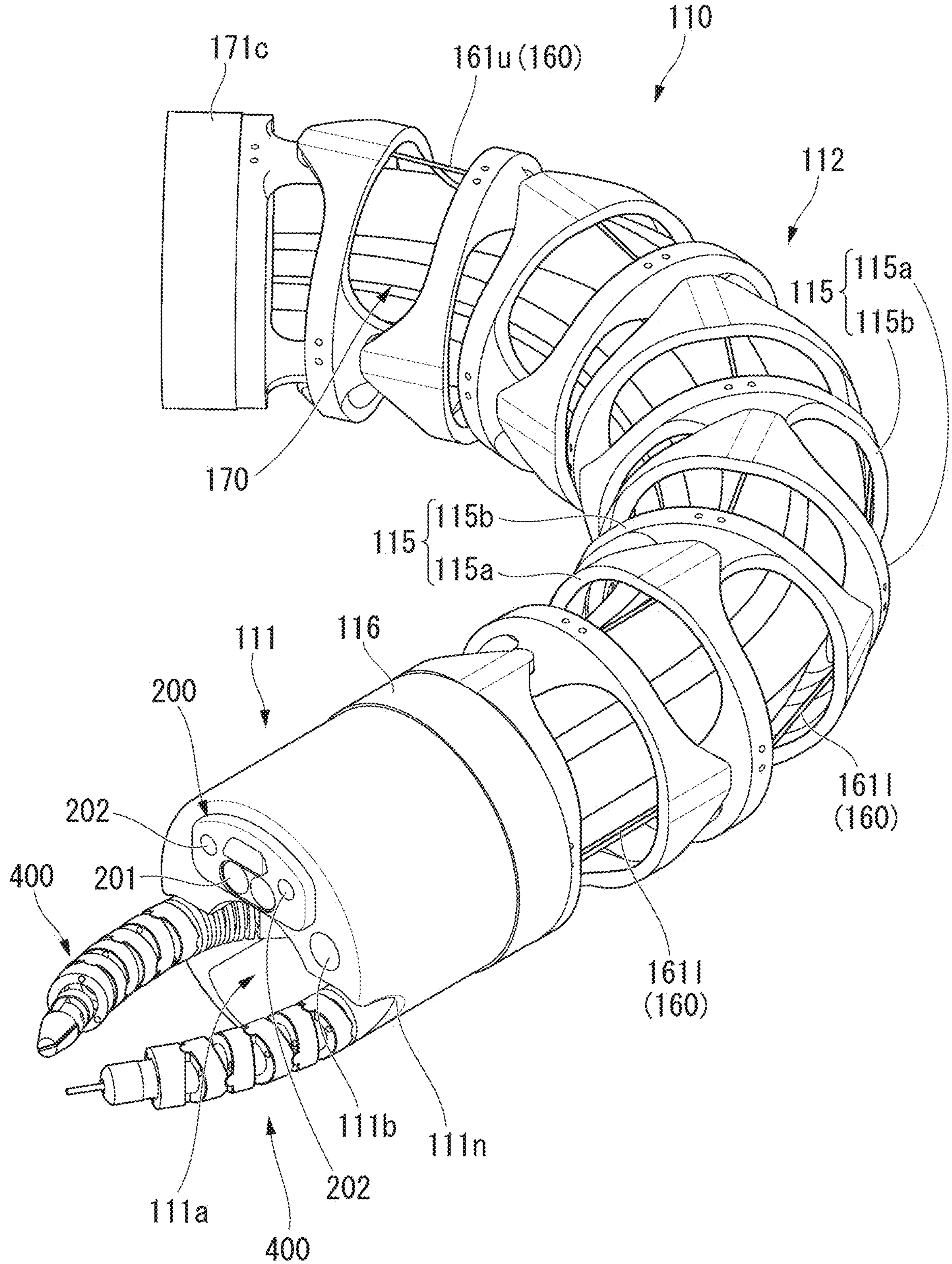

110
171c
161u (160)
112
115
115a
115b
170
115
115b
115a
111
116
200
202
400
201
161l
(160)
161l
(160)
111b
111n
202
111a
400

FIG. 4

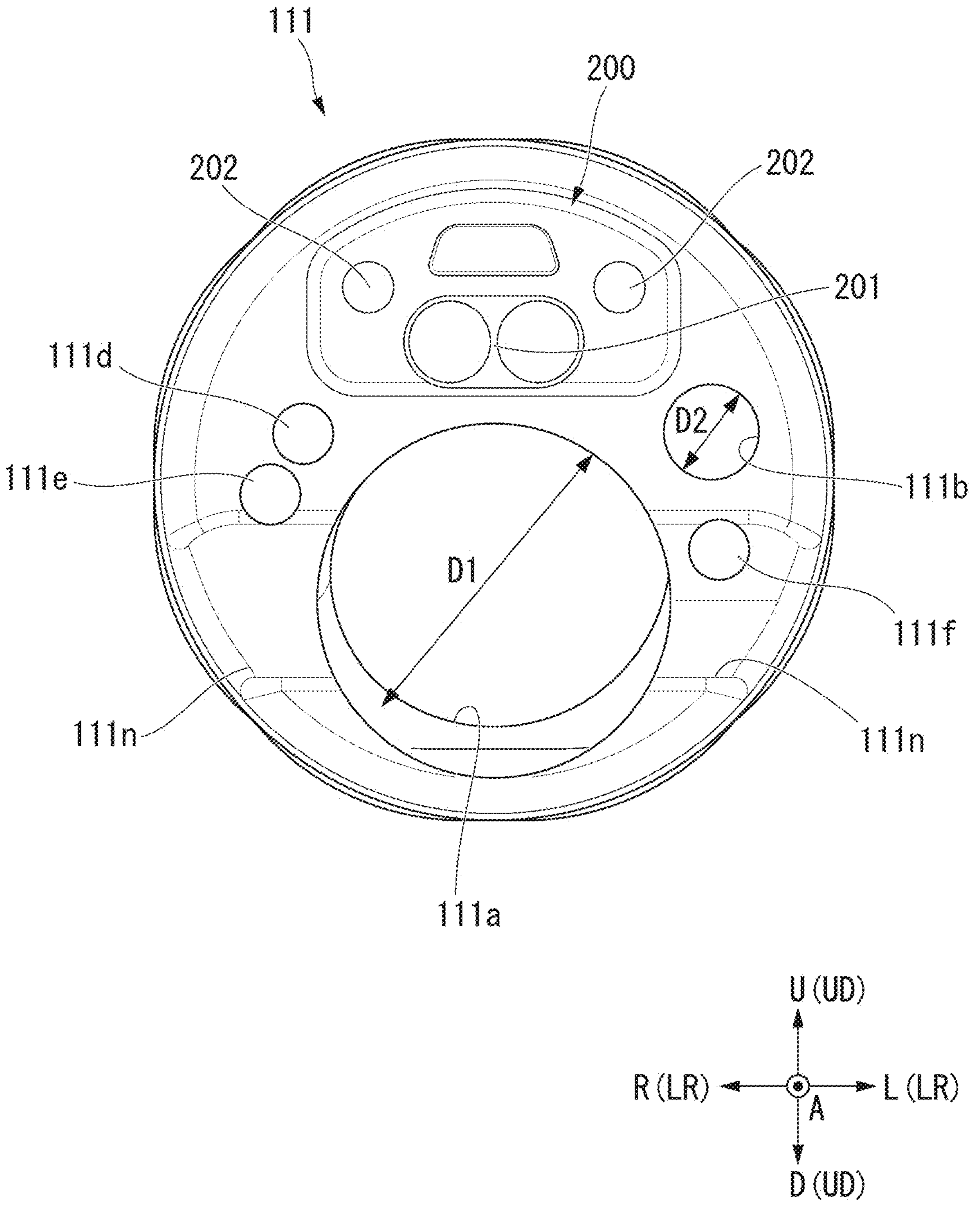
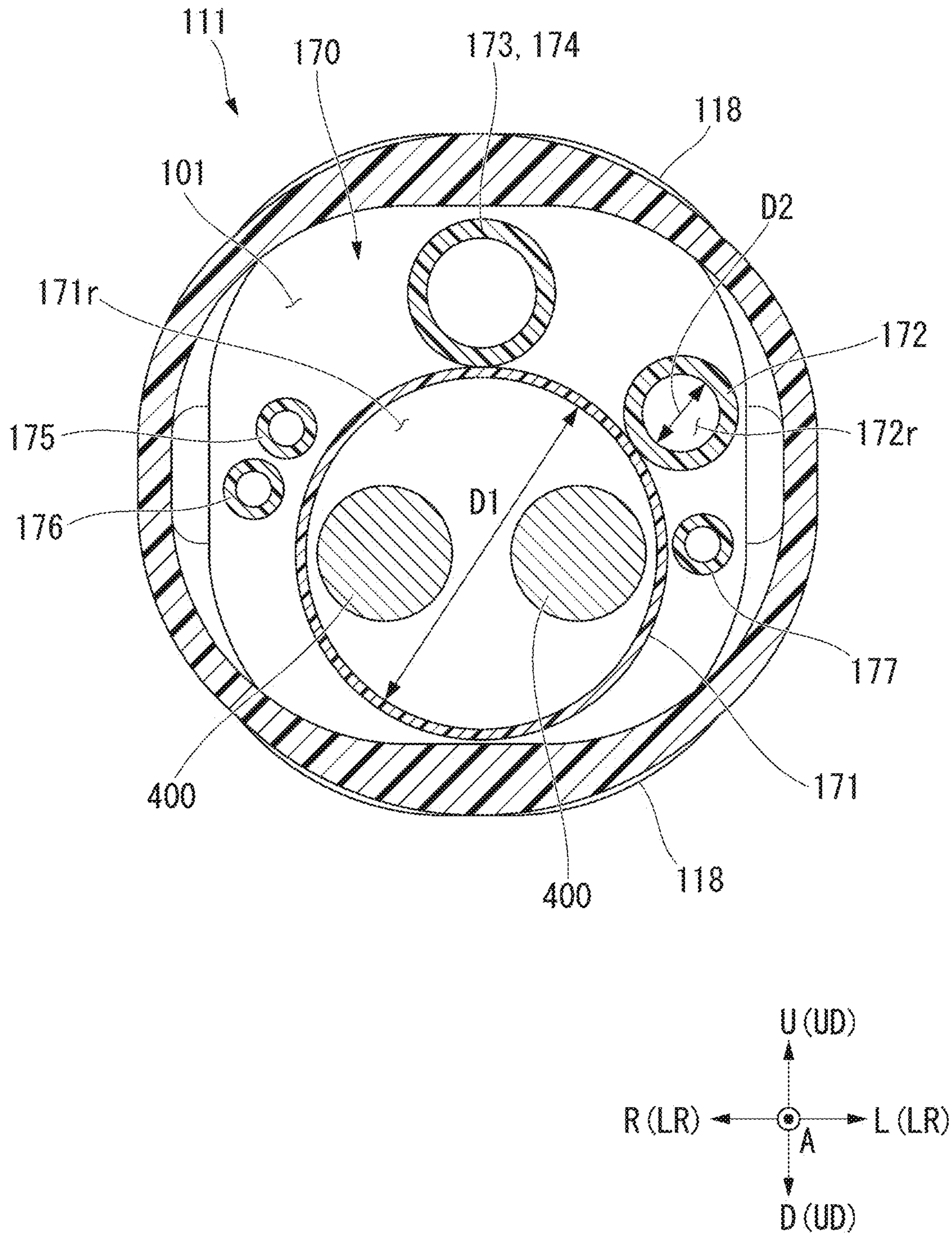

111
200
202
202
201
111d
D2
111e
111b
D1
111f
111n
111n
111a
U (UD)
R (LR)
L (LR)
A
D (UD)

FIG. 5


111
170
173, 174
118
101
D2
171r
172
175
172r
176
D1
177
400
171
400
118
U (UD)
R (LR)
L (LR)
A
D (UD)

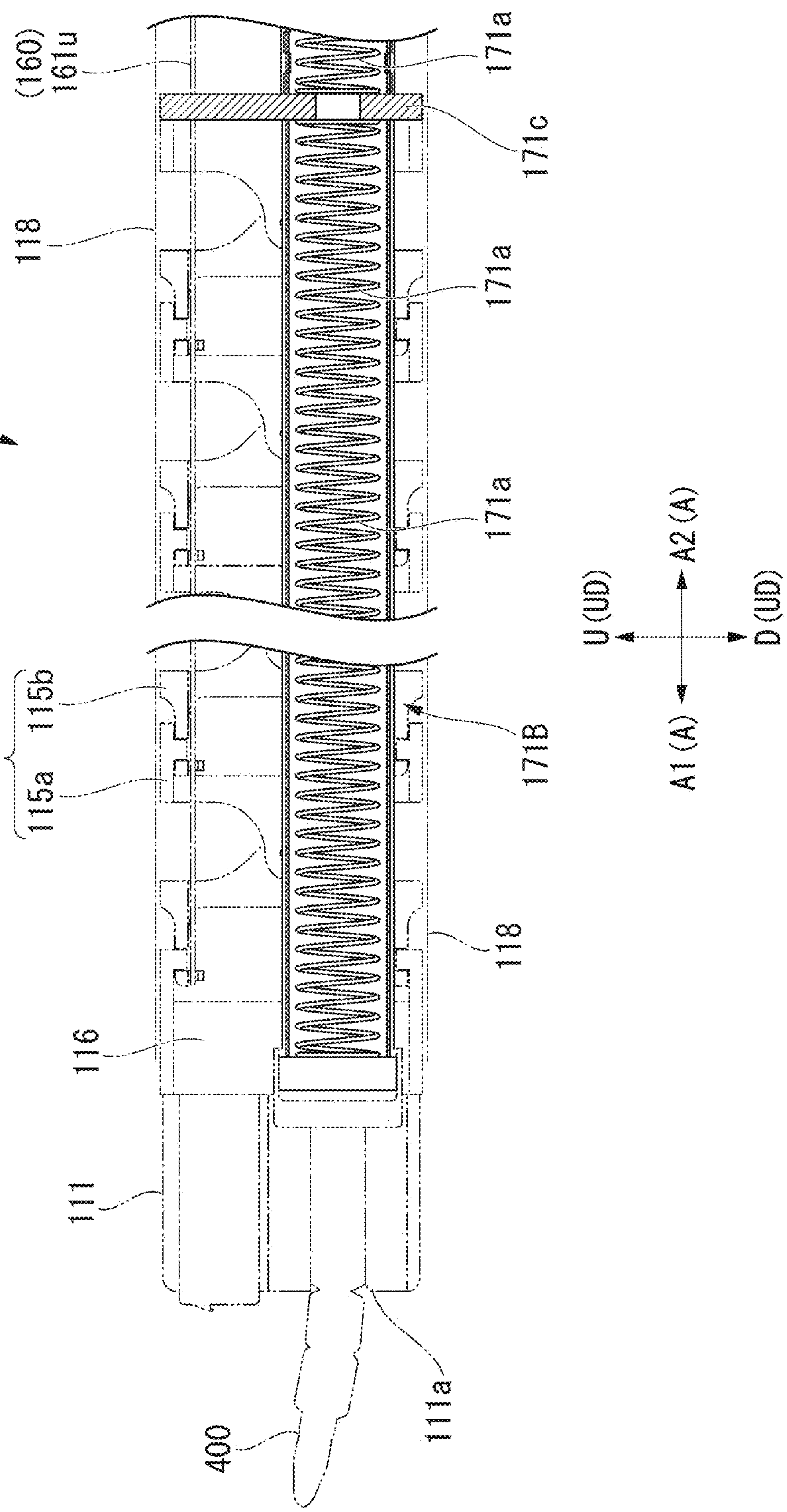
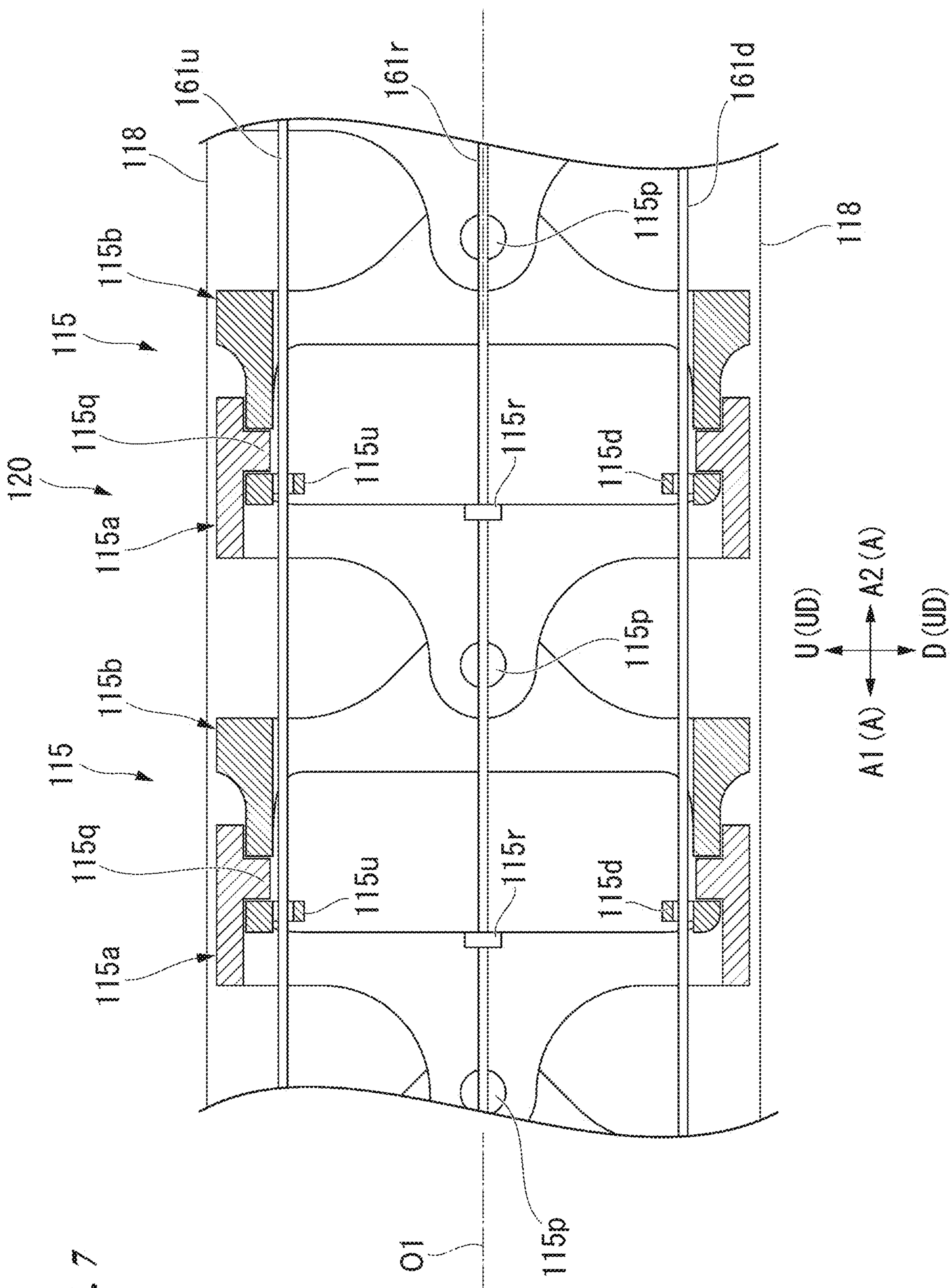
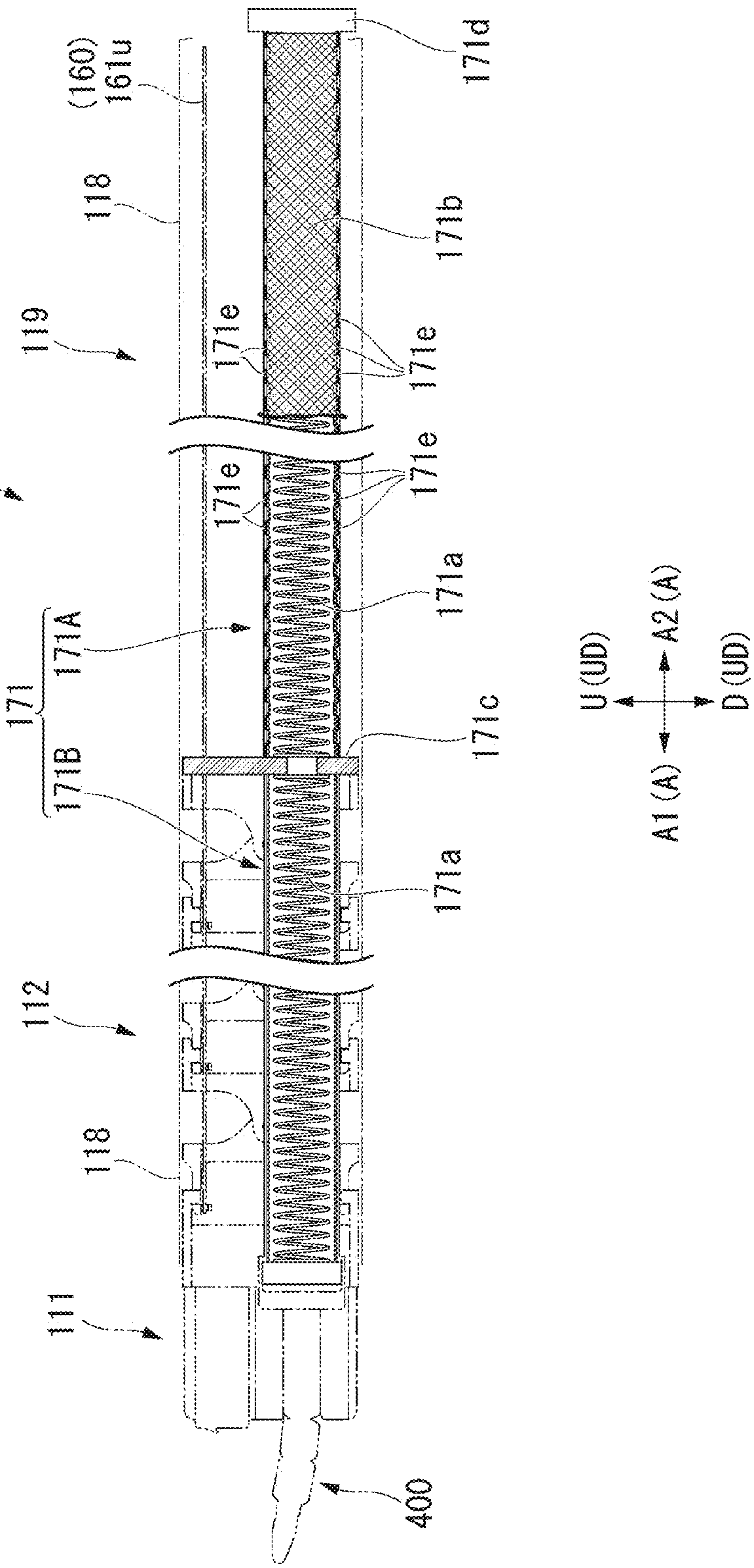

(160)
161u
171a
171c
118
171a
112
171a
U (UD)
A2 (A)
D (UD)
A1 (A)
115
115b
115a
171B
118
116
111
400
111a

FIG. 6

FIG. 7

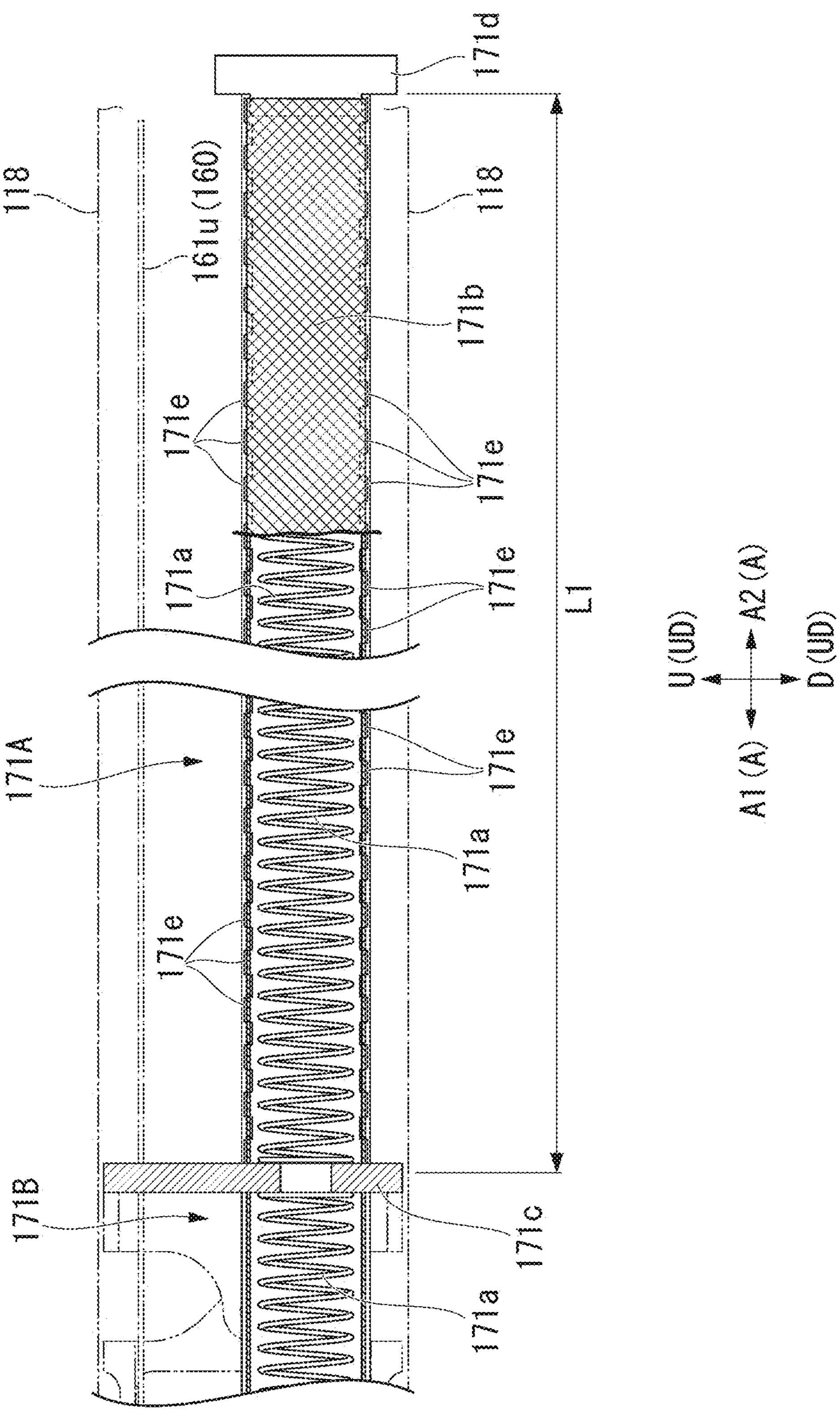

161u
161r
161d
118
118
115p
115p
115p
115b
115b
115
115
115q
115q
120
115a
115a
115u
115u
115r
115r
115d
115d
U (UD)
D (UD)
A1 (A)
A2 (A)
O1

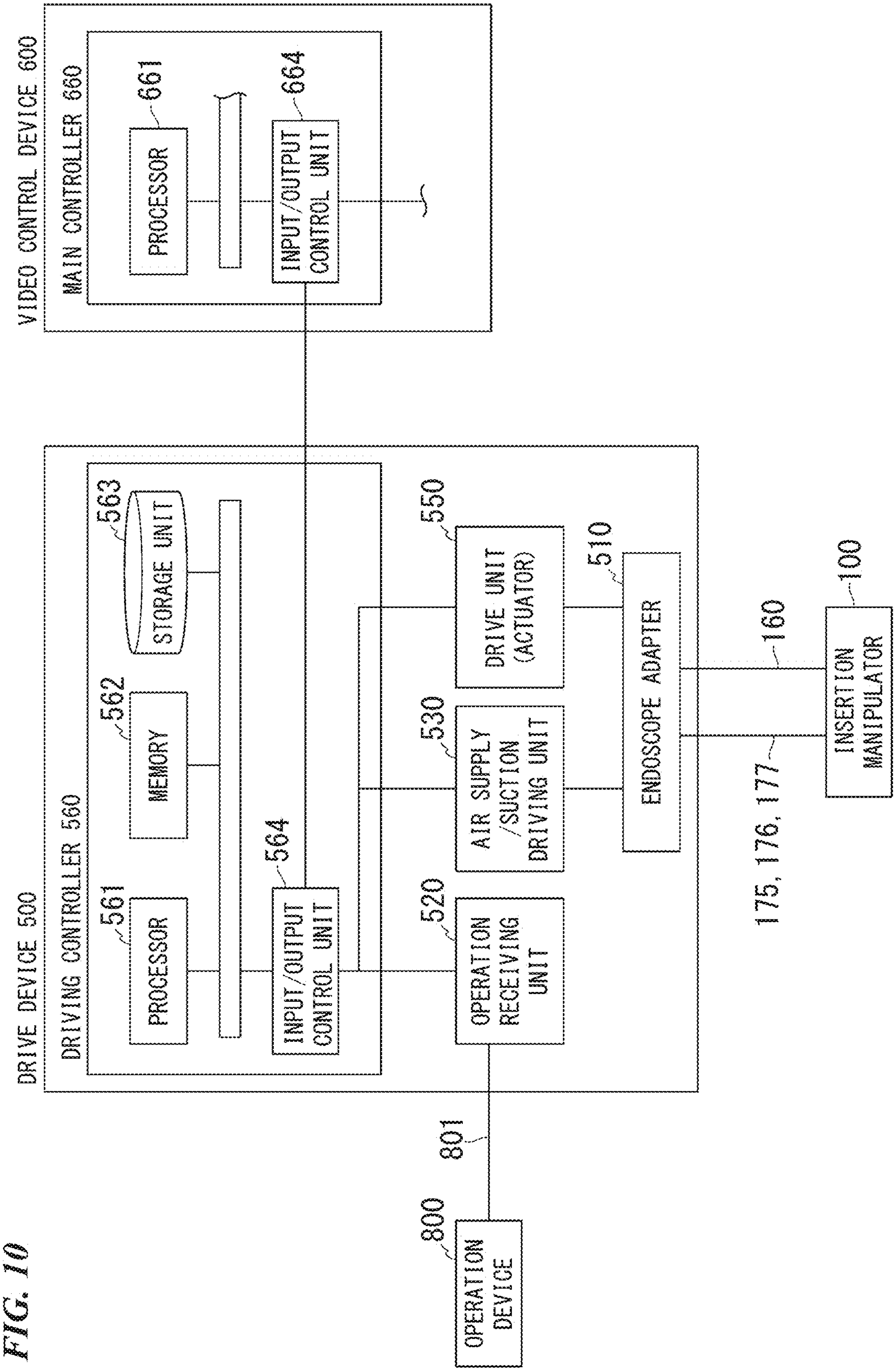


171d
(160)
161u
118
171b
119
171e
171e
110
171e
171e
171a
171A
171
171B
171c
171a
112
118
111
400
U (UD)
D (UD)
A2 (A)
A1 (A)

FIG. 8

FIG. 9


171d
118
161u (160)
118
171b
171e
171e
171a
171e
L1
U (UD)
A2 (A)
D (UD)
A1 (A)
171A
171e
171a
171e
171B
171c
171a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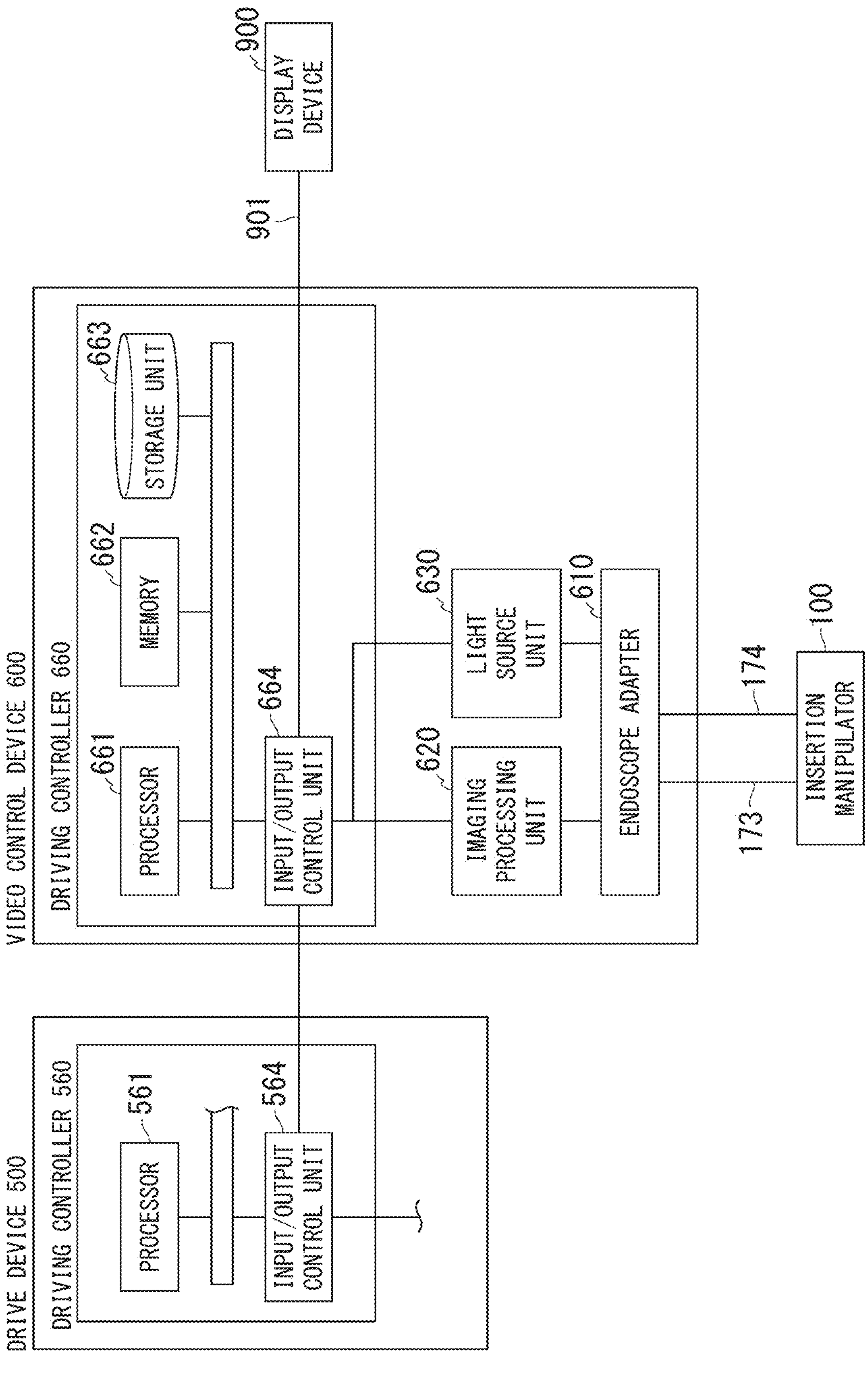
MAIN CONTROLLER 660
PROCESSOR
661
INPUT/OUTPUT CONTROL UNIT
664
DRIVE DEVICE 500
DRIVING CONTROLLER 560
PROCESSOR
561
MEMORY
562
STORAGE UNIT
563
INPUT/OUTPUT CONTROL UNIT
564
OPERATION RECEIVING UNIT
520
AIR SUPPLY /SUCTION DRIVING UNIT
530
DRIVE UNIT (ACTUATOR)
550
ENDOSCOPE ADAPTER
510
160
175, 176, 177
INSERTION MANIPULATOR
100
801
OPERATION DEVICE
800

FIG. 10

FIG. 11


DISPLAY DEVICE
900
901
VIDEO CONTROL DEVICE 600
DRIVING CONTROLLER 660
PROCESSOR
661
MEMORY
662
STORAGE UNIT
663
INPUT/OUTPUT CONTROL UNIT
664
IMAGING PROCESSING UNIT
620
LIGHT SOURCE UNIT
630
ENDOSCOPE ADAPTER
610
173
174
INSERTION MANIPULATOR
100
DRIVE DEVICE 500
DRIVING CONTROLLER 560
PROCESSOR
561
INPUT/OUTPUT CONTROL UNIT
564

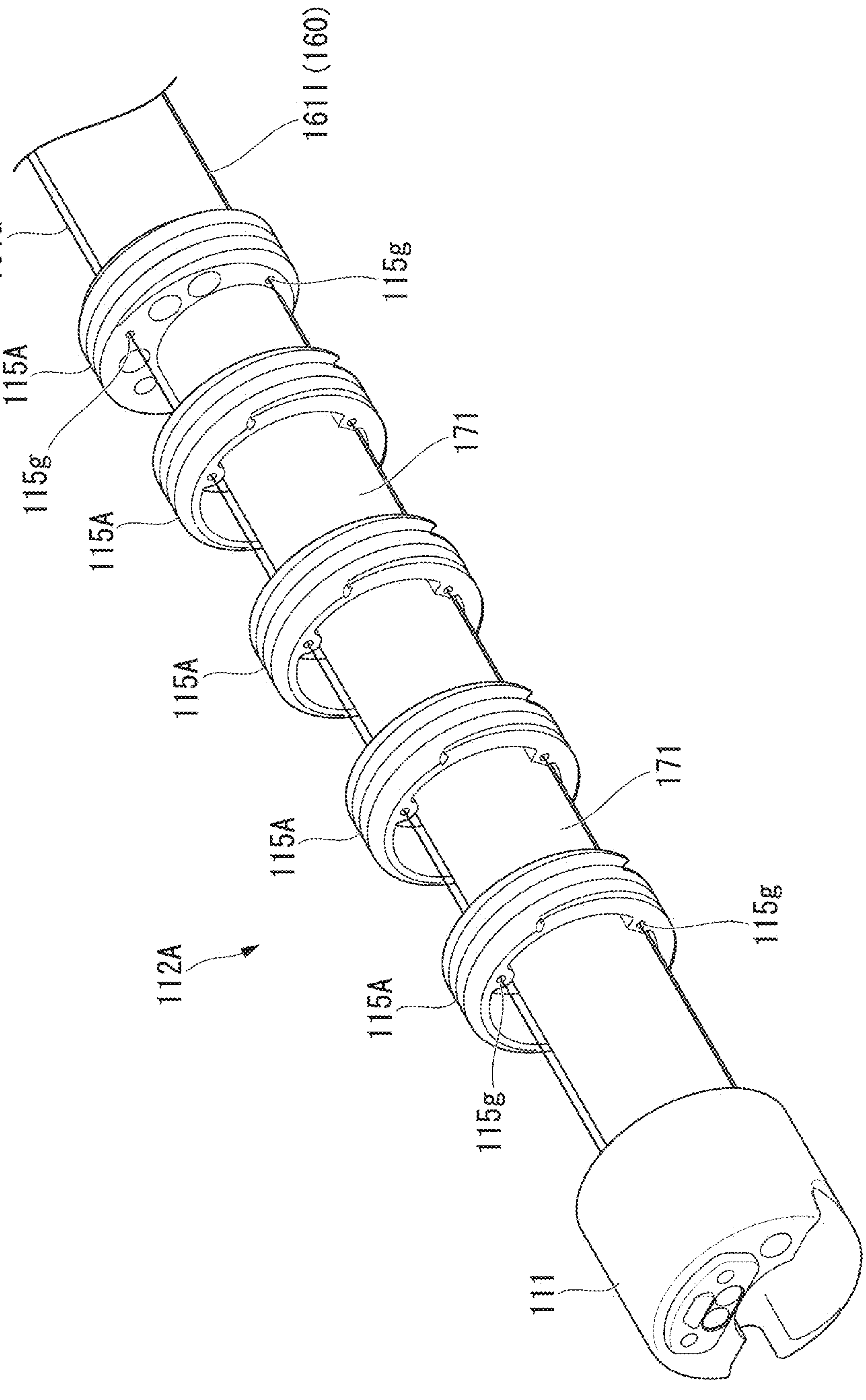
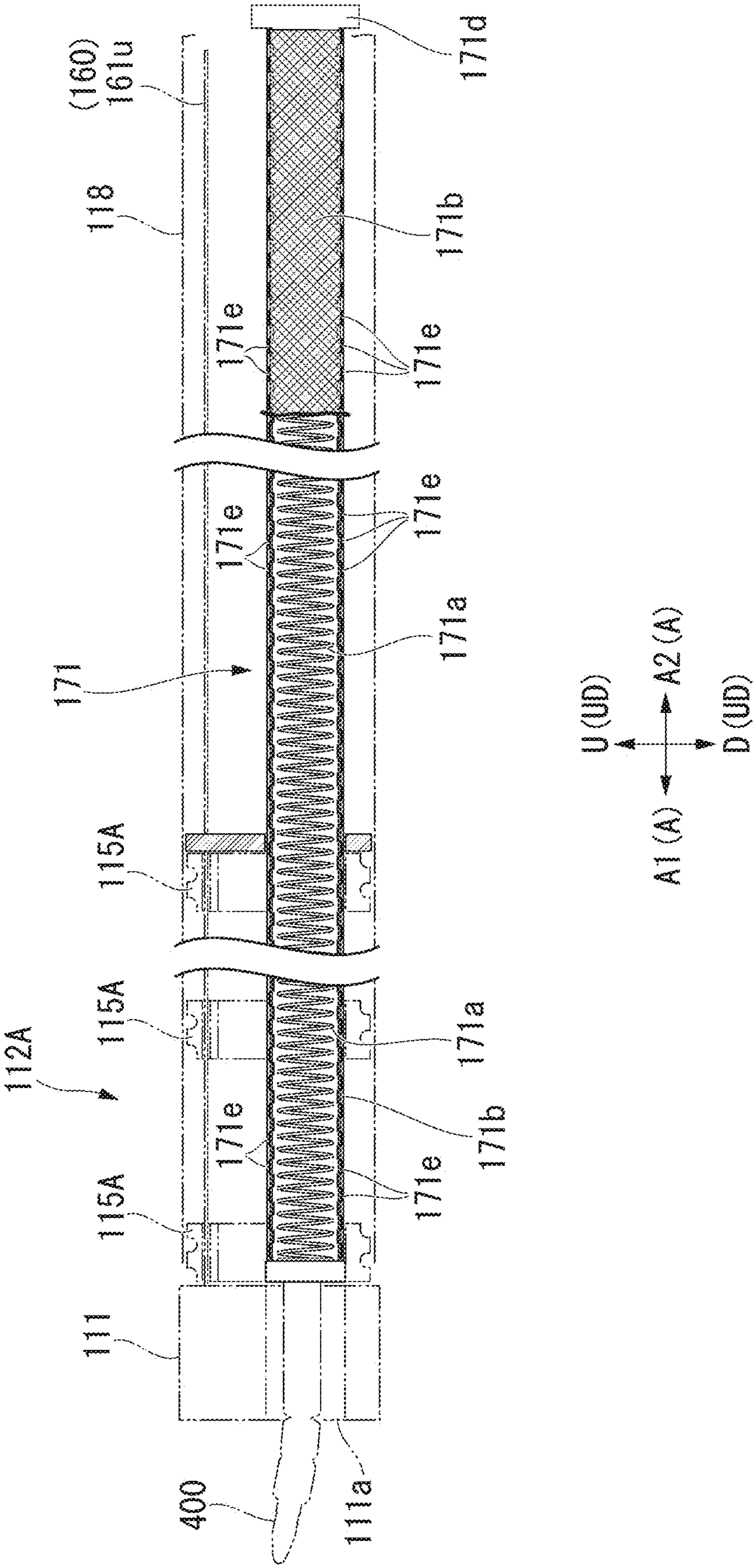

161l (160)
(160)
161u
115g
115A
115g
171
115A
115A
171
115A
115g
112A
115A
115g
111

FIG. 12

FIG. 13



(160)
161u
171d
118
171b
171e
171e
171e
171e
171
171a
U (UD)
A2 (A)
D (UD)
A1 (A)
115A
115A
112A
171a
171b
171e
171e
115A
111
400
111a

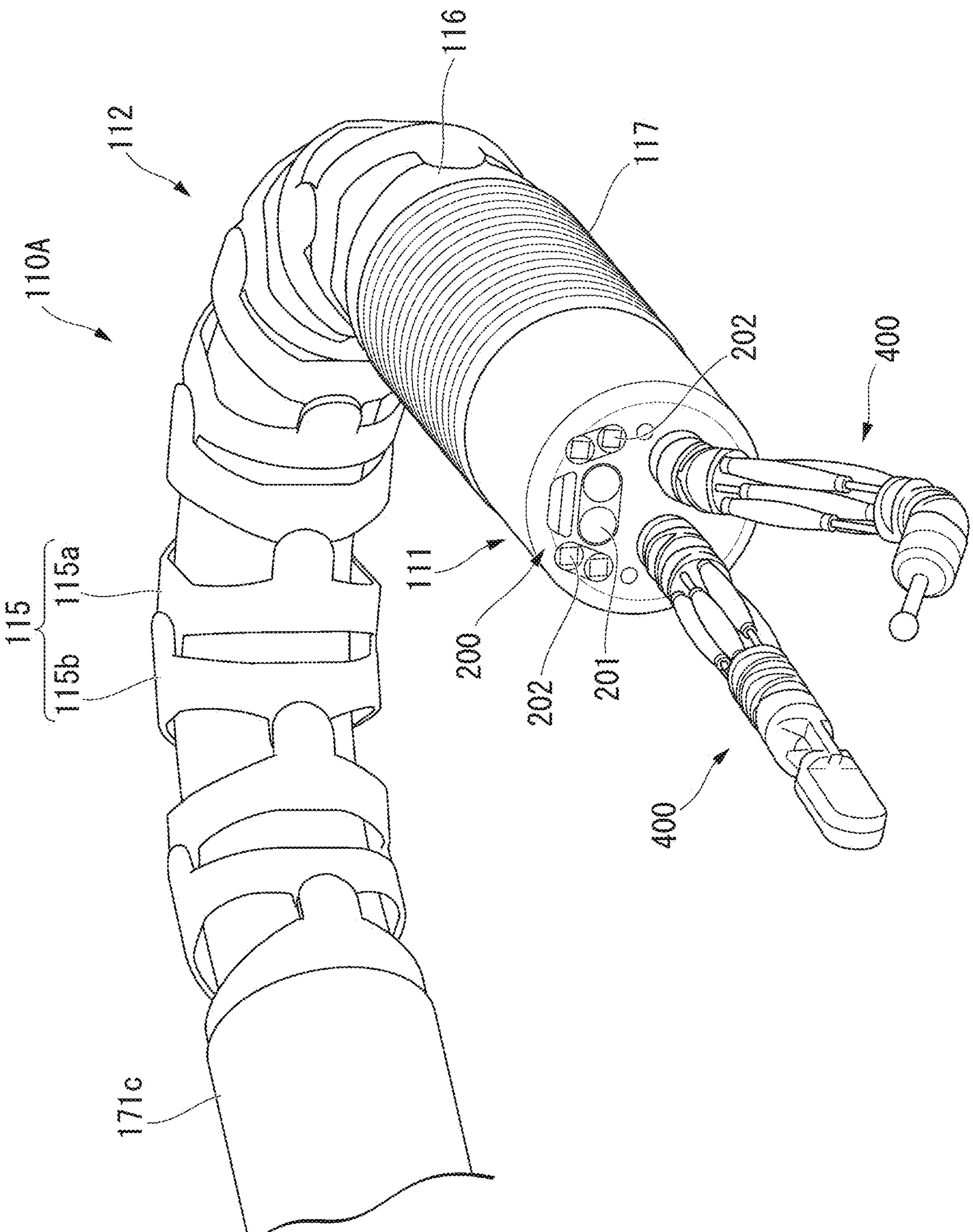


FIG. 14

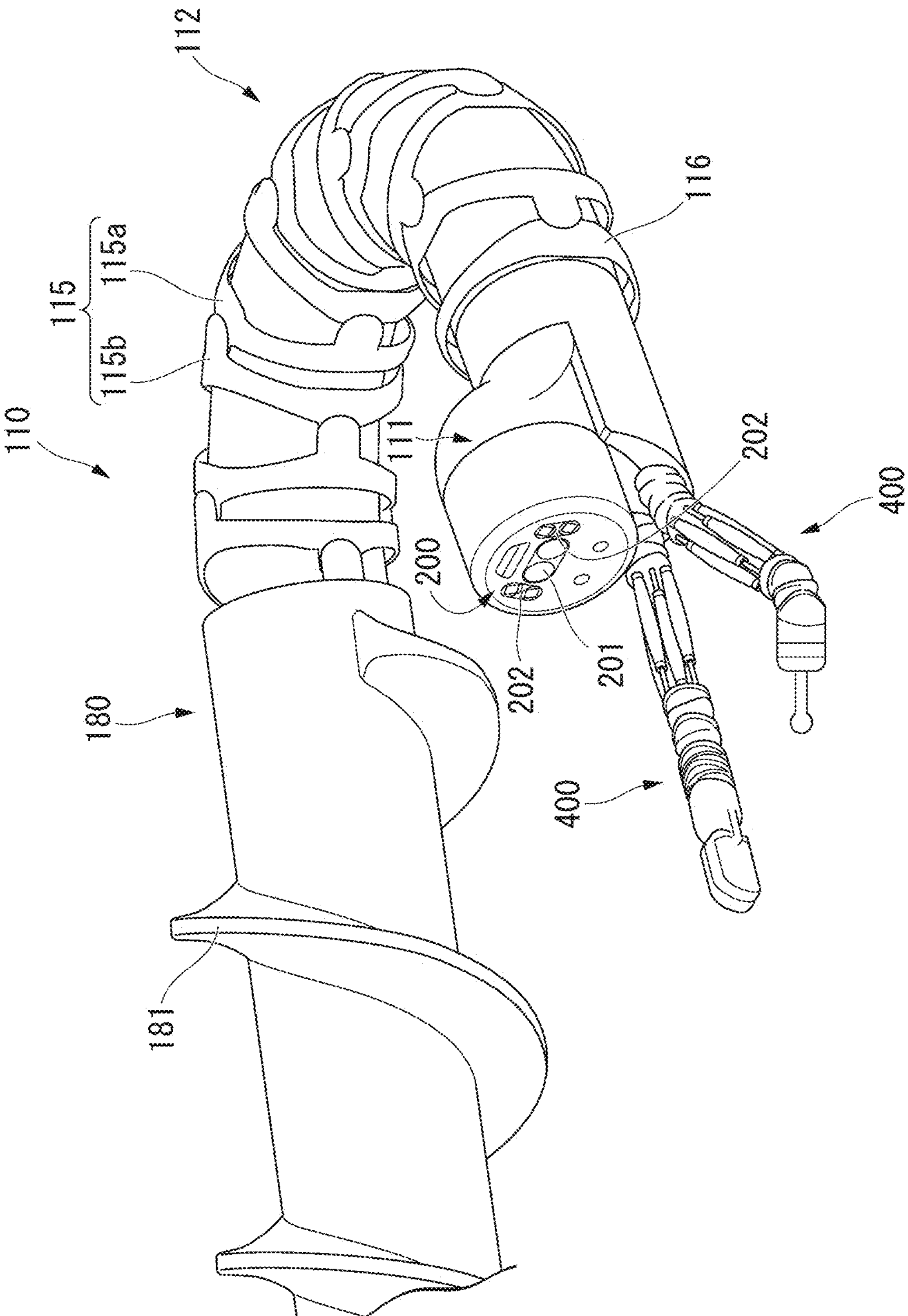


FIG. 15

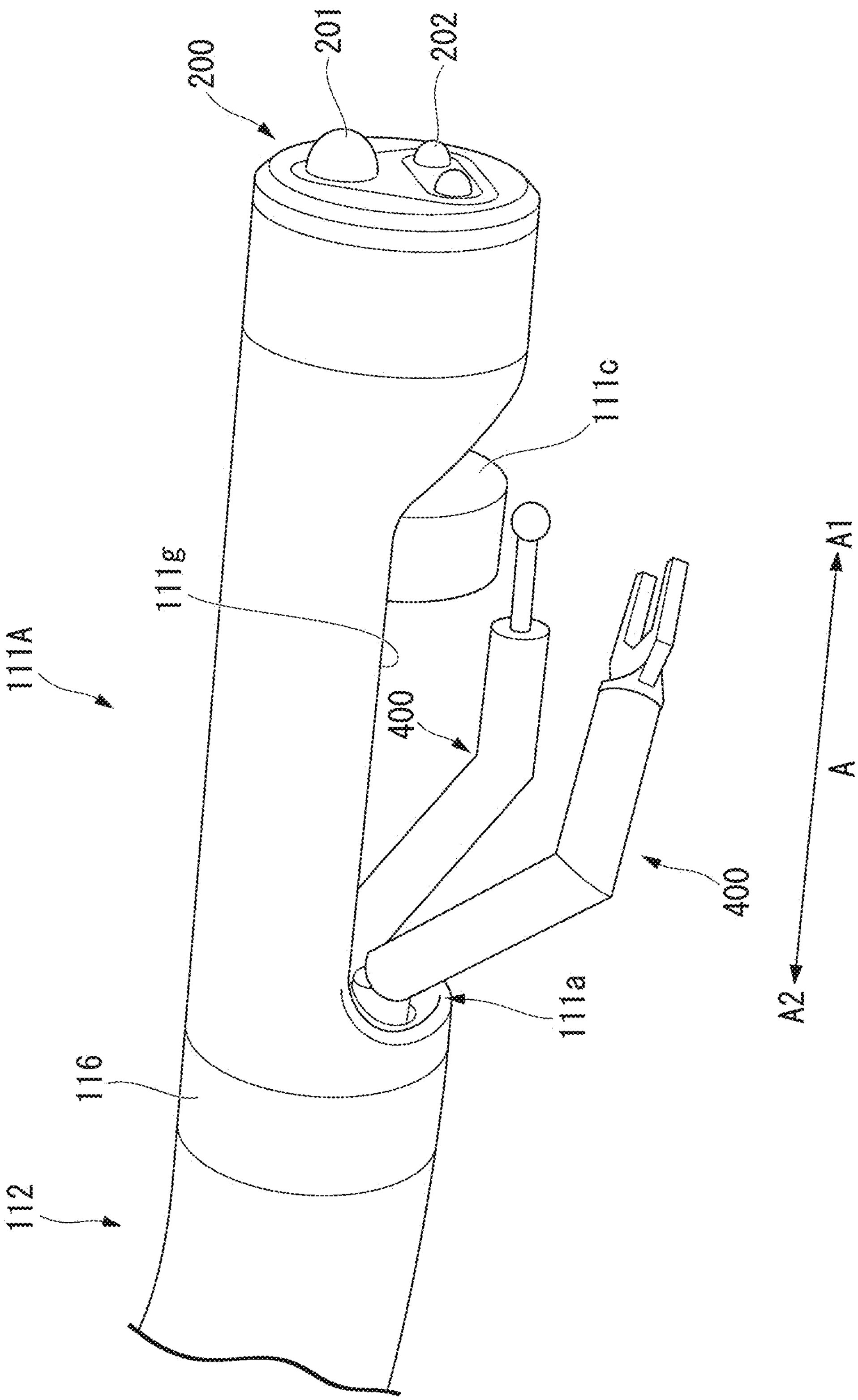
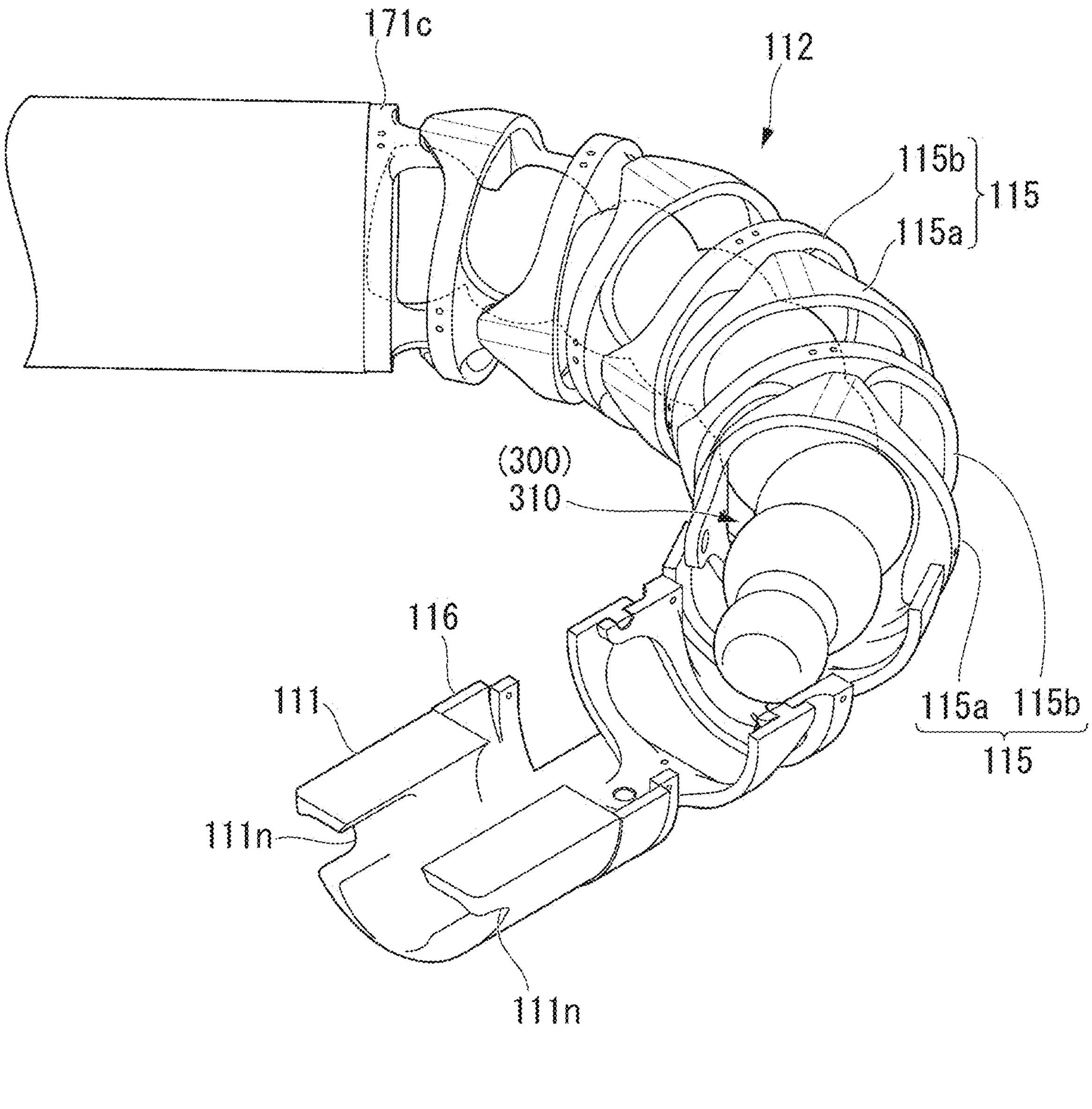


FIG. 16

FIG. 17



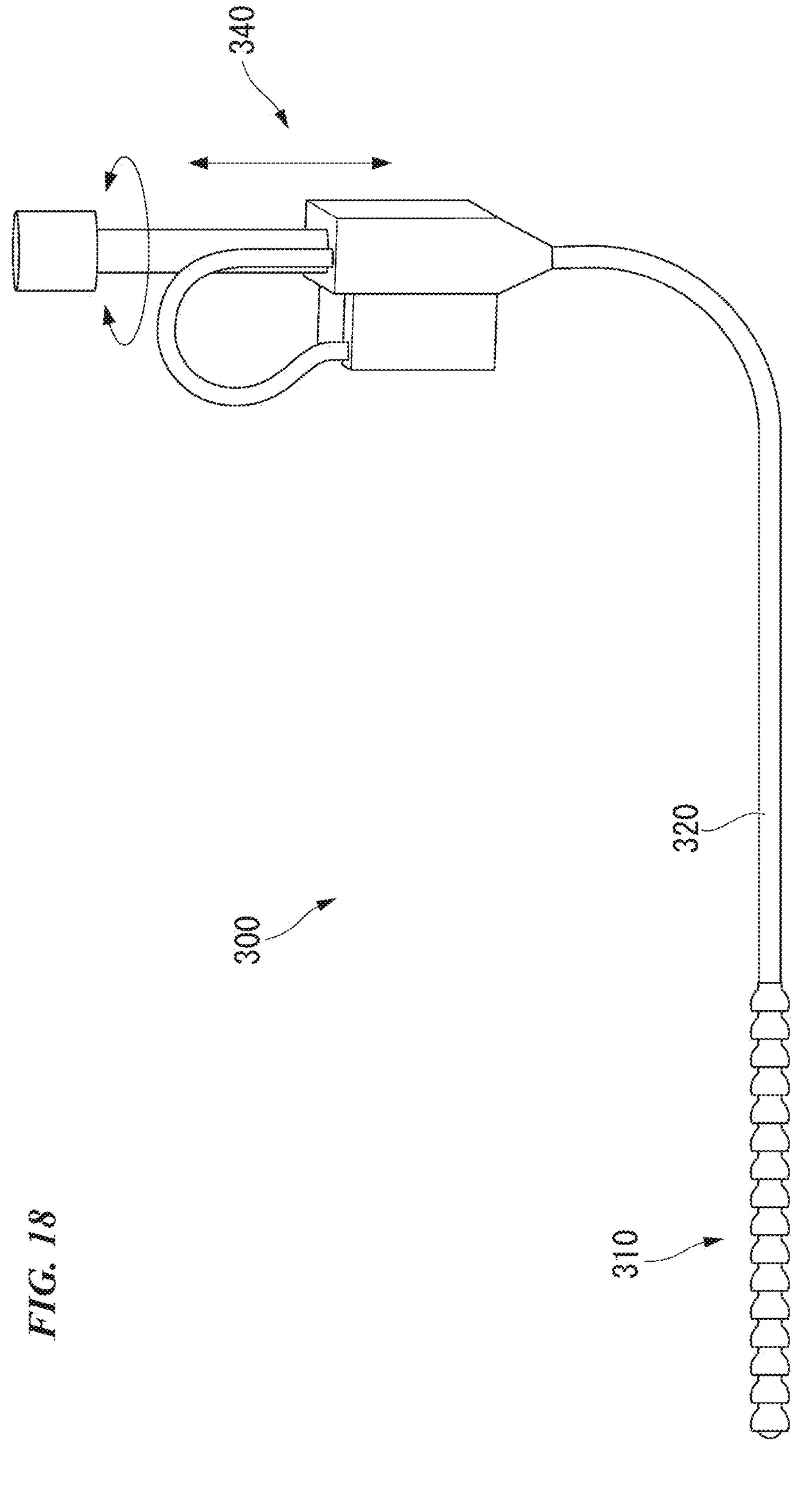


FIG. 18

FIG. 19

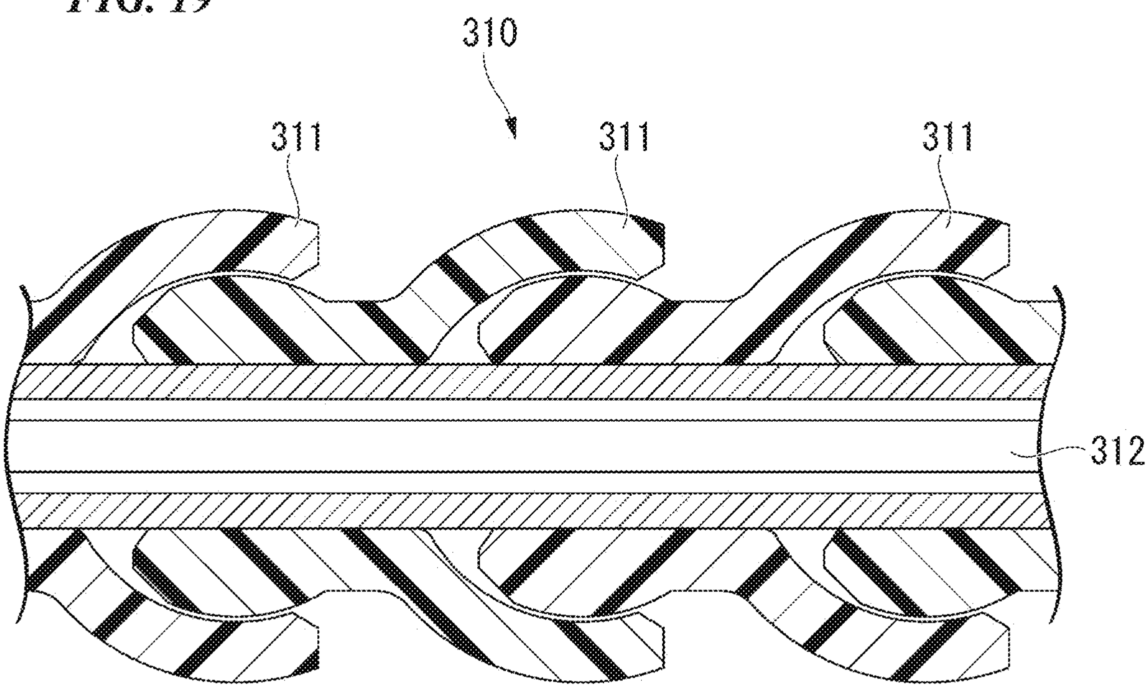


FIG. 20

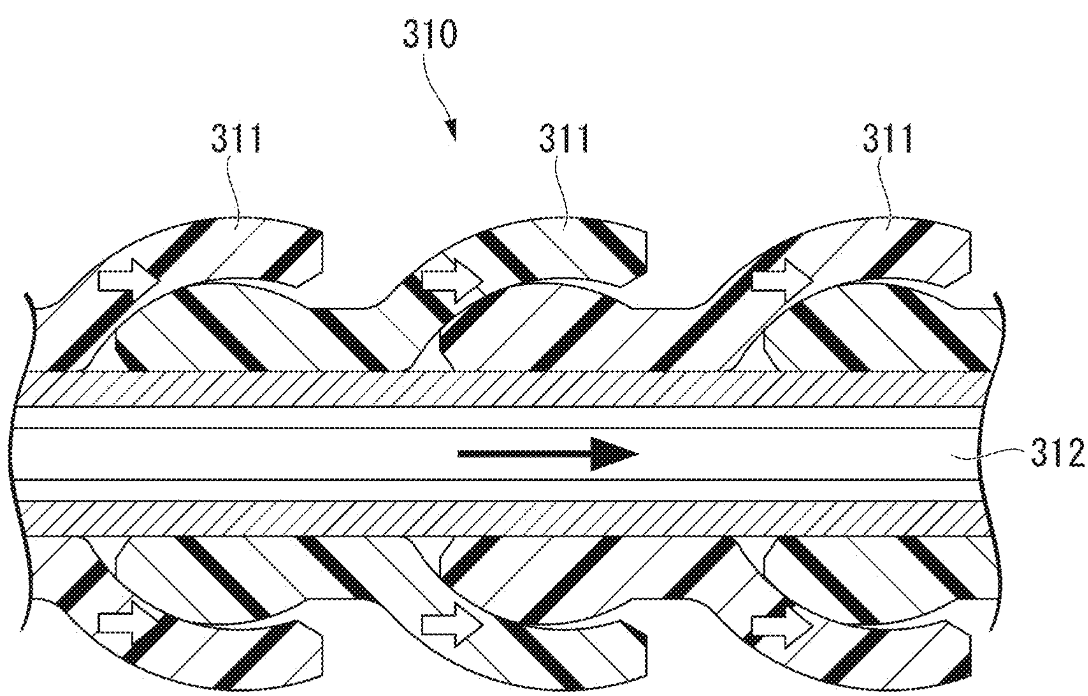


FIG. 21

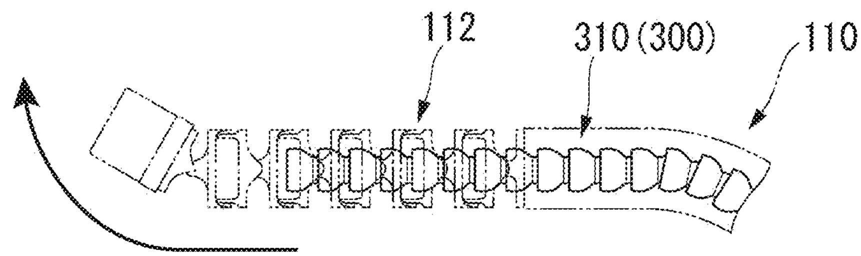


FIG. 22

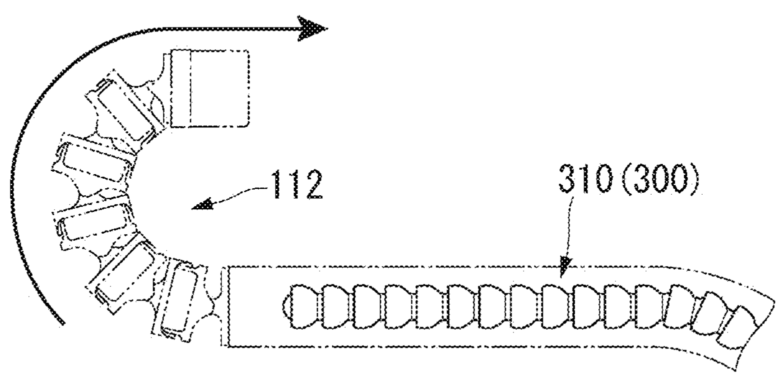


FIG. 23

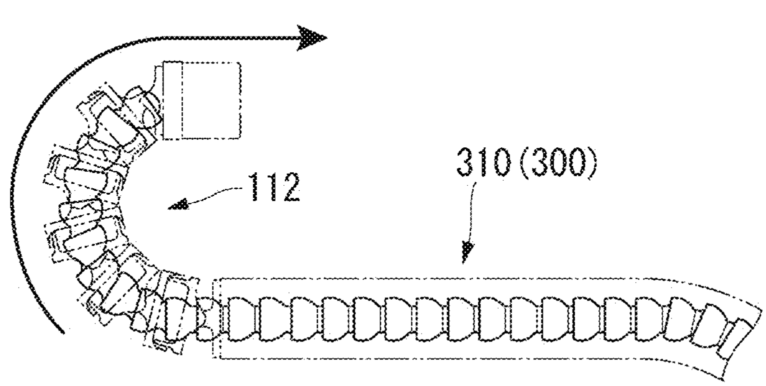


FIG. 24

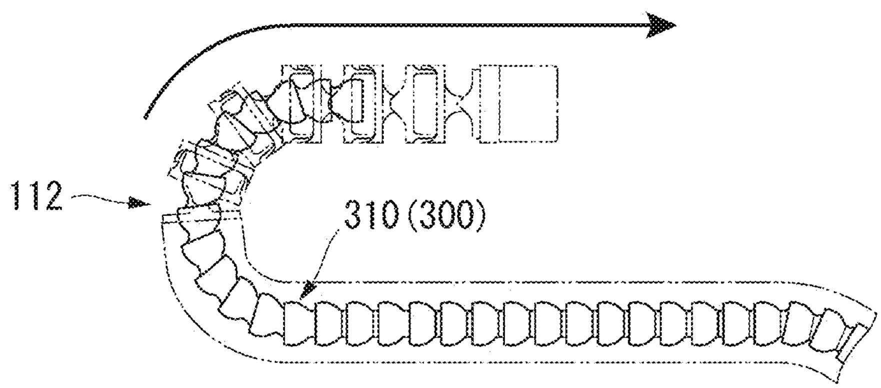
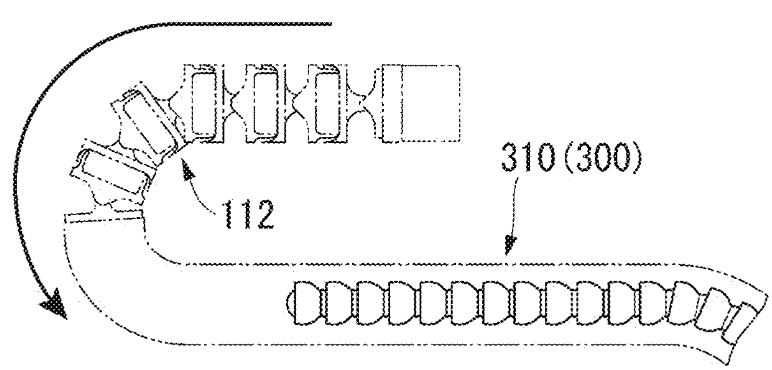


FIG. 25



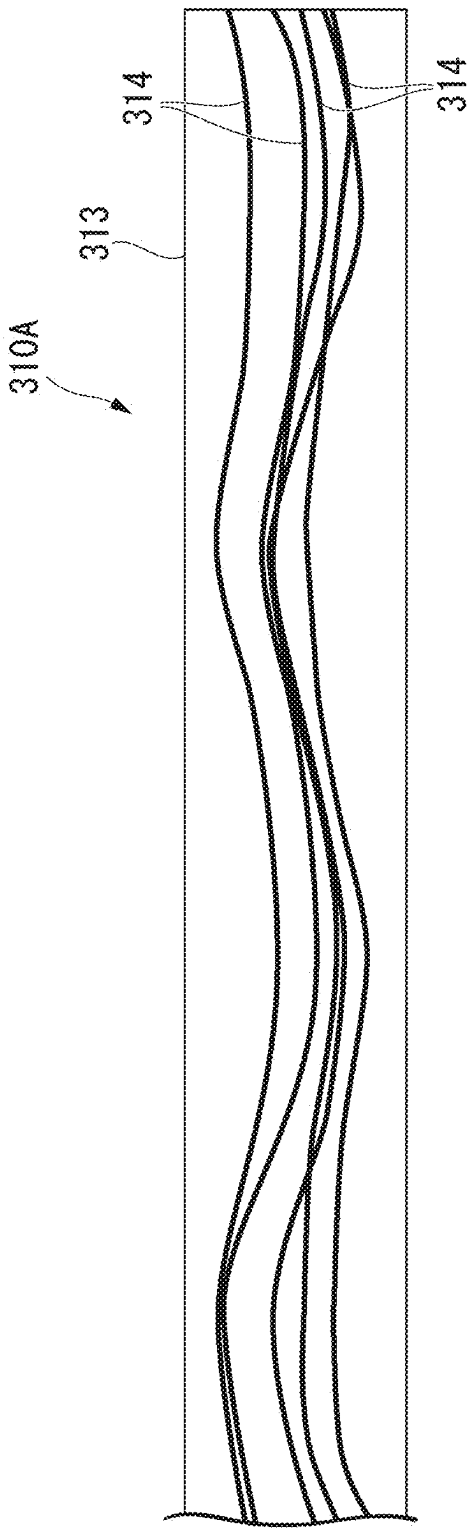


FIG. 26

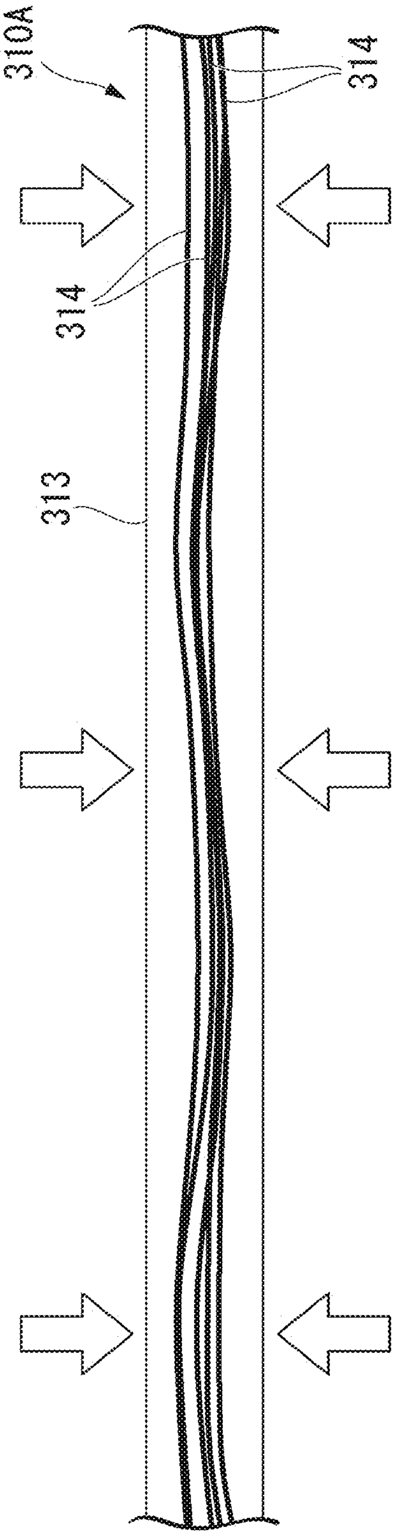


FIG. 27

FIG. 28

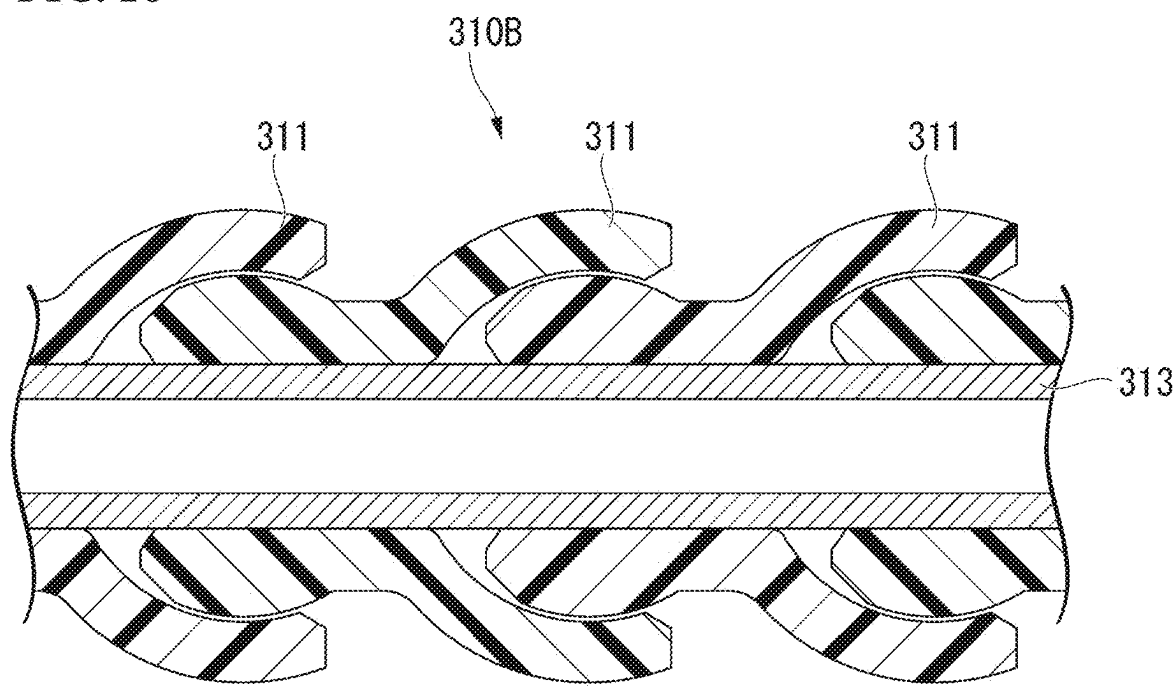


FIG. 29

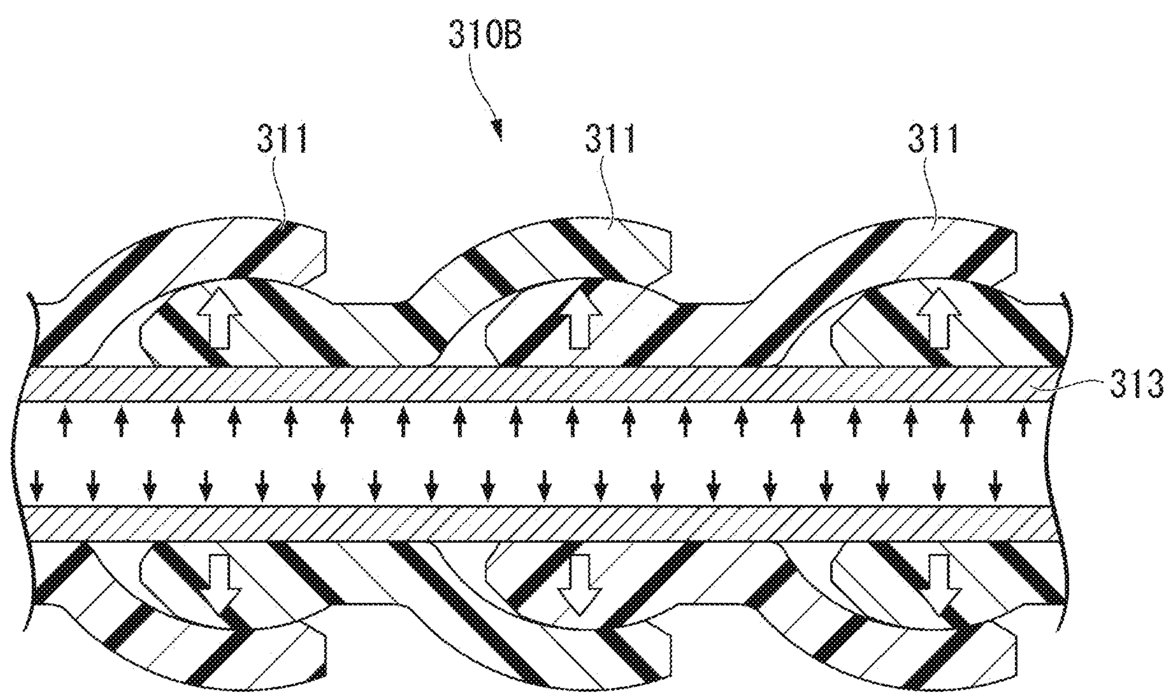


FIG. 30

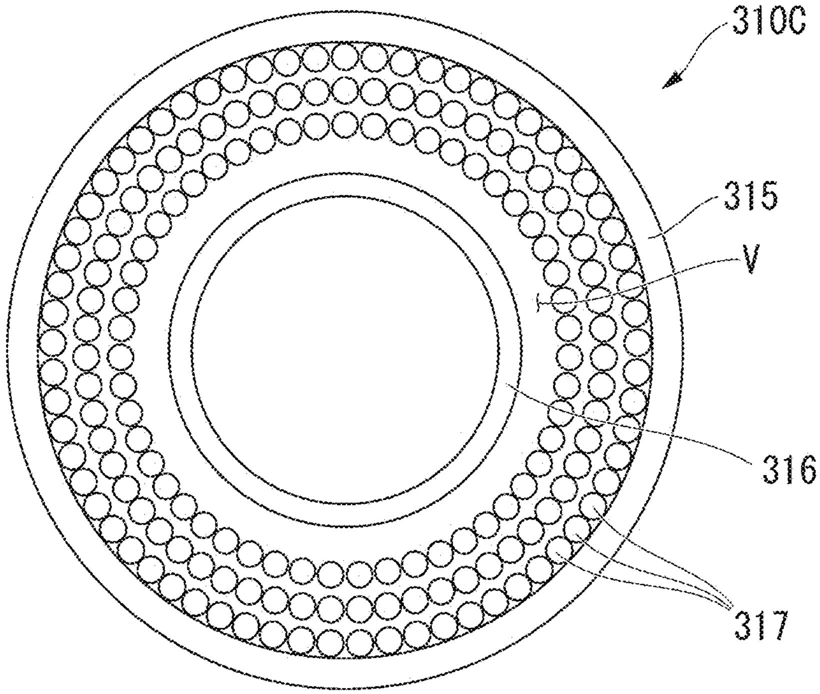
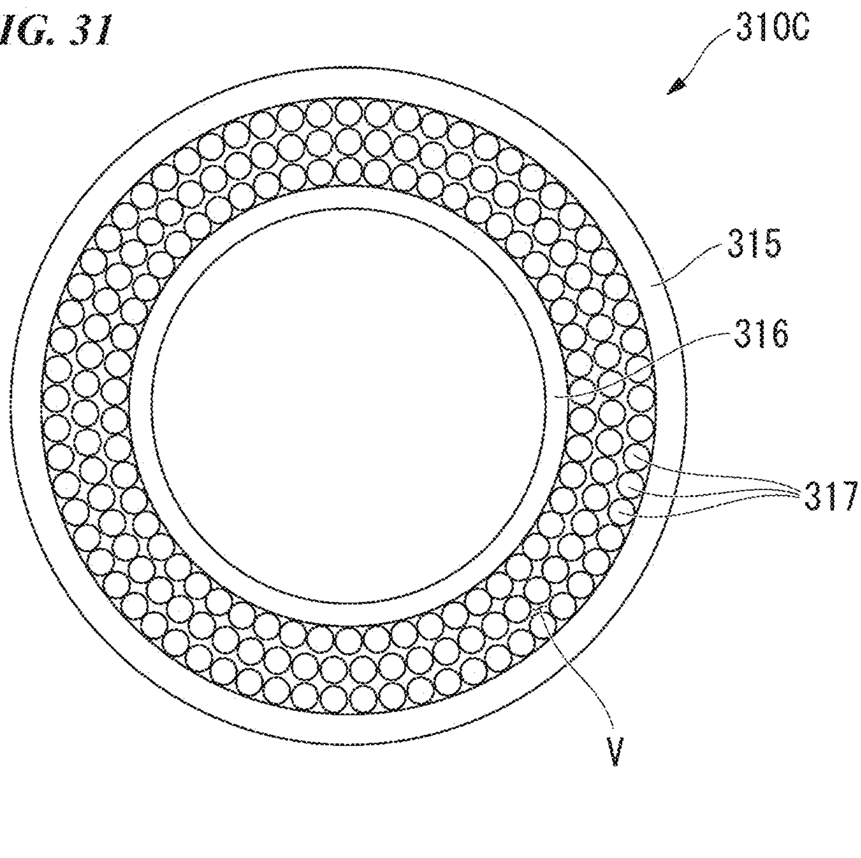


FIG. 31



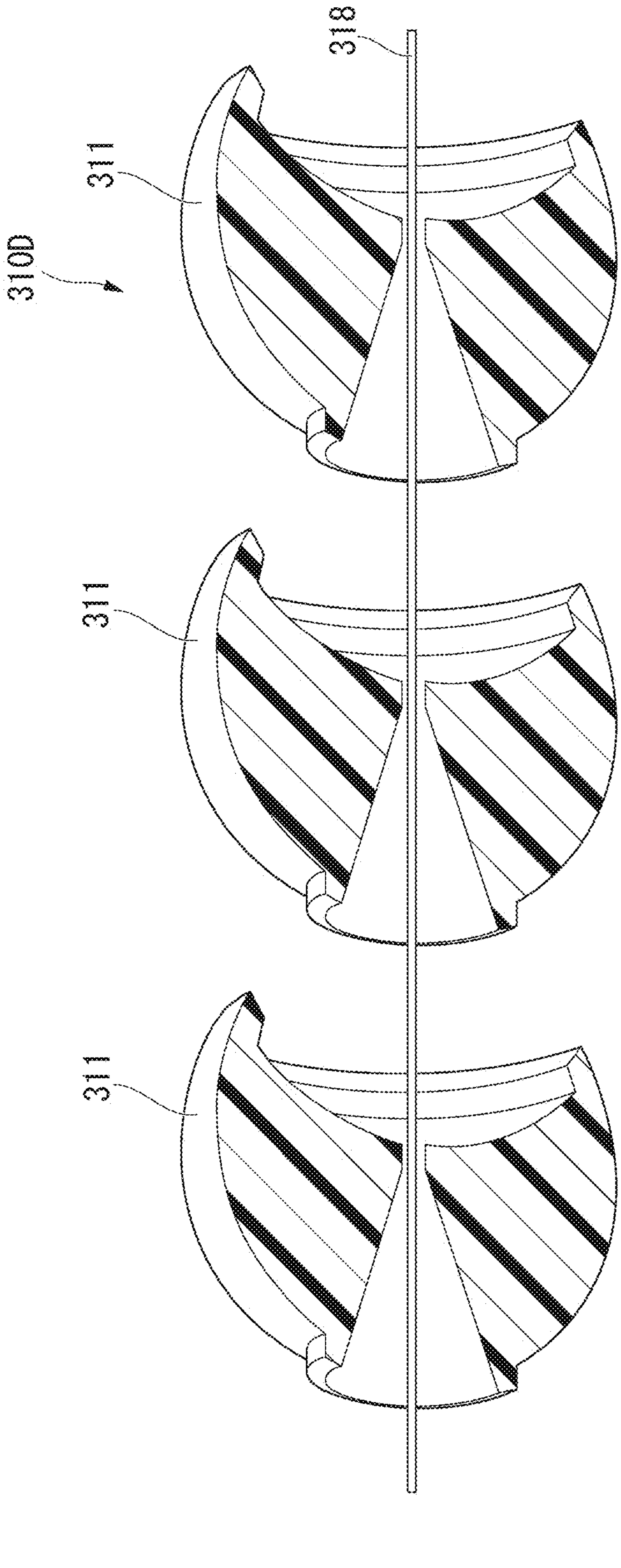


FIG. 32

FIG. 33

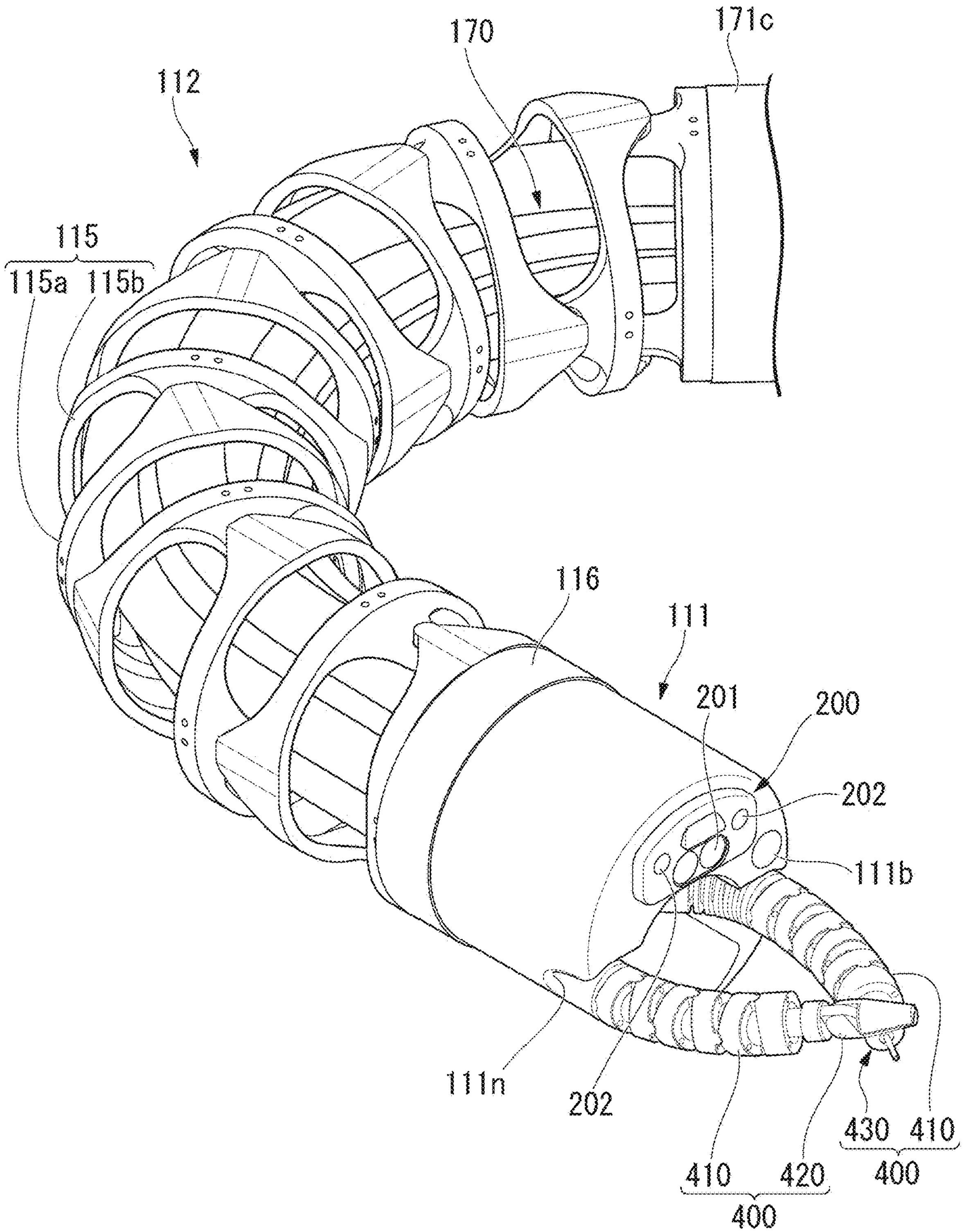

171c
170
112
115
115a 115b
116
111
201
200
202
111b
111n
202
410
420
400
430 410
400

FIG. 34

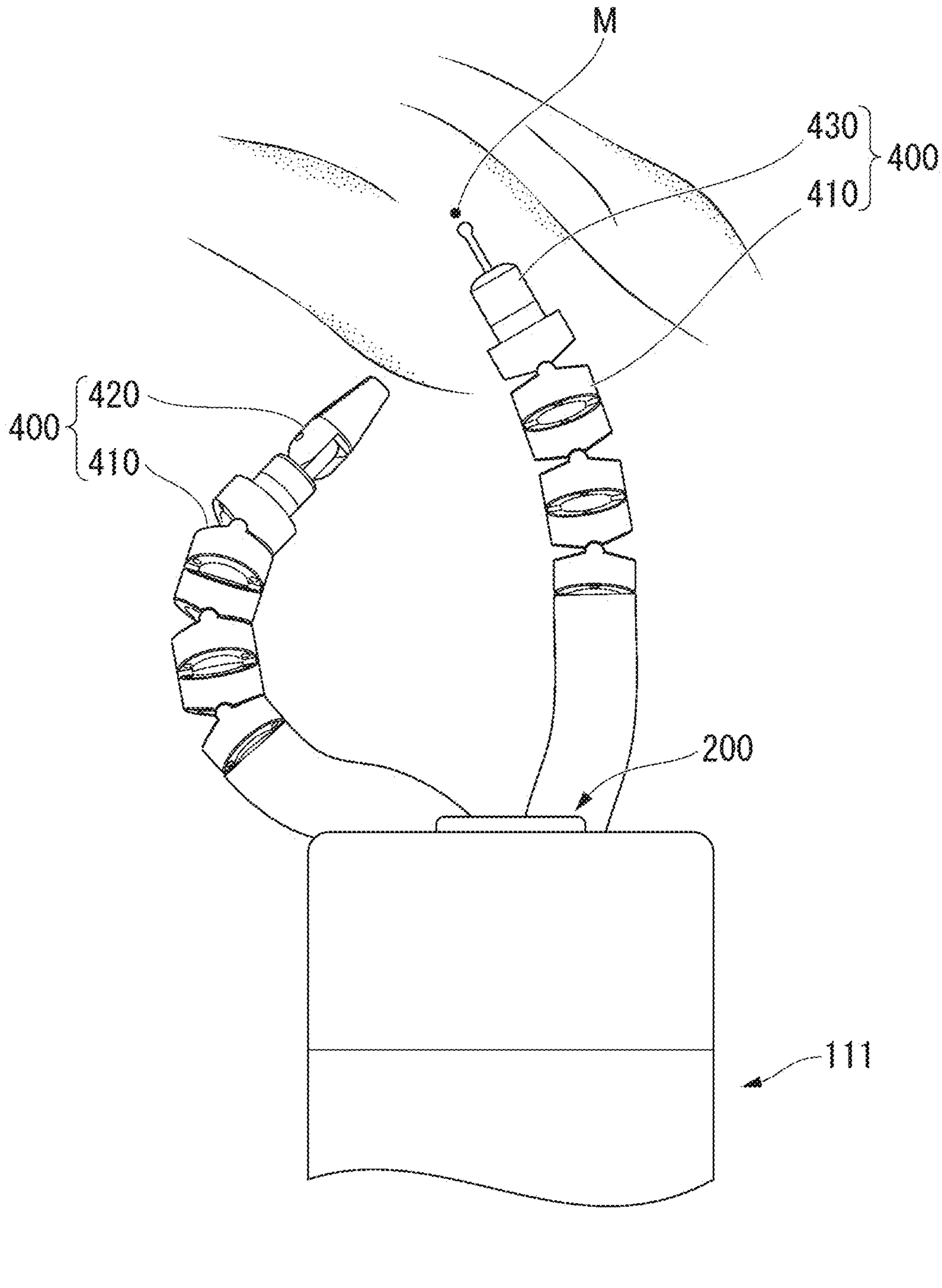


FIG. 35

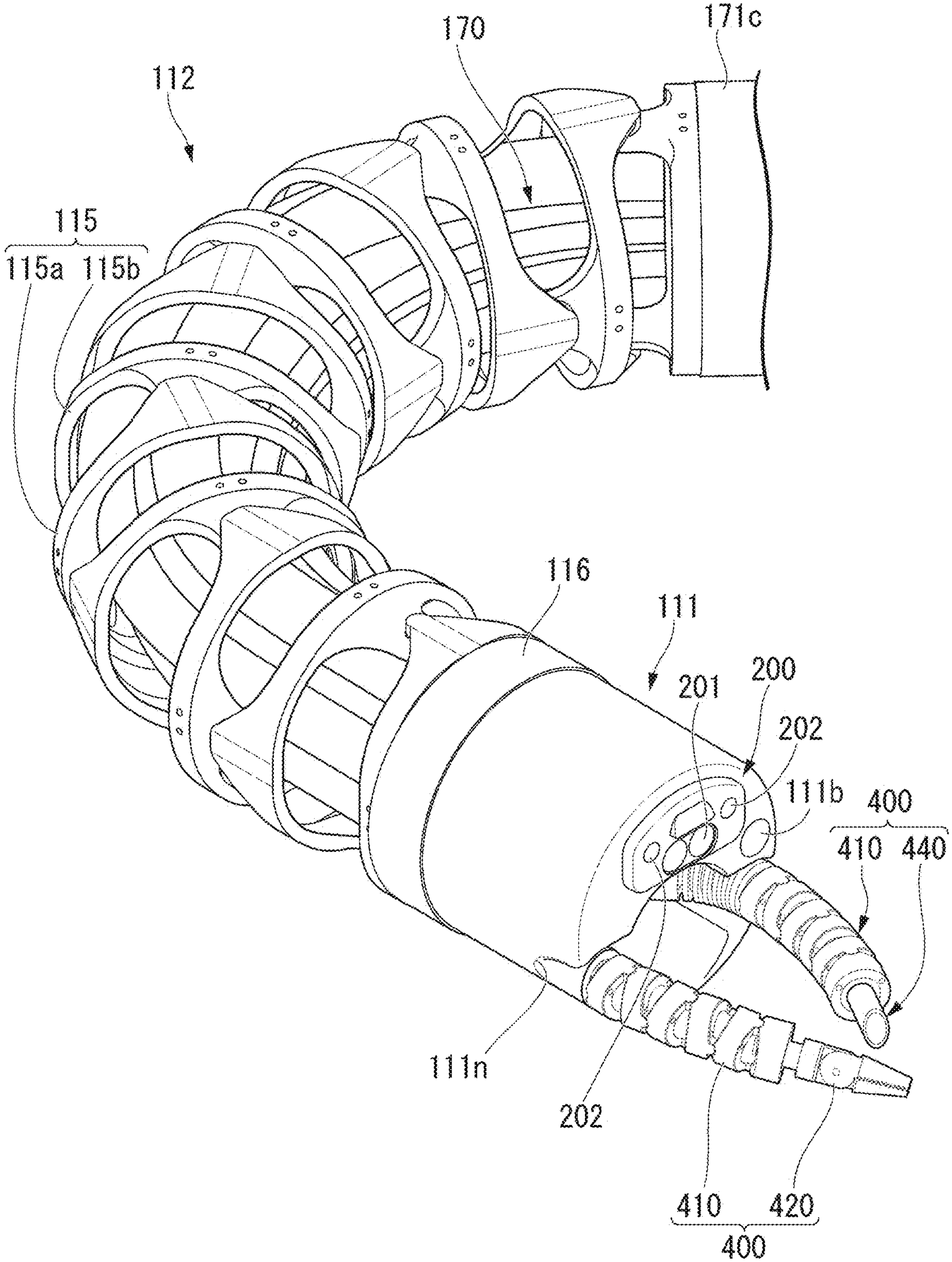

170
171c
112
115
115a 115b
116
111
201
200
202
111b
400
410 440
111n
202
410 420
400

FIG. 36

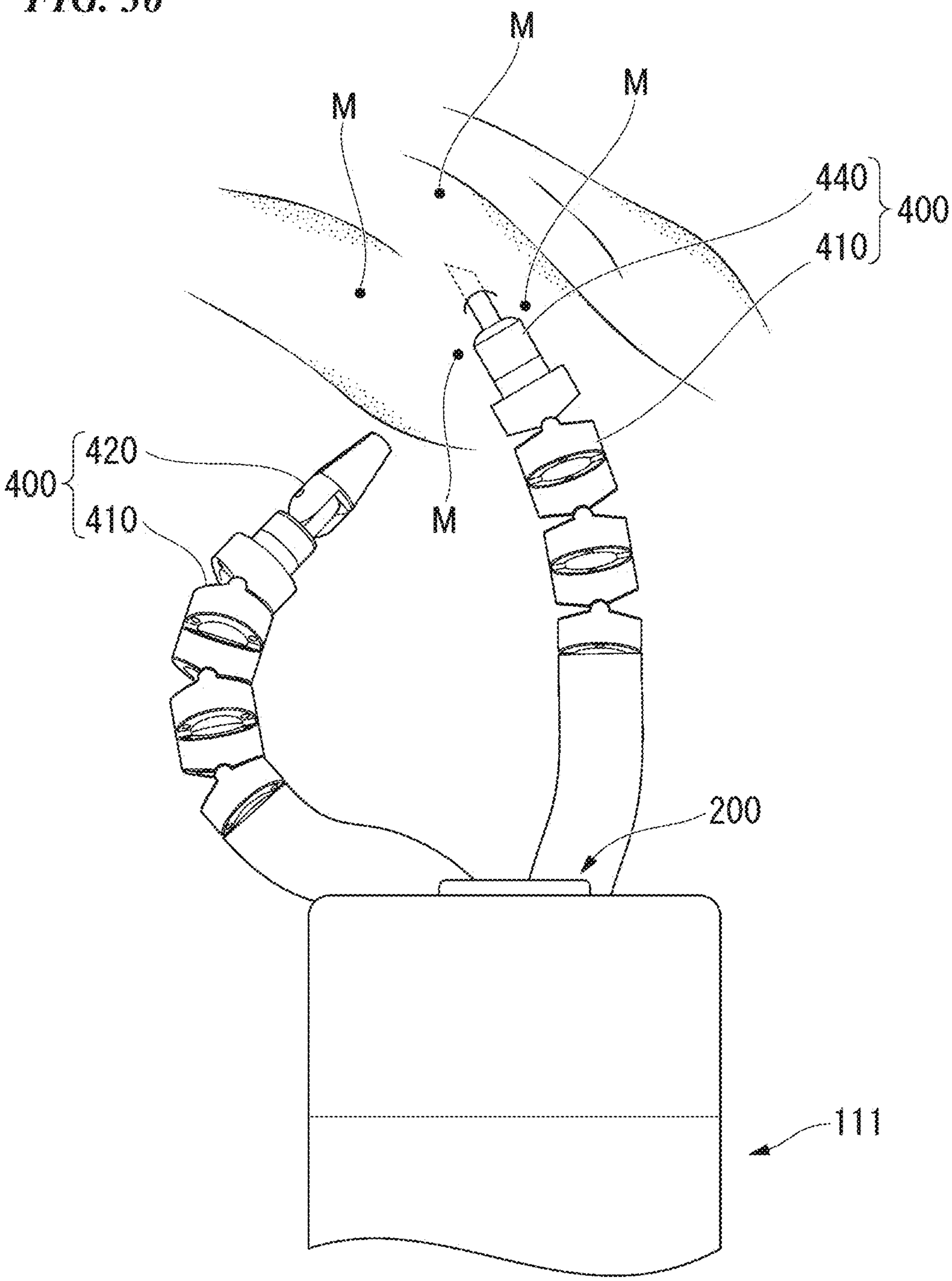


FIG. 37

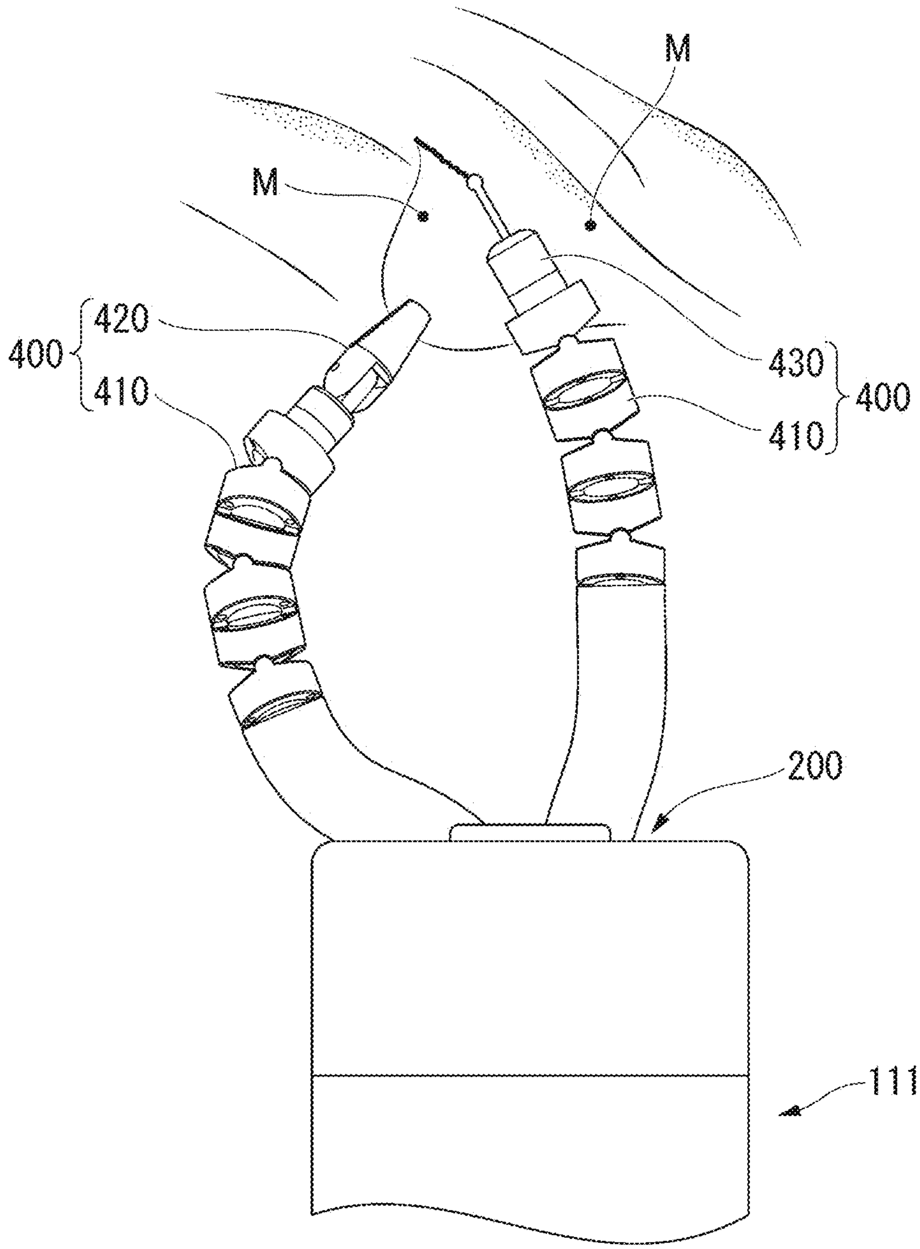


FIG. 38

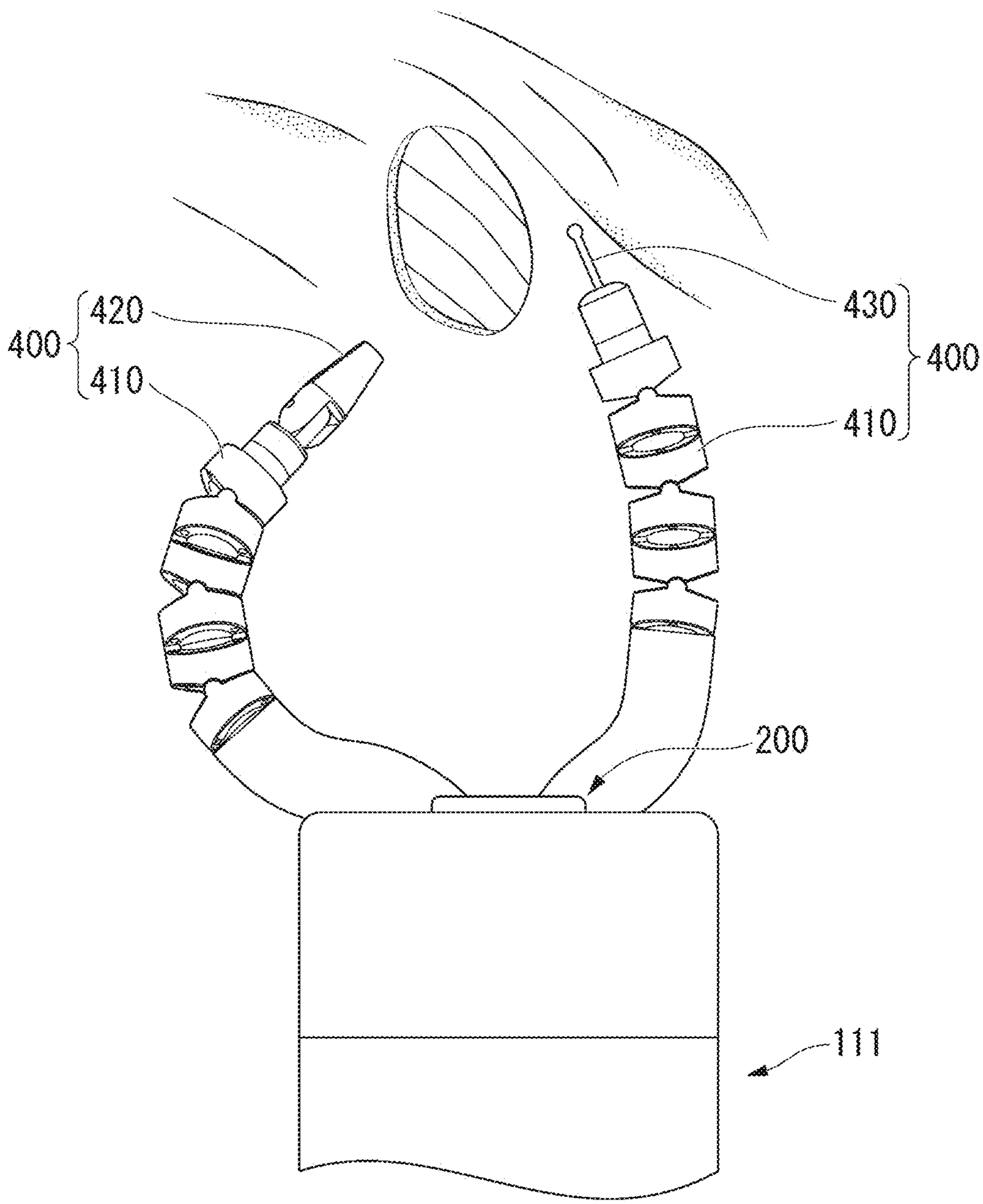


FIG. 39

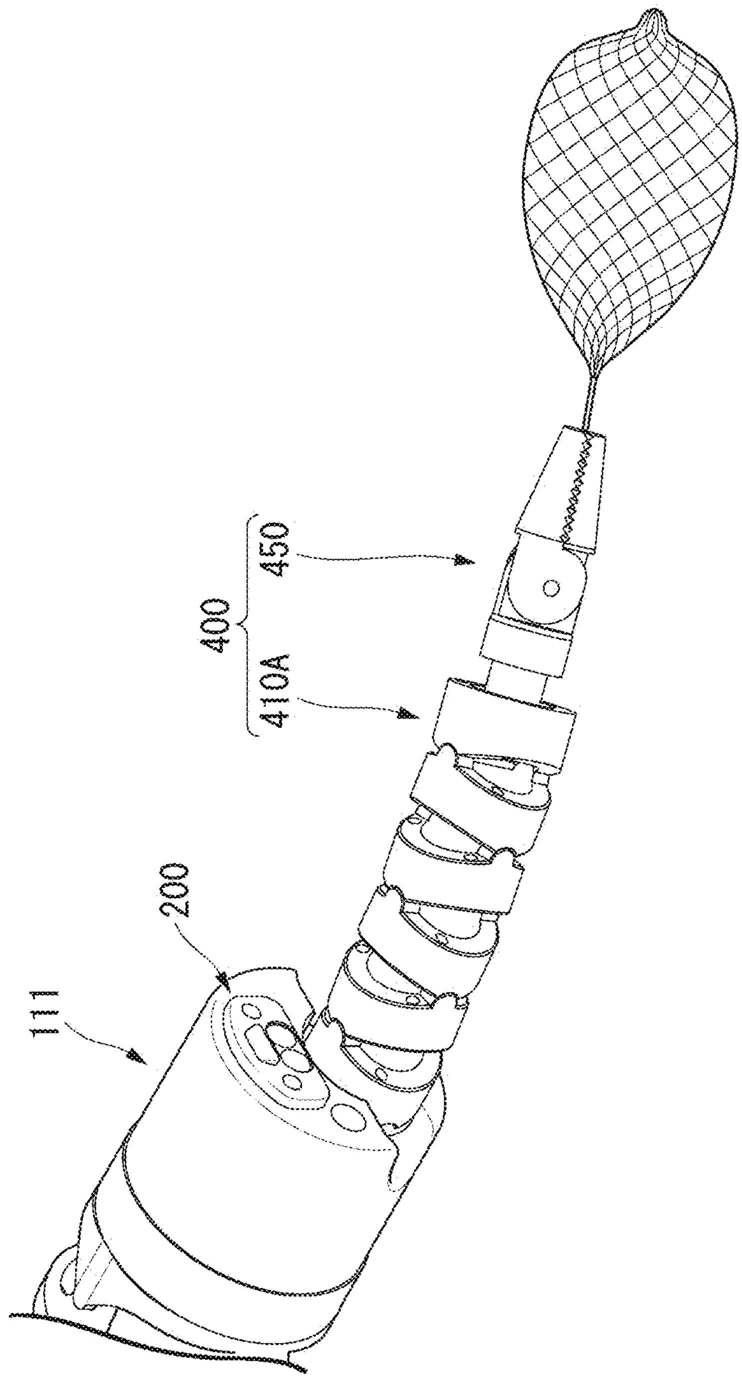
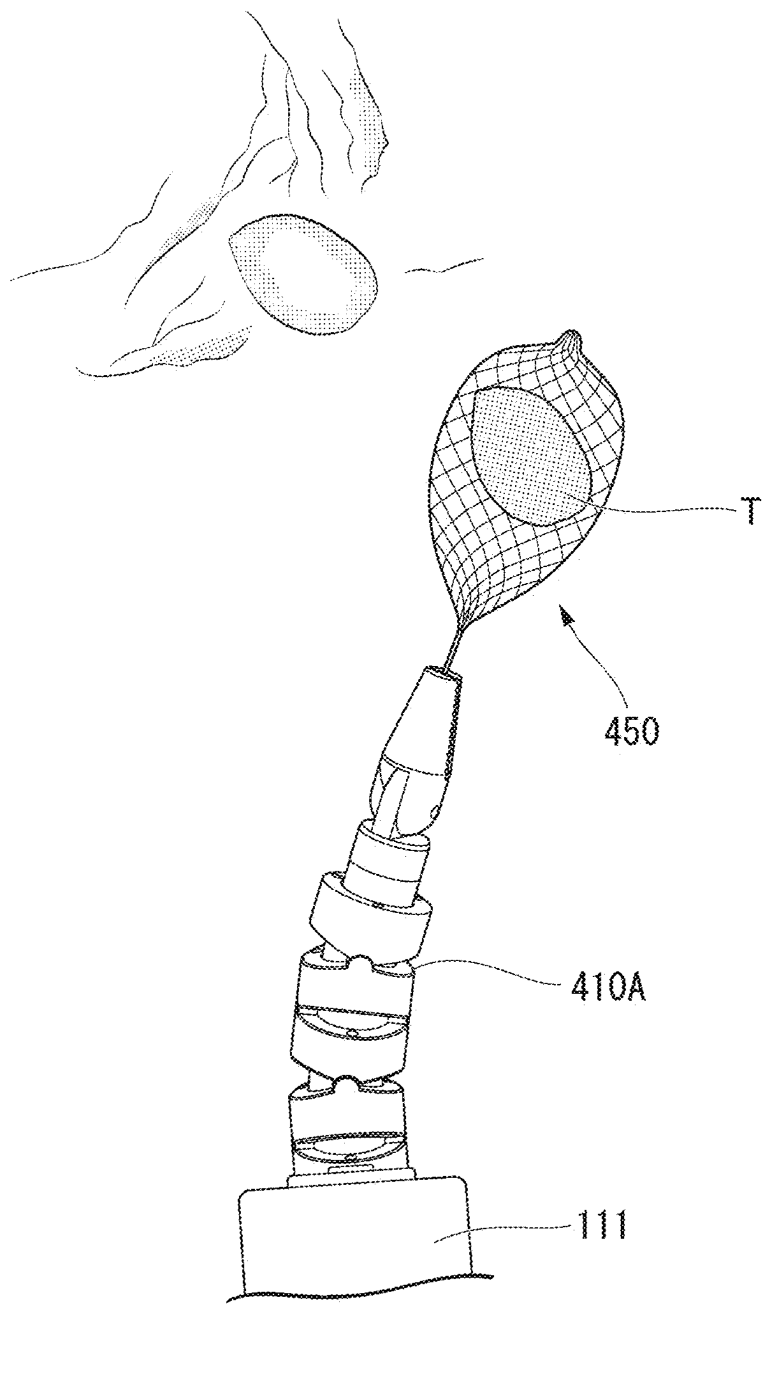
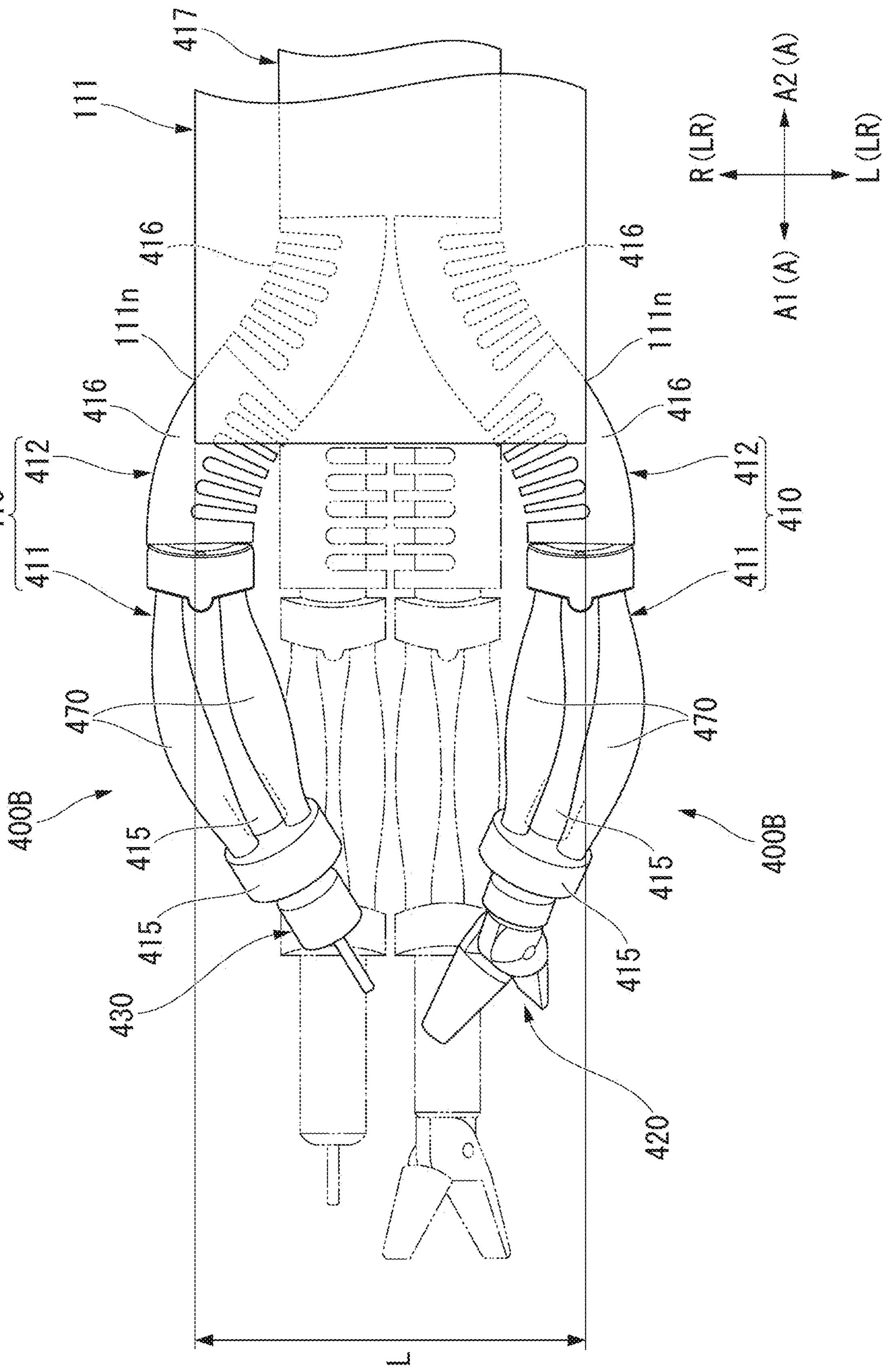


FIG. 40





417
111
416
111n
412
410
411
470
400B
415
430
420
L
R (LR)
L (LR)
A1 (A)
A2 (A)

FIG. 41

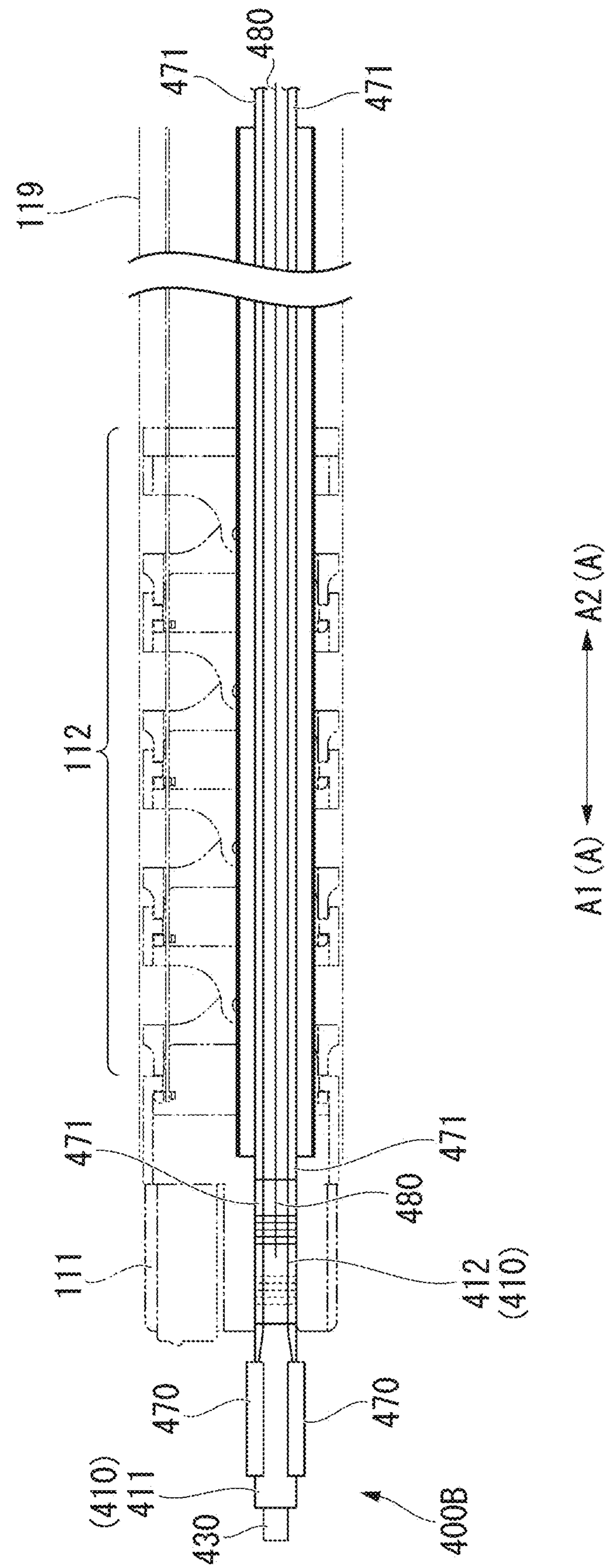

480
471
471
119
112
A2 (A)
A1 (A)
471
471
480
111
412
(410)
470
470
(410)
411
430
400B

FIG. 42

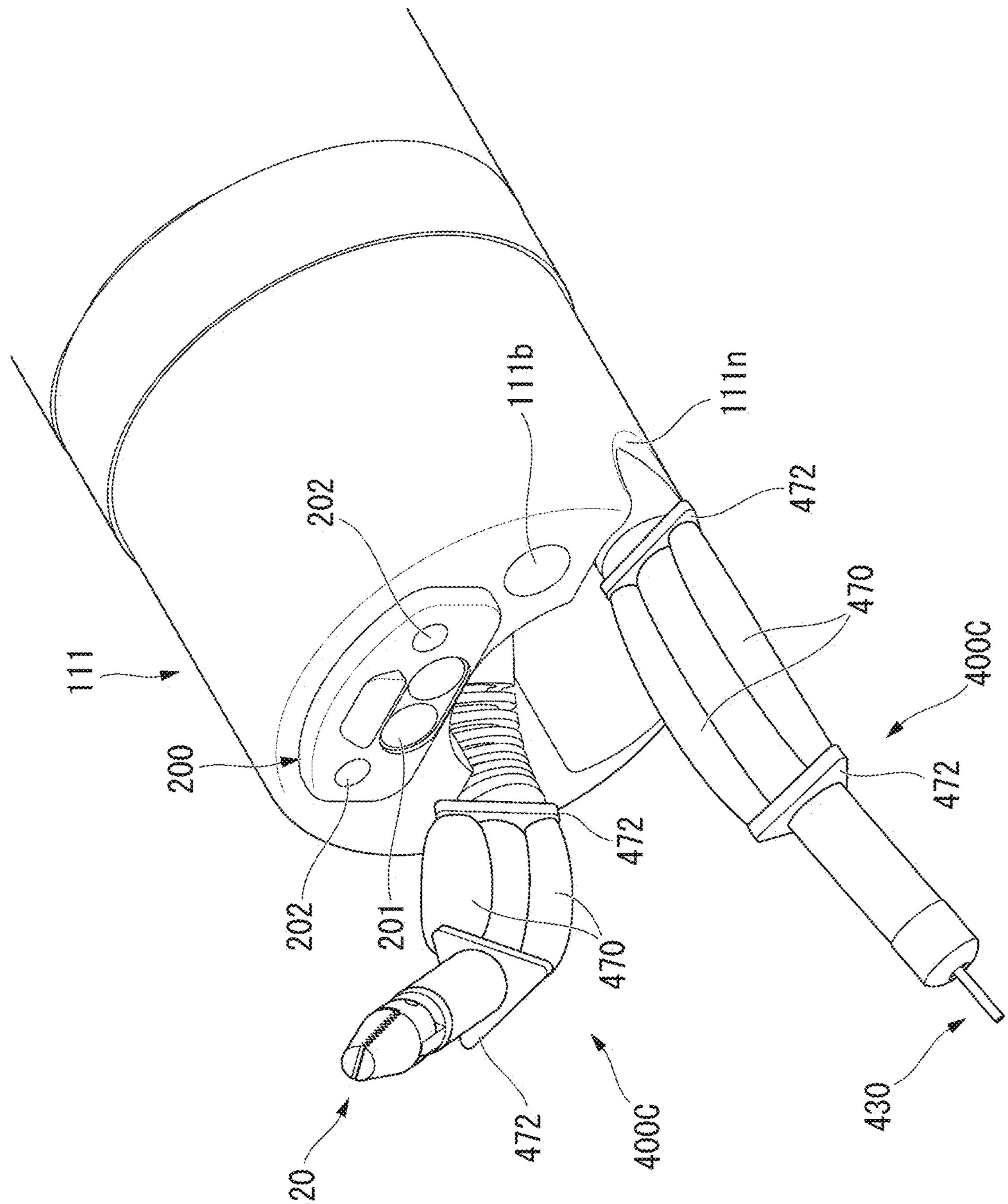
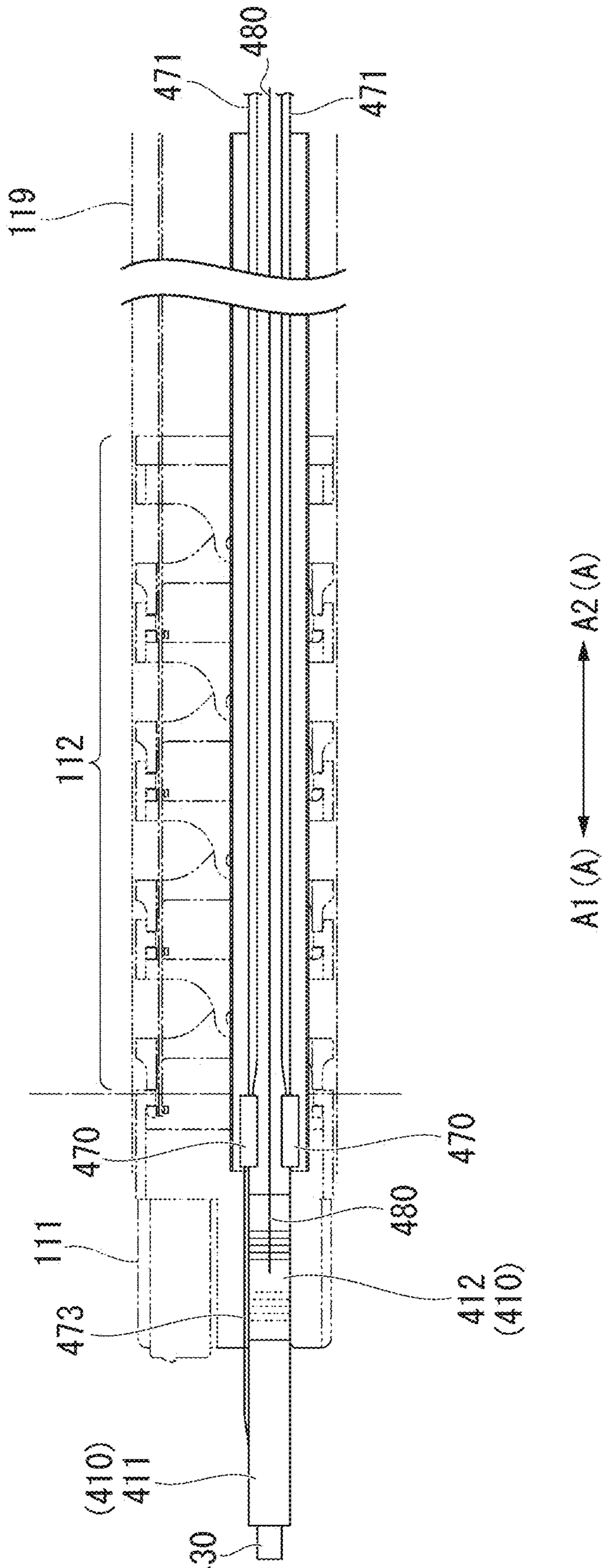

111
200
202
201
202
111b
111n
472
470
472
472
470
472
400C
400C
420
430

FIG. 43

FIG. 44


480
471
471
119
112
A2 (A)
A1 (A)
470
470
480
111
412
(410)
473
(410)
411
430

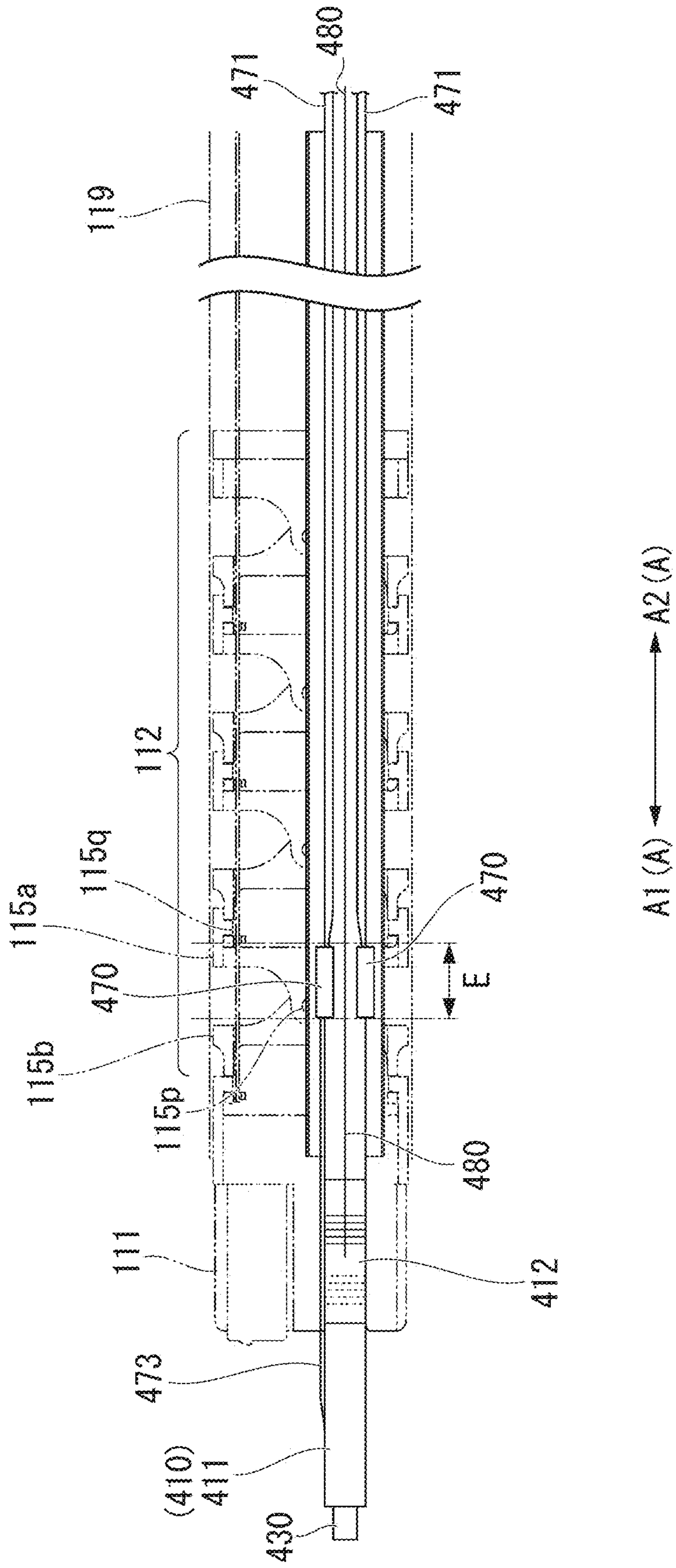

480
471
471
119
112
A2 (A)
A1 (A)
115q
115a
470
470
E
115b
115p
480
111
412
473
(410)
411
430

FIG. 45

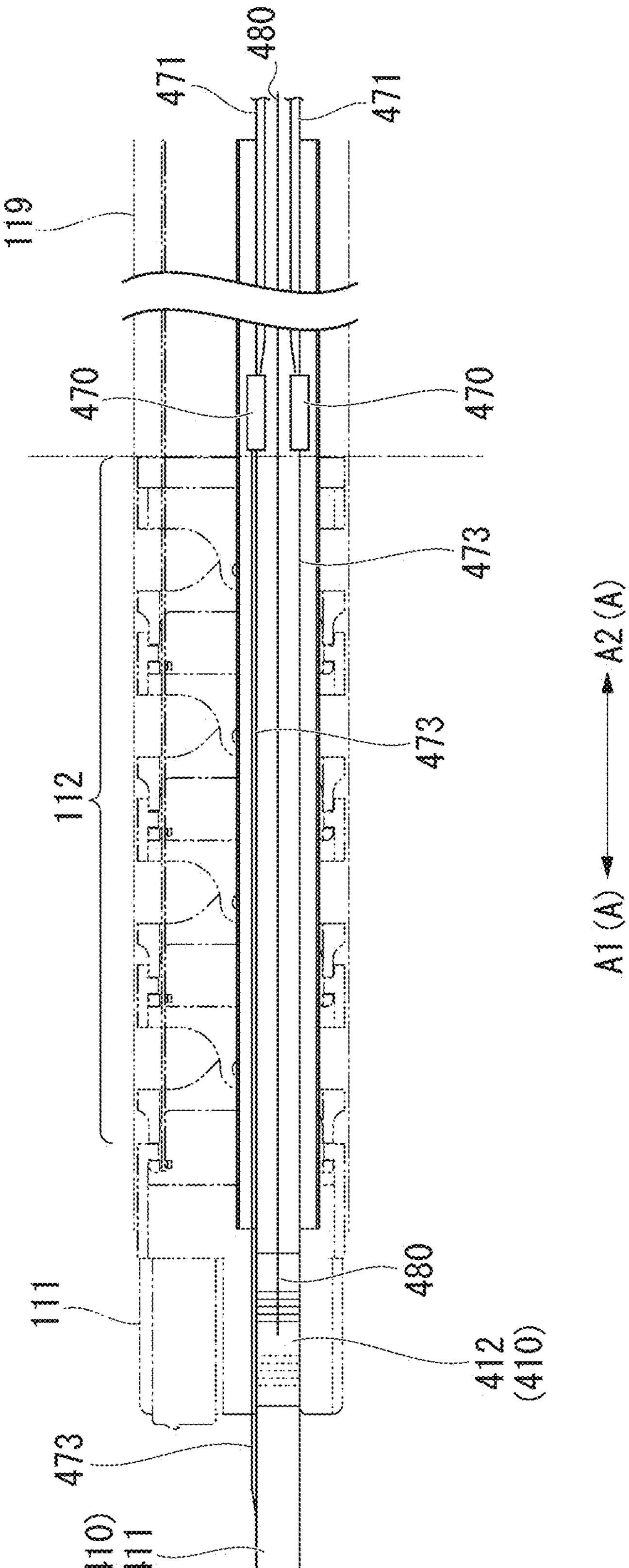

480
471
471
119
470
470
473
473
112
A2 (A)
A1 (A)
111
480
412
(410)
473
(410)
411
430

FIG. 46

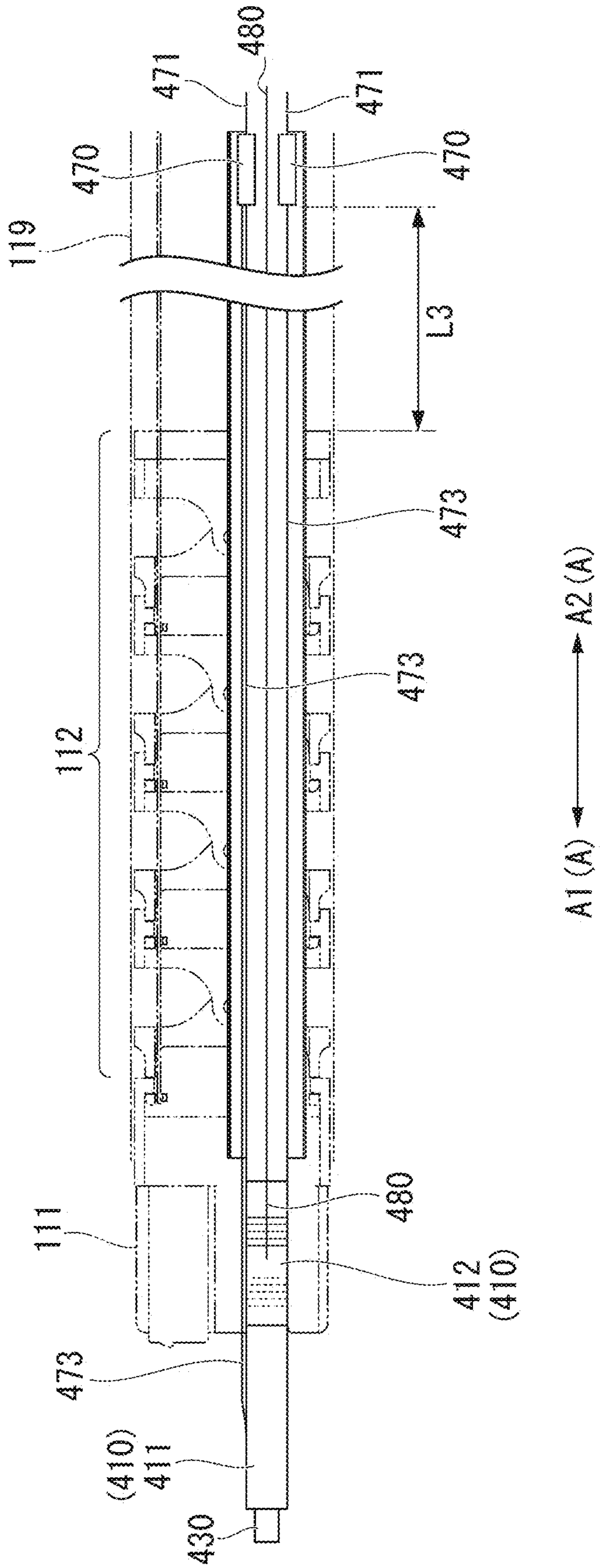
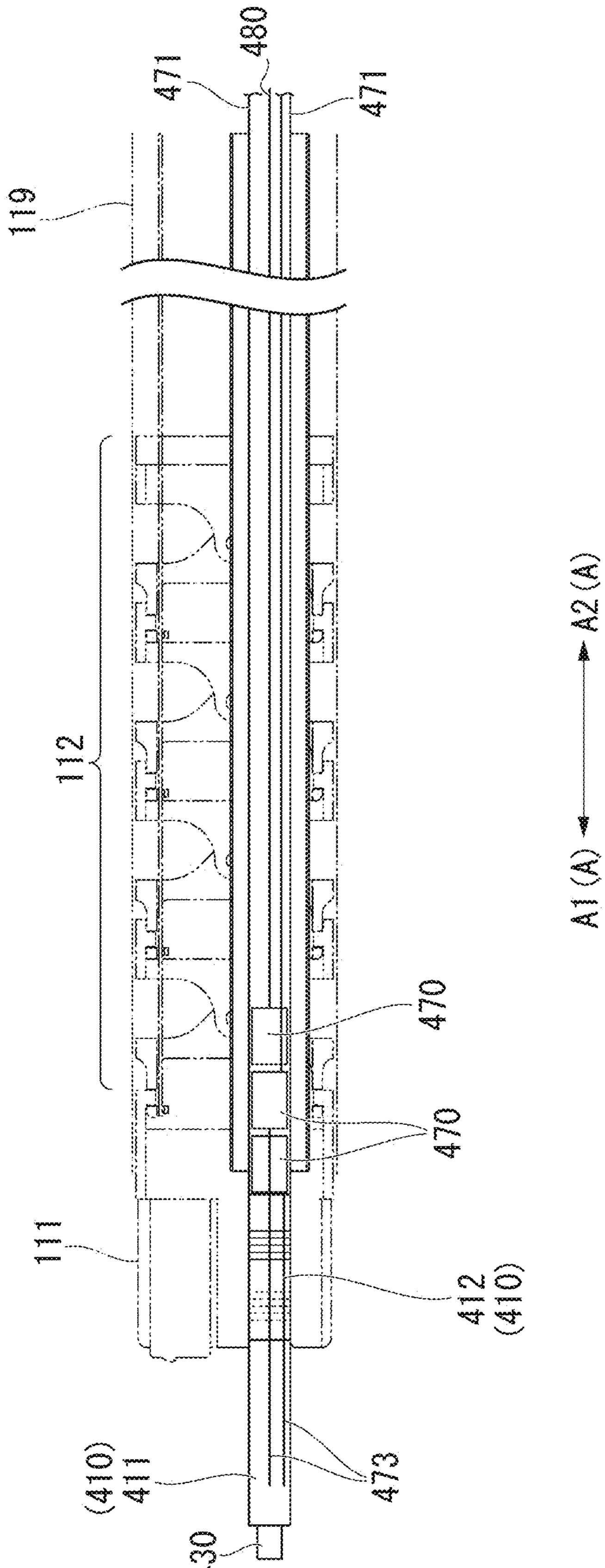

480
471
471
470
470
119
L3
473
473
112
A2 (A)
A1 (A)
111
480
412
(410)
473
(410)
411
430

FIG. 47

FIG. 48



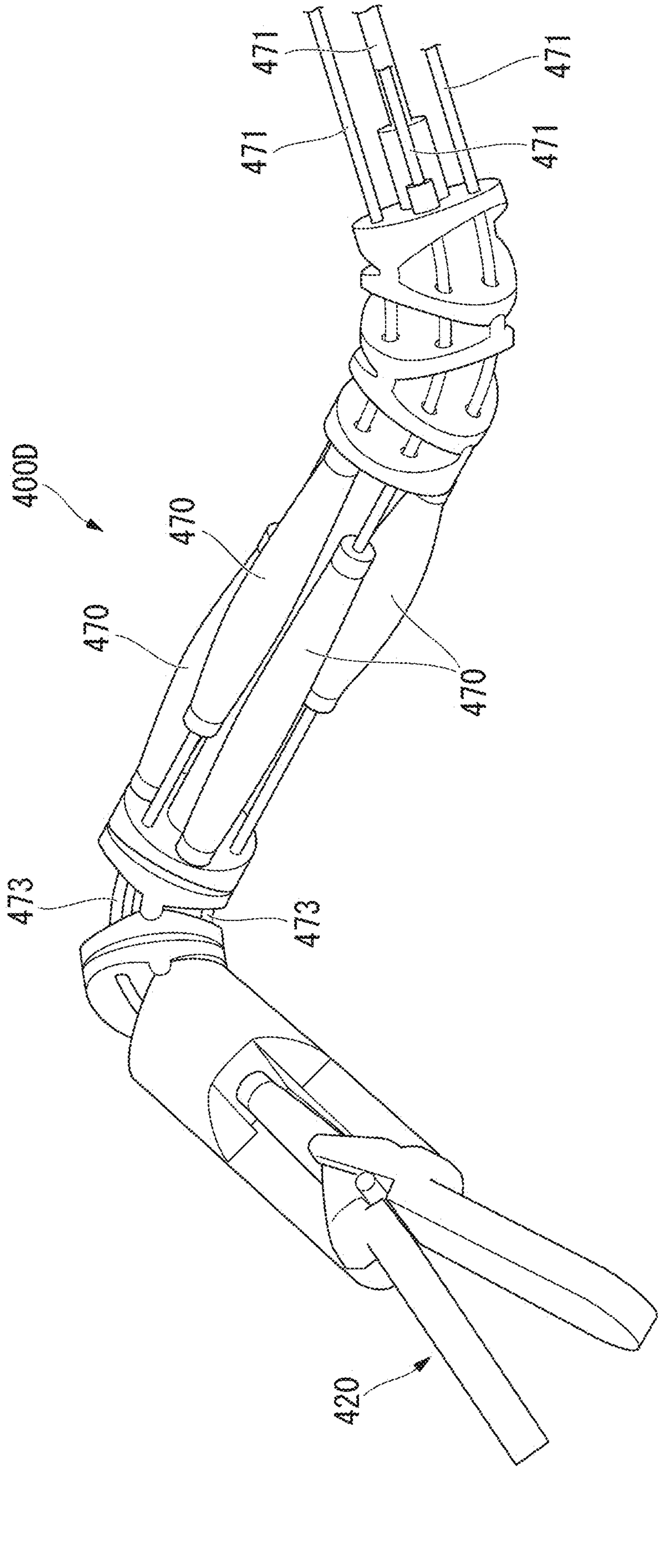


FIG. 49

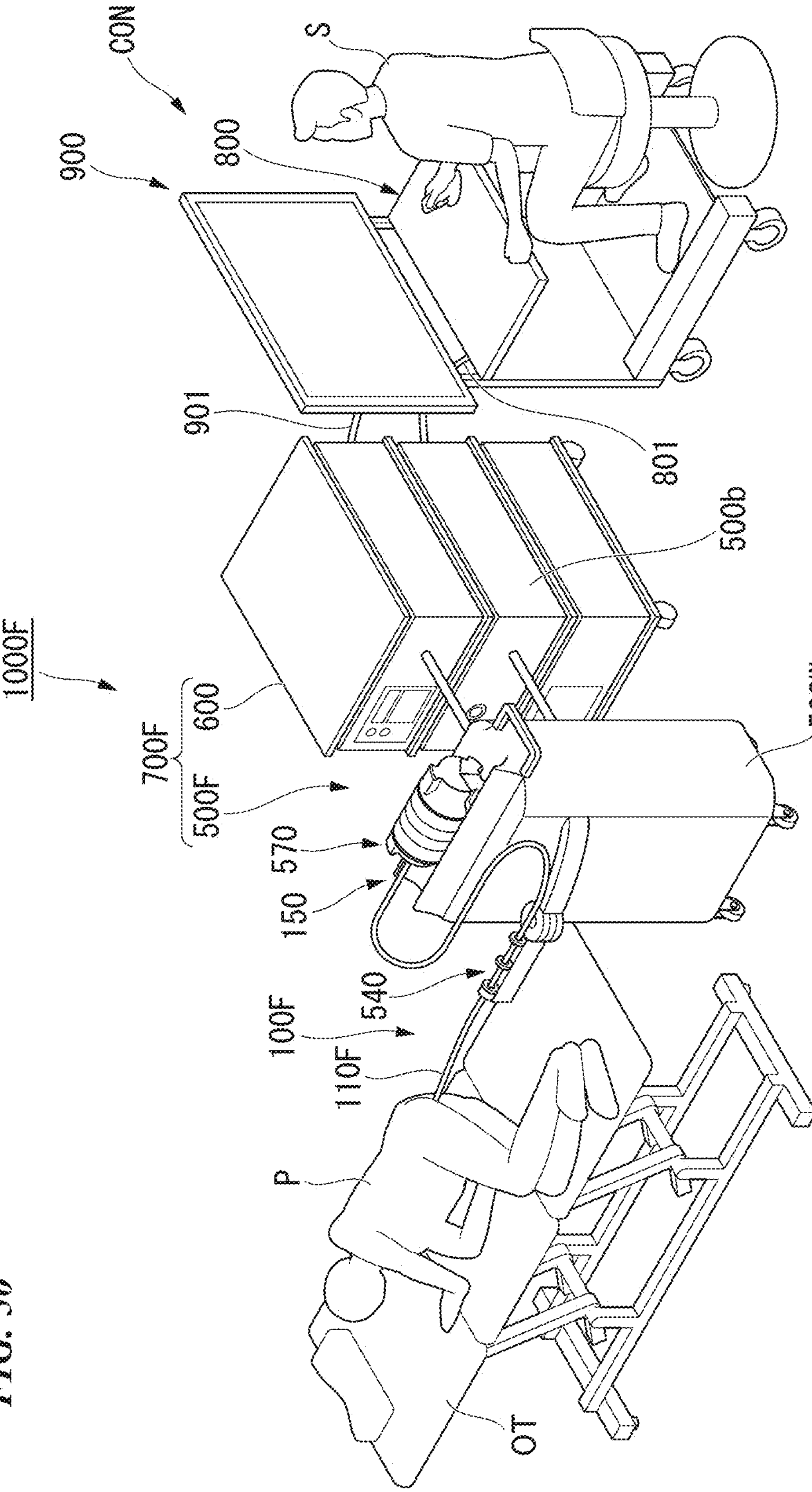

CON
S
900
800
901
801
500b
1000F
700F
600
500F
500W
570
150
540
100F
110F
P
OT

FIG. 50

FIG. 51

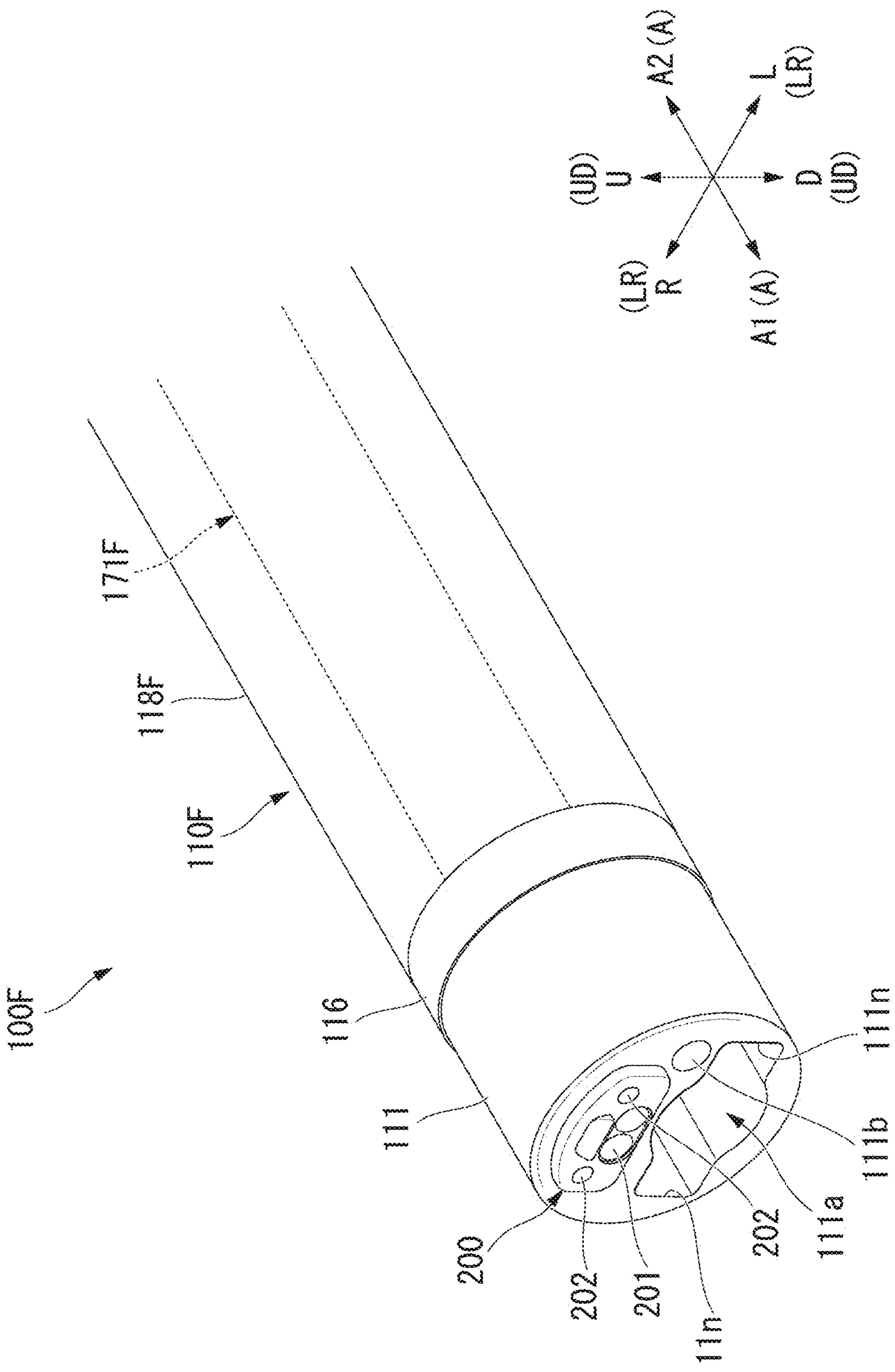
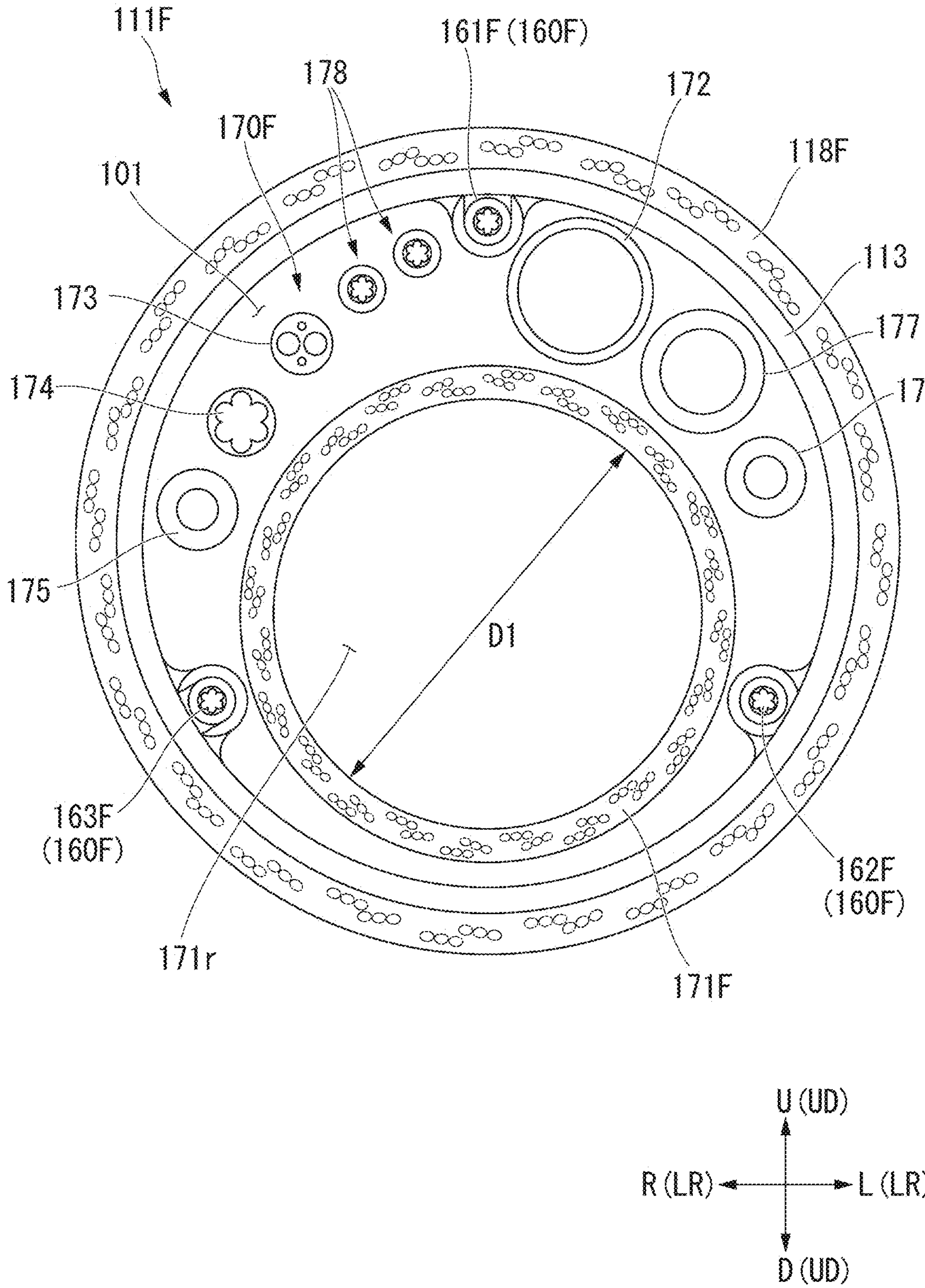

100F
110F
118F
171F
116
111
200
202
201
111n
202
111a
111b
111n
A2 (A)
L
(LR)
(UD)
U
D
(UD)
(LR)
R
A1 (A)

FIG. 52


111F
161F (160F)
178
172
170F
118F
101
113
173
177
174
176
175
D1
163F
(160F)
162F
(160F)
171r
171F
U (UD)
R (LR)
L (LR)
D (UD)

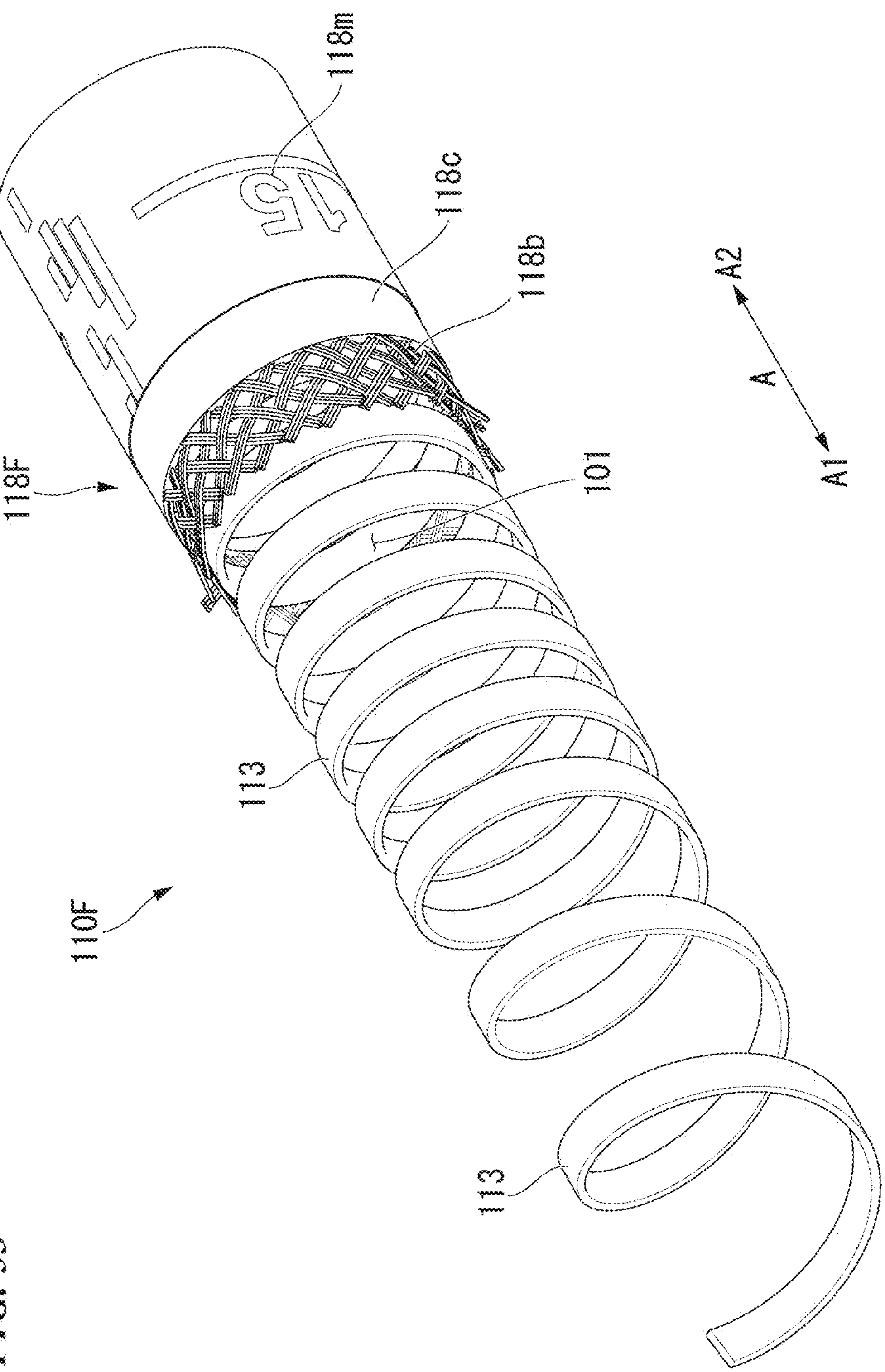


FIG. 53

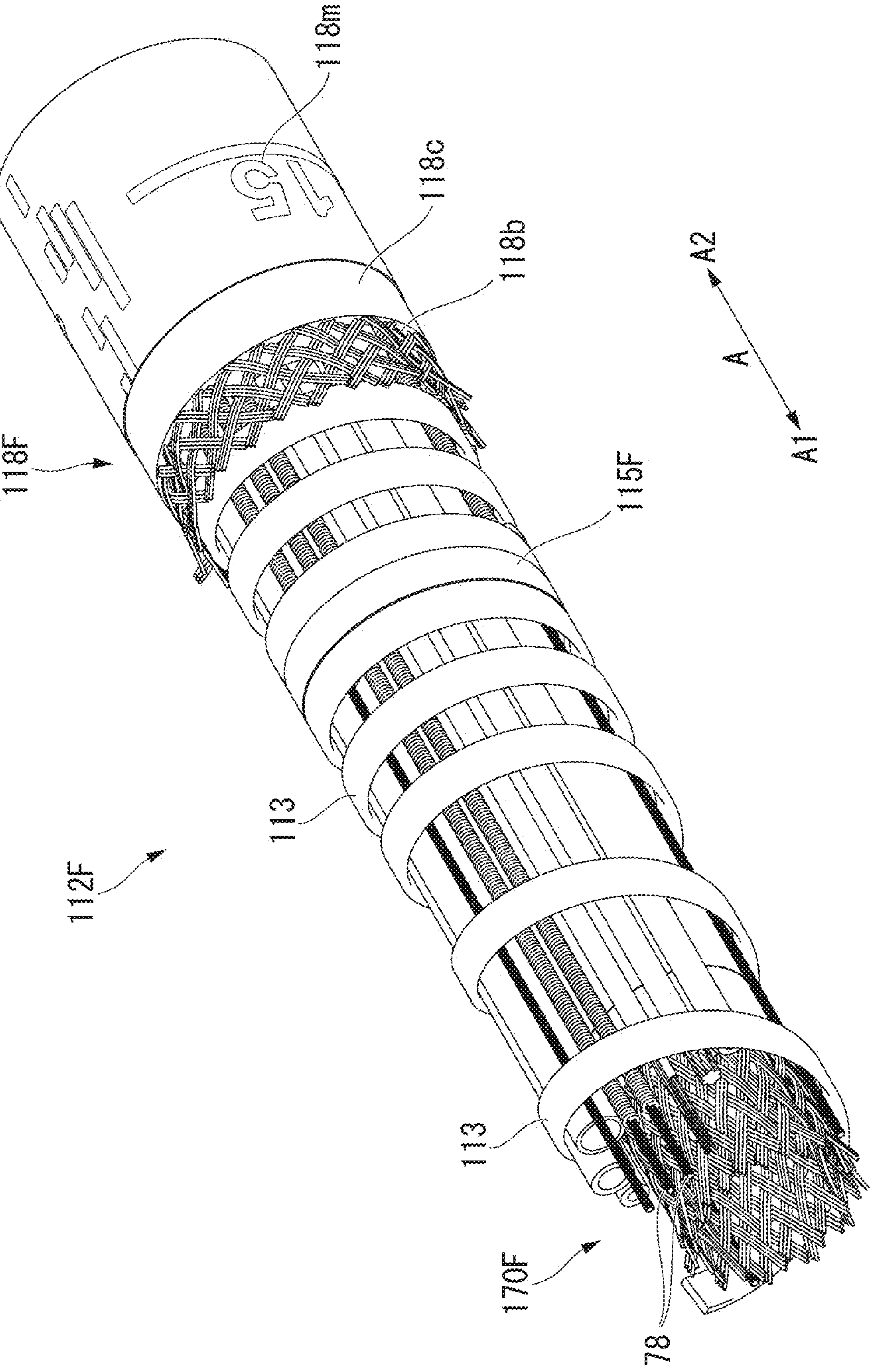


FIG. 54

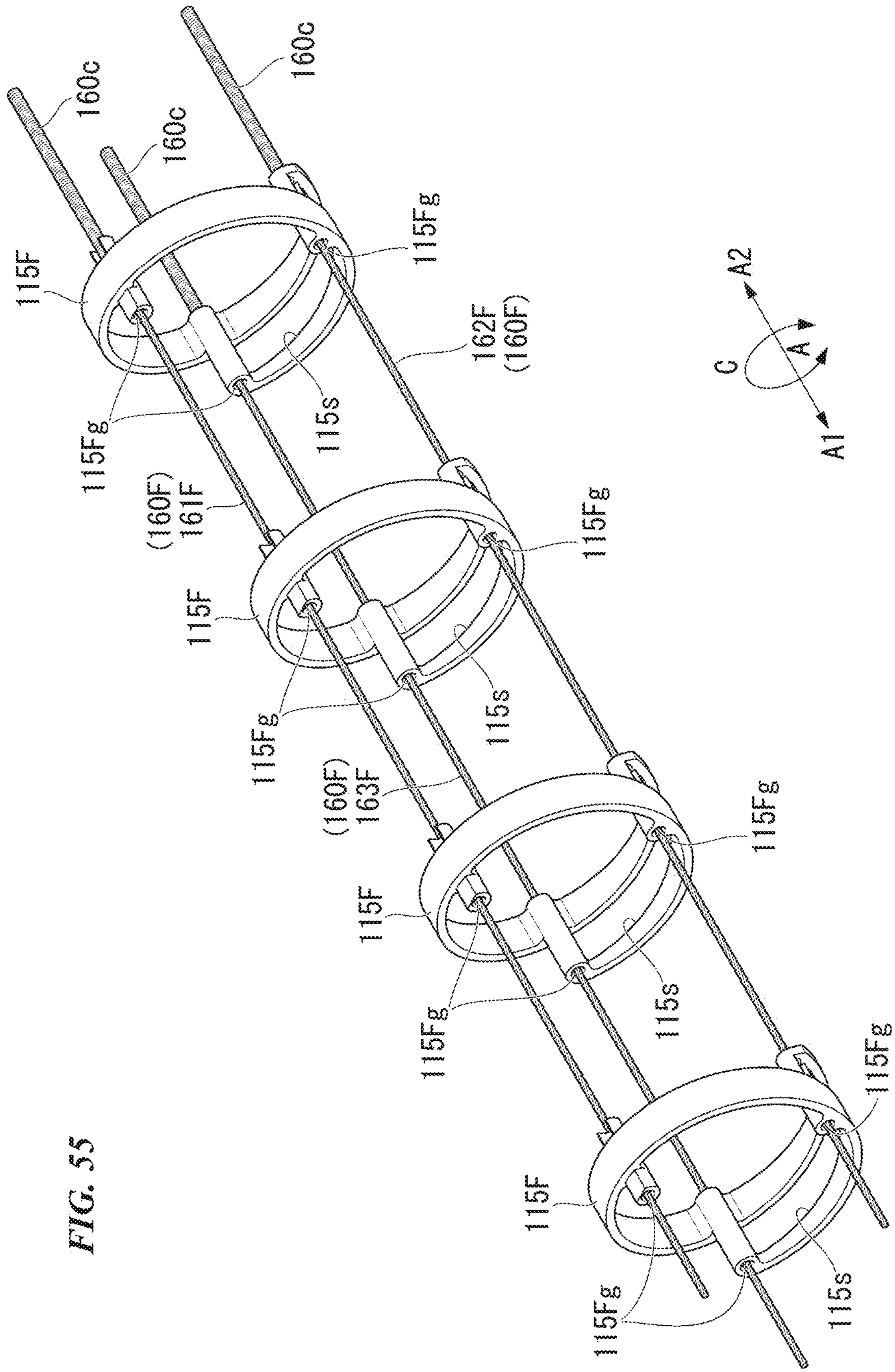
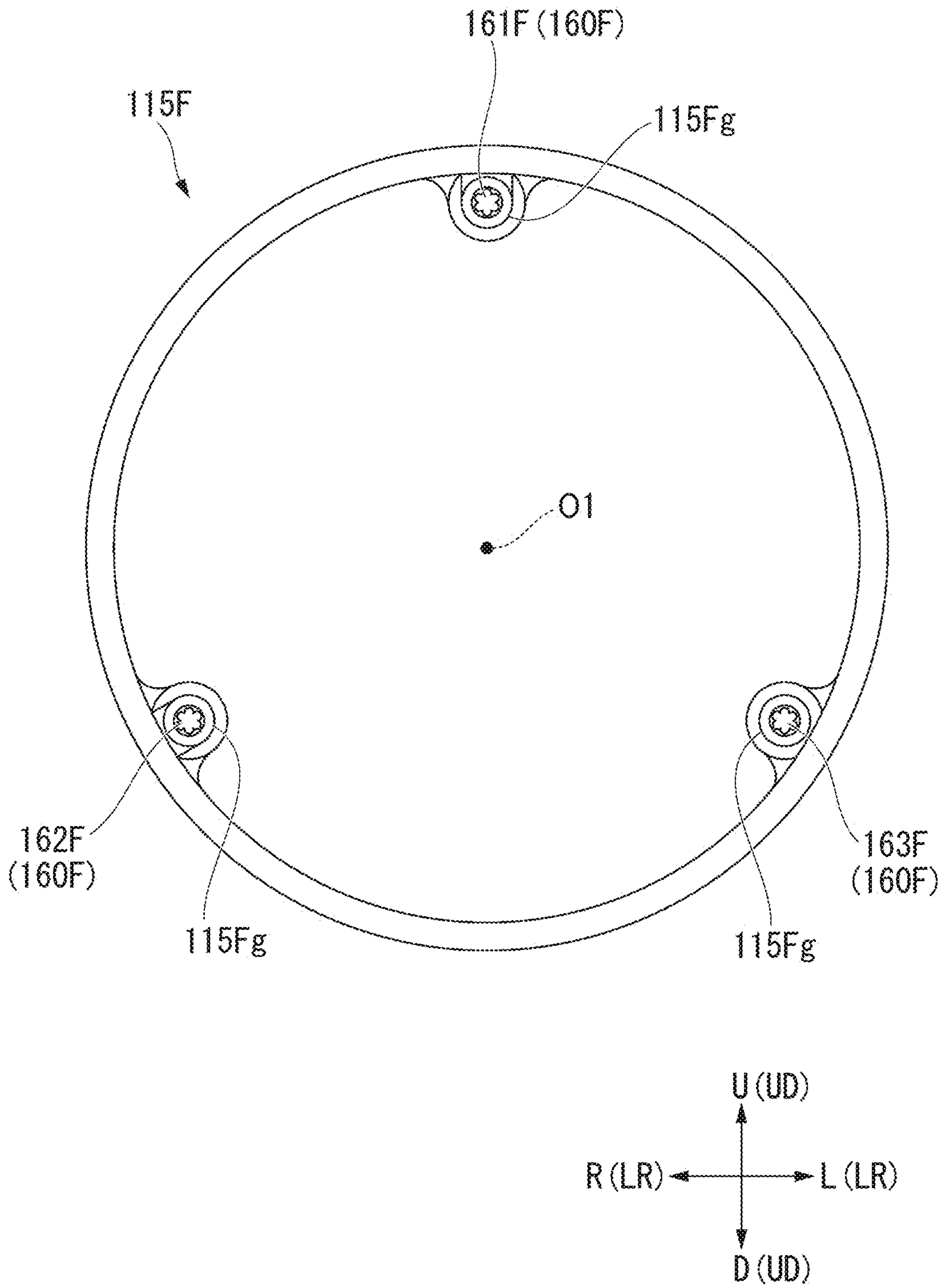

160c
160c
160c
115F
115Fg
115Fg
162F
(160F)
115s
A2
A1
A
C
(160F)
161F
115F
115Fg
115Fg
115s
(160F)
163F
115F
115Fg
115Fg
115s
115F
115Fg
115Fg
115s

FIG. 55

FIG. 56


161F (160F)
115F
115Fg
O1
162F
(160F)
115Fg
163F
(160F)
115Fg
U (UD)
R (LR)
L (LR)
D (UD)

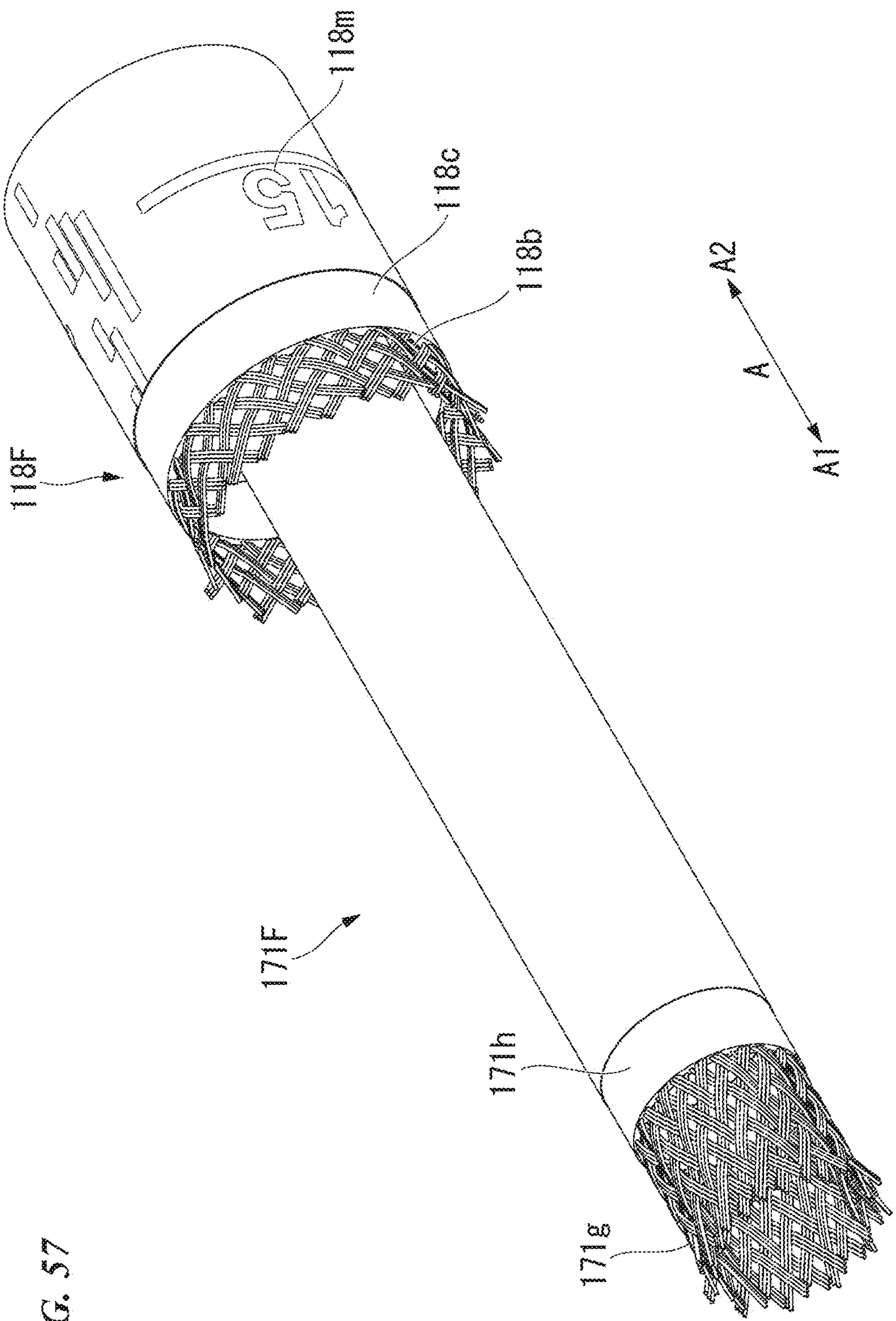
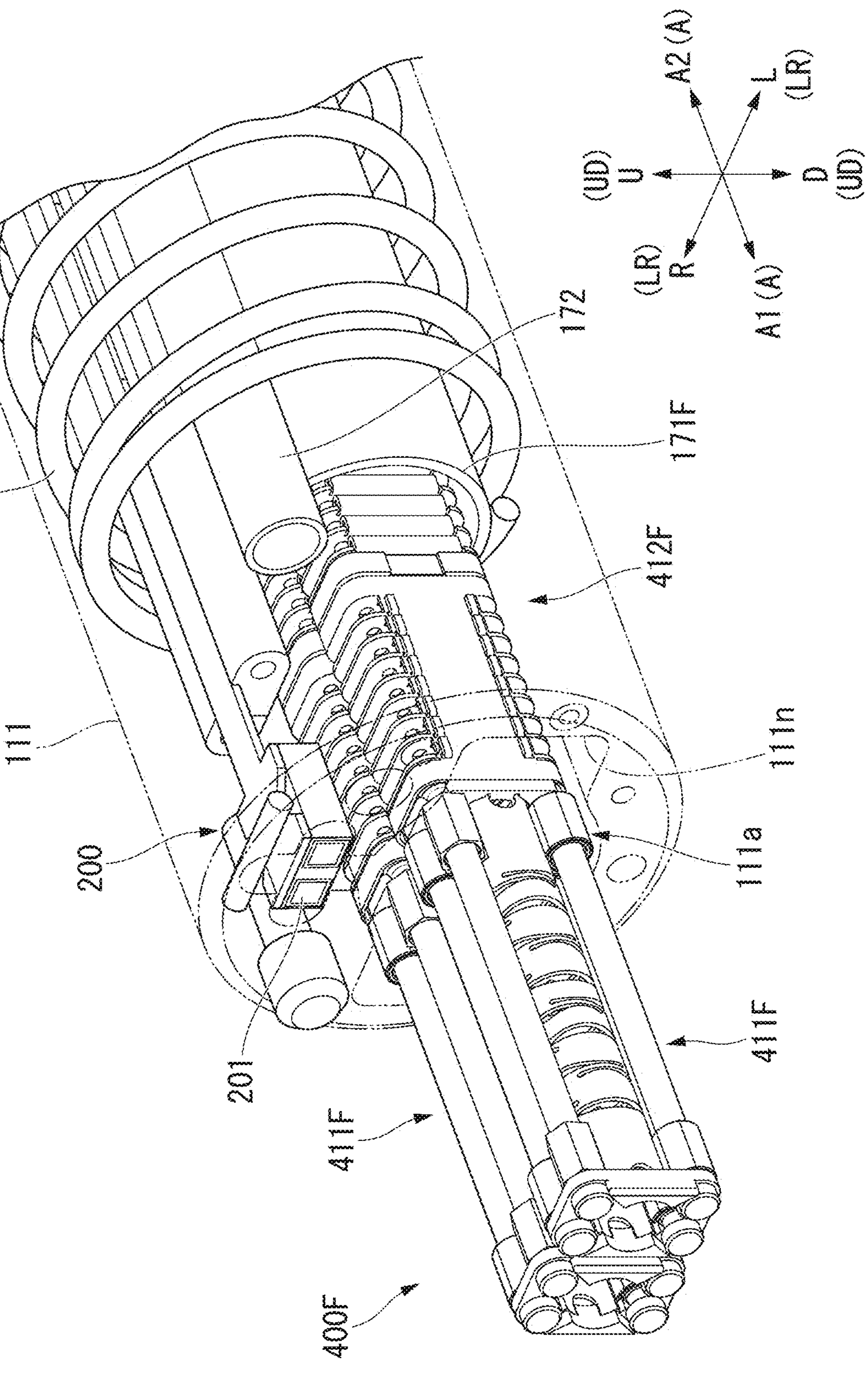


FIG. 57



113
111
200
201
172
171F
412F
111n
111a
411F
411F
400F
A2 (A)
L
(LR)
(UD)
U
D
(UD)
(LR)
R
A1 (A)

FIG. 58

FIG. 59

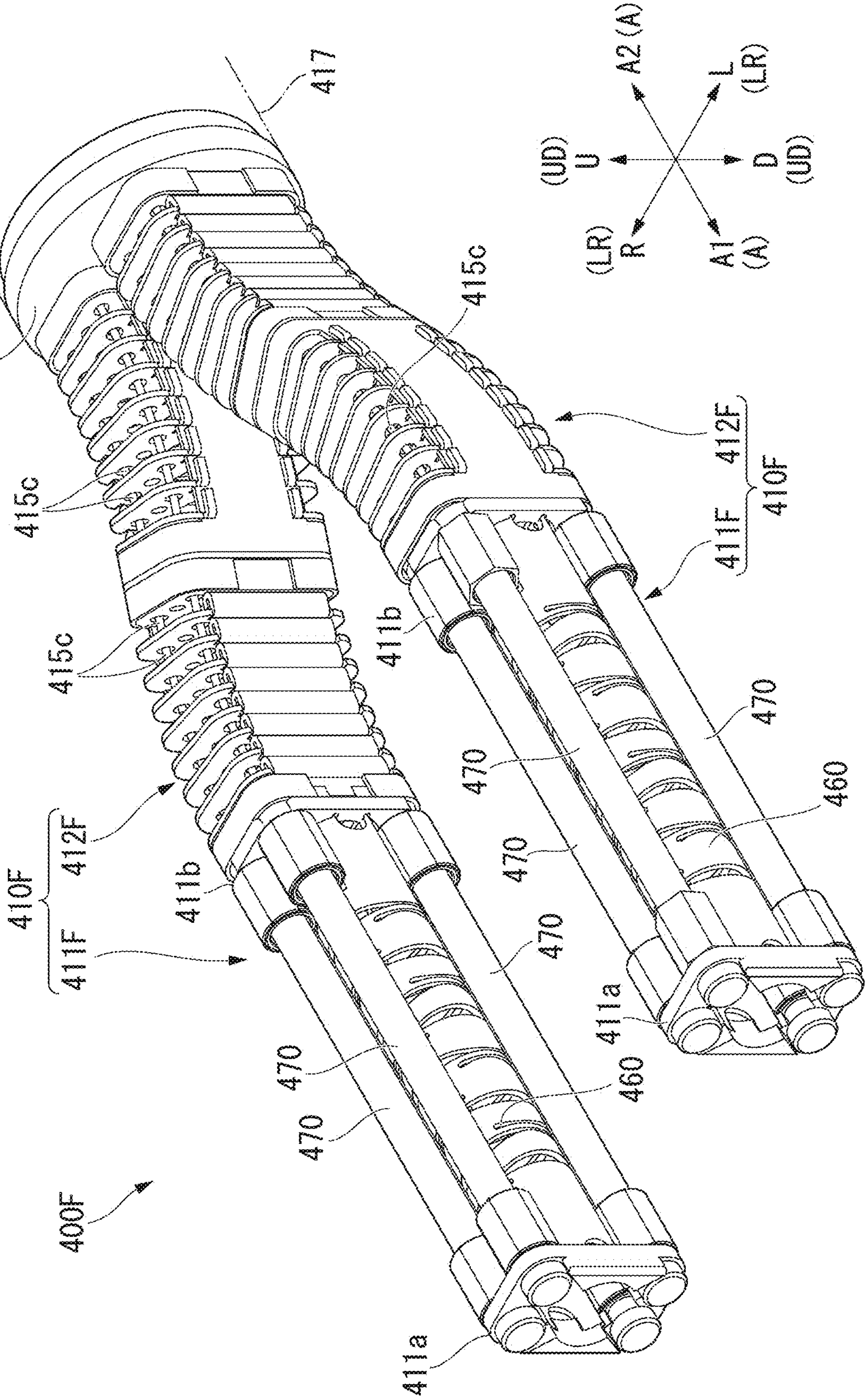

400F
411a
470
470
460
410F
411F
412F
411b
415c
415c
417a
417
415c
412F
410F
411F
411b
470
470
470
460
411a
A2 (A)
L (LR)
U (UD)
D (UD)
R (LR)
A1 (A)

FIG. 60

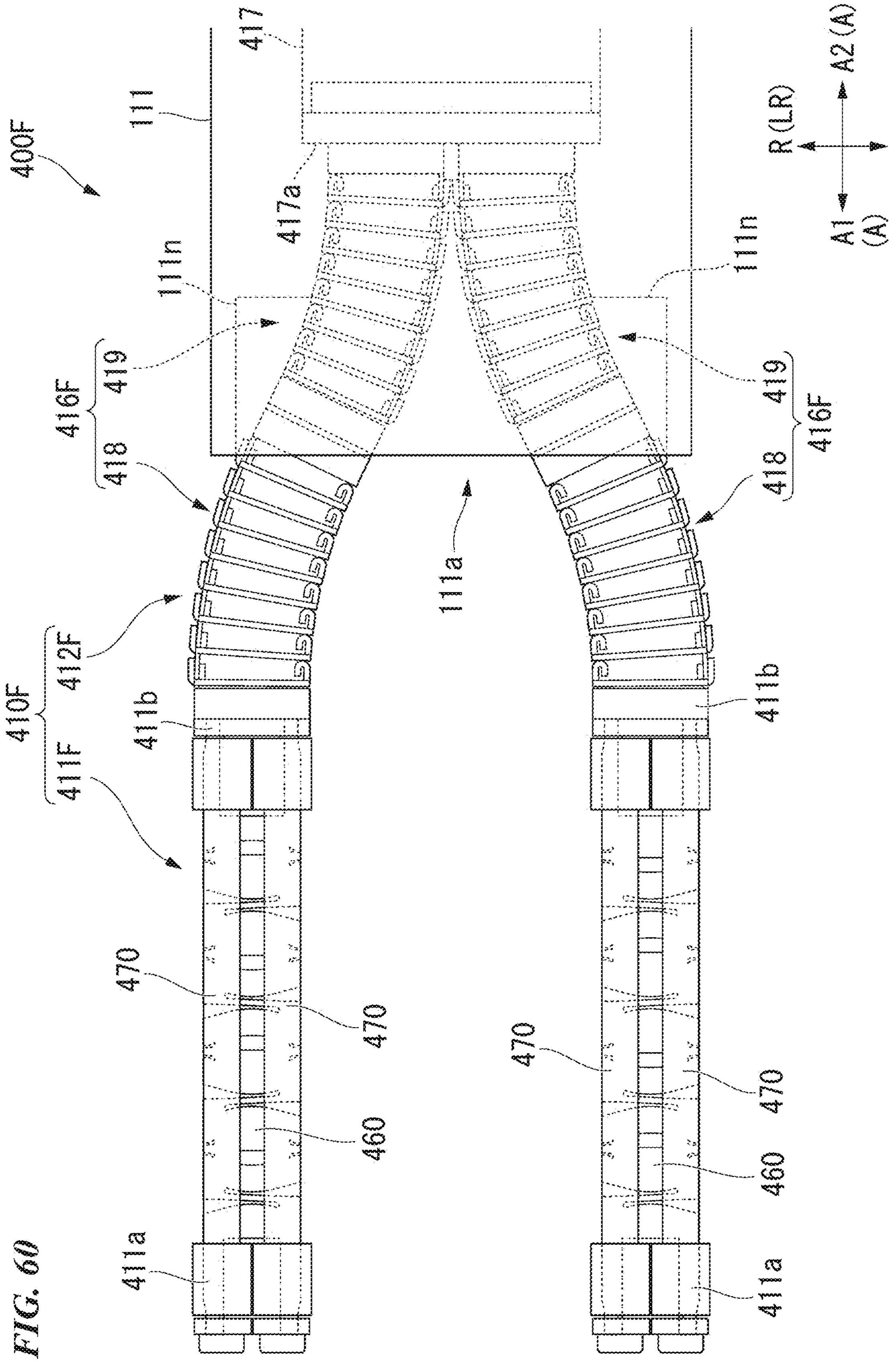
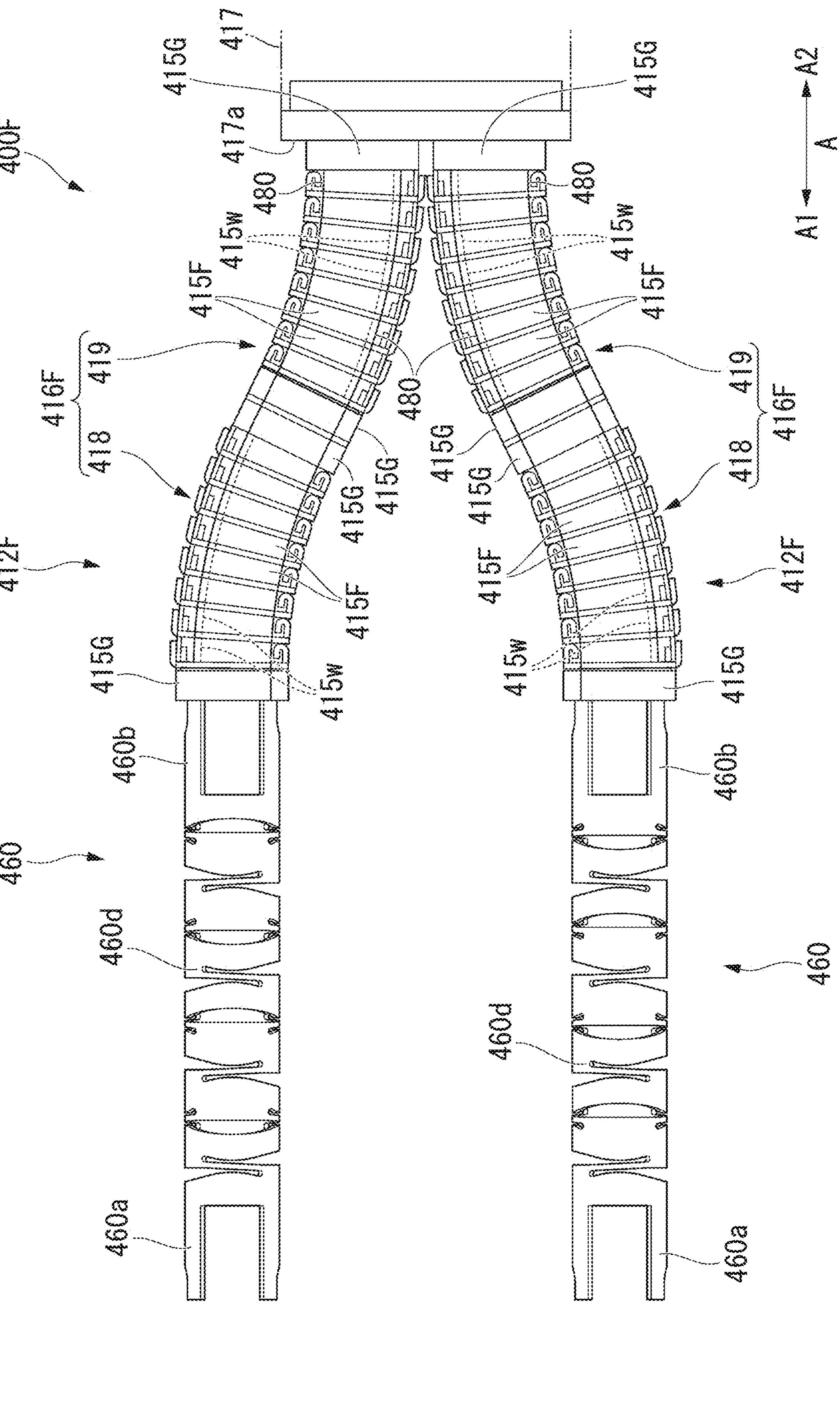

400F
111
417
417a
111n
111n
419
419
416F
416F
418
418
111a
412F
410F
411F
411b
411b
470
470
470
470
460
460
411a
411a
R (LR)
A2 (A)
A1 (A)

FIG. 61


400F
417
415G
417a
480
415w
415F
419
416F
418
412F
460b
460
460d
460a
A2
A
A1

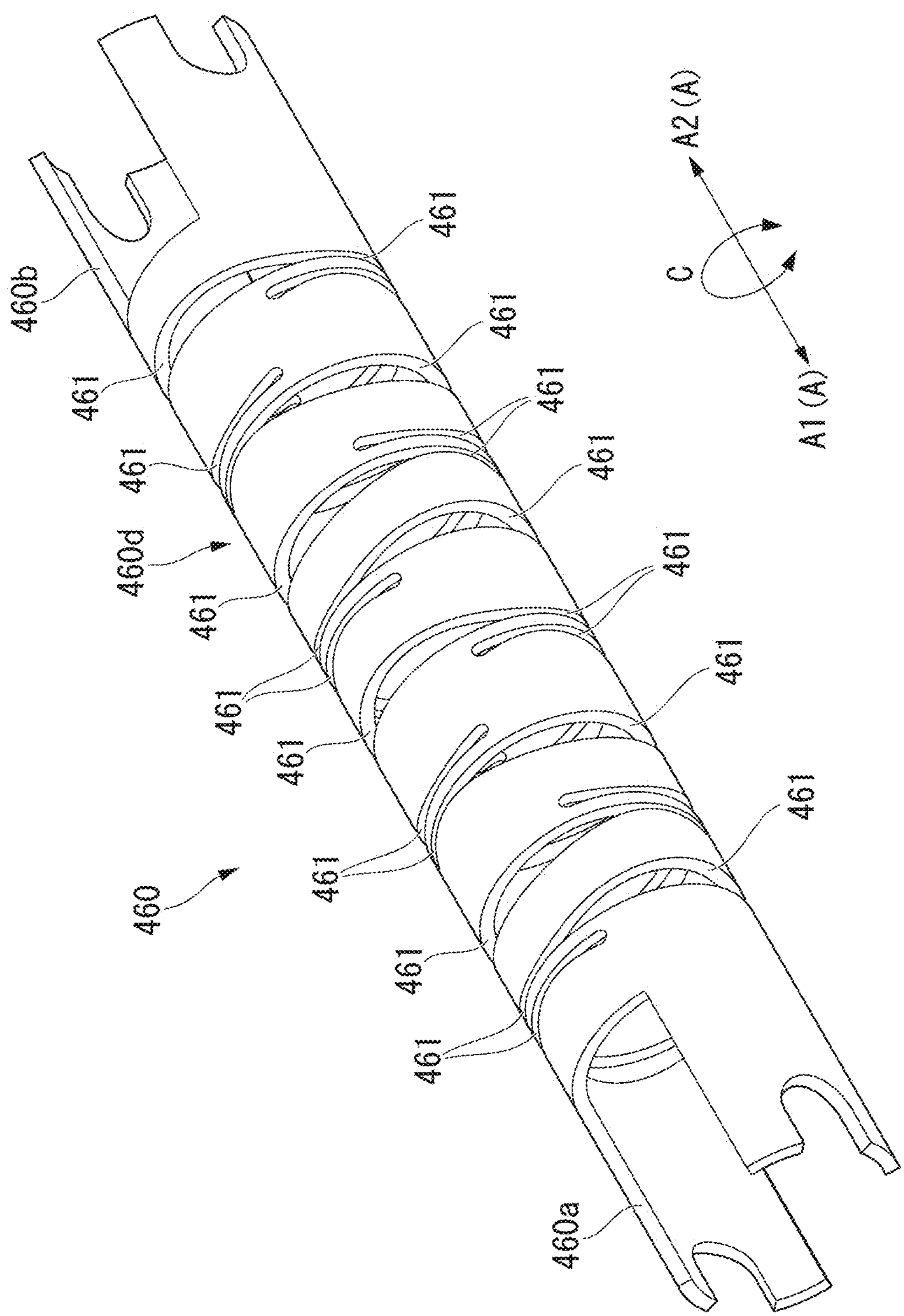


FIG. 62

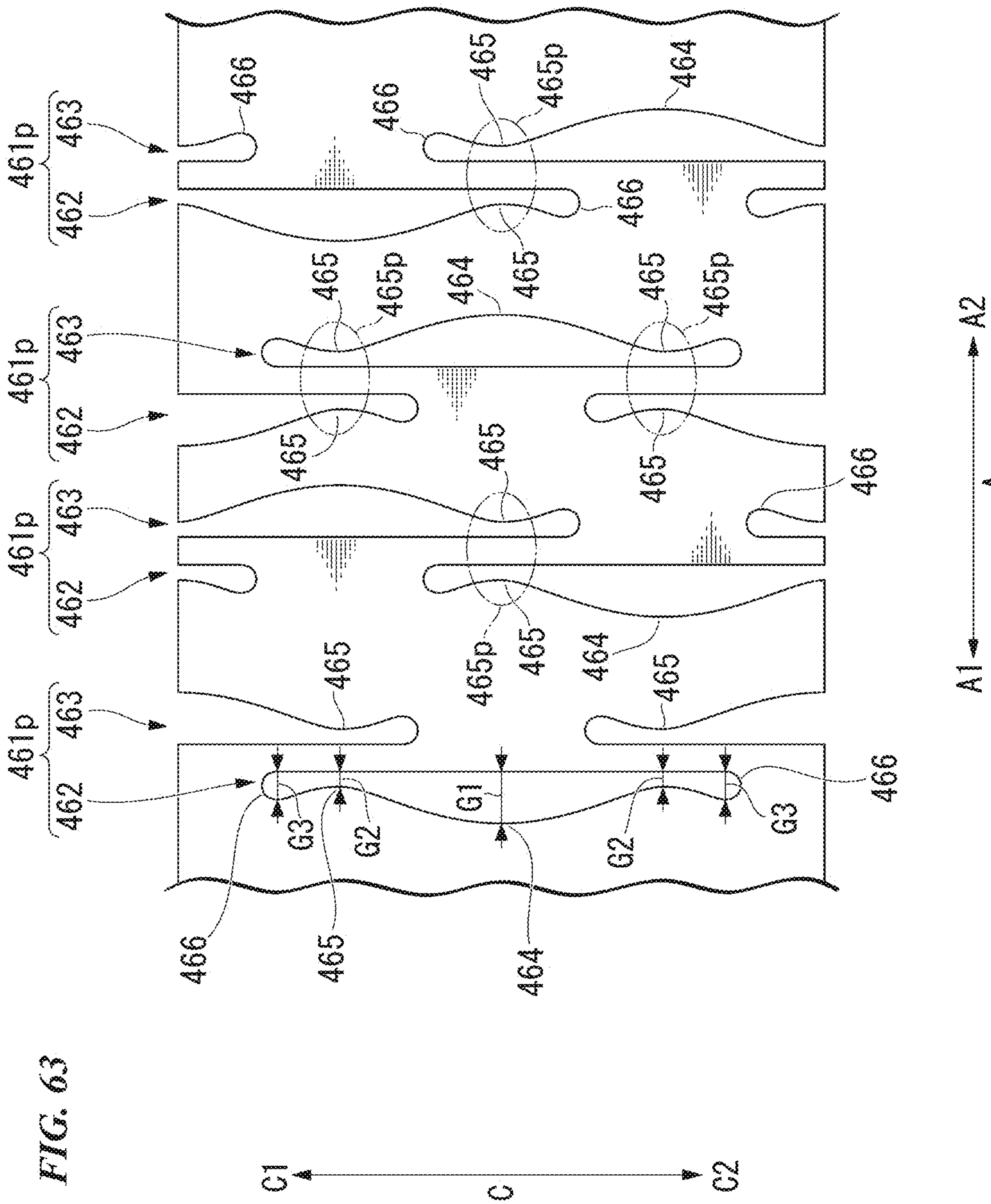
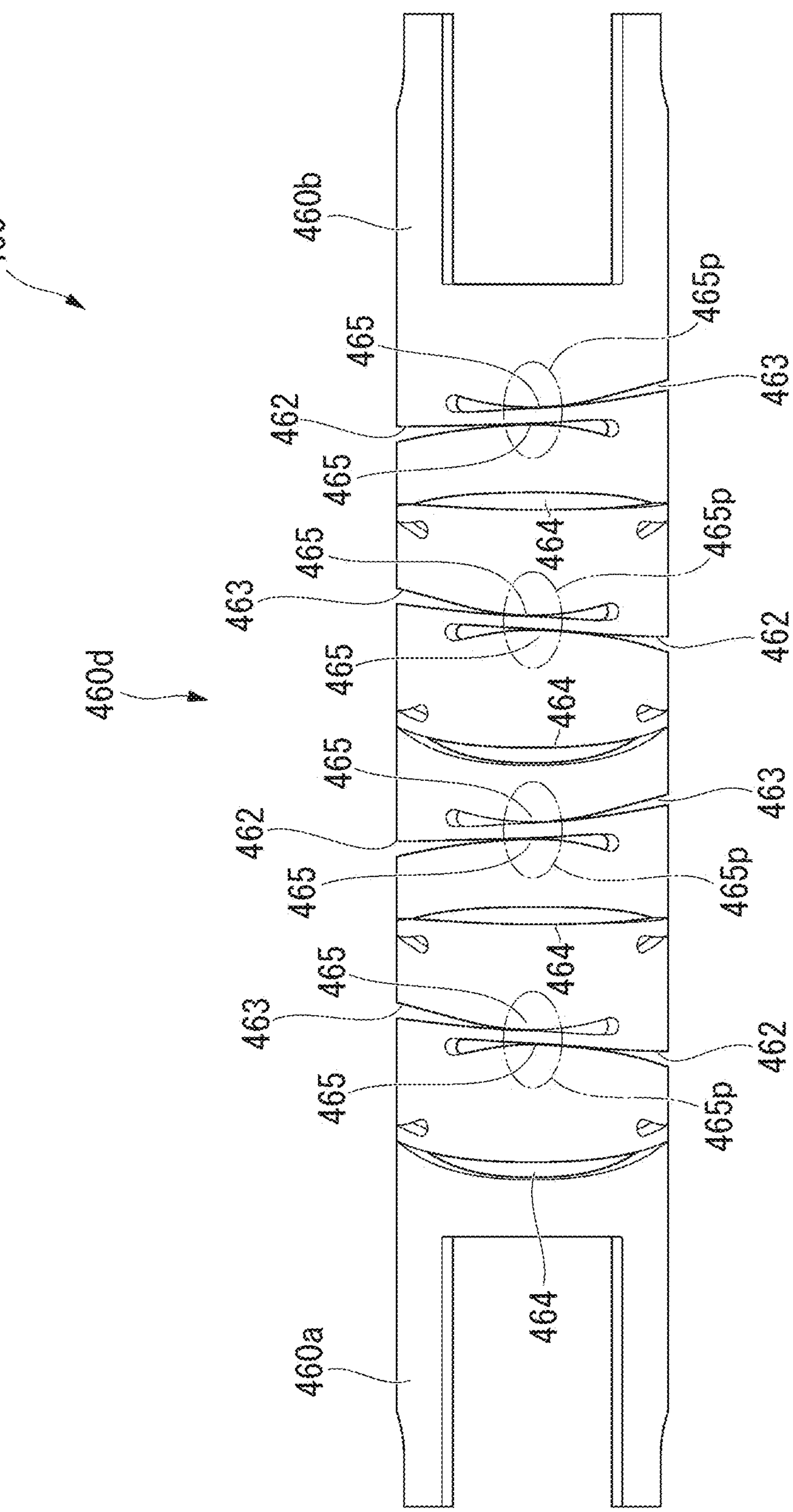
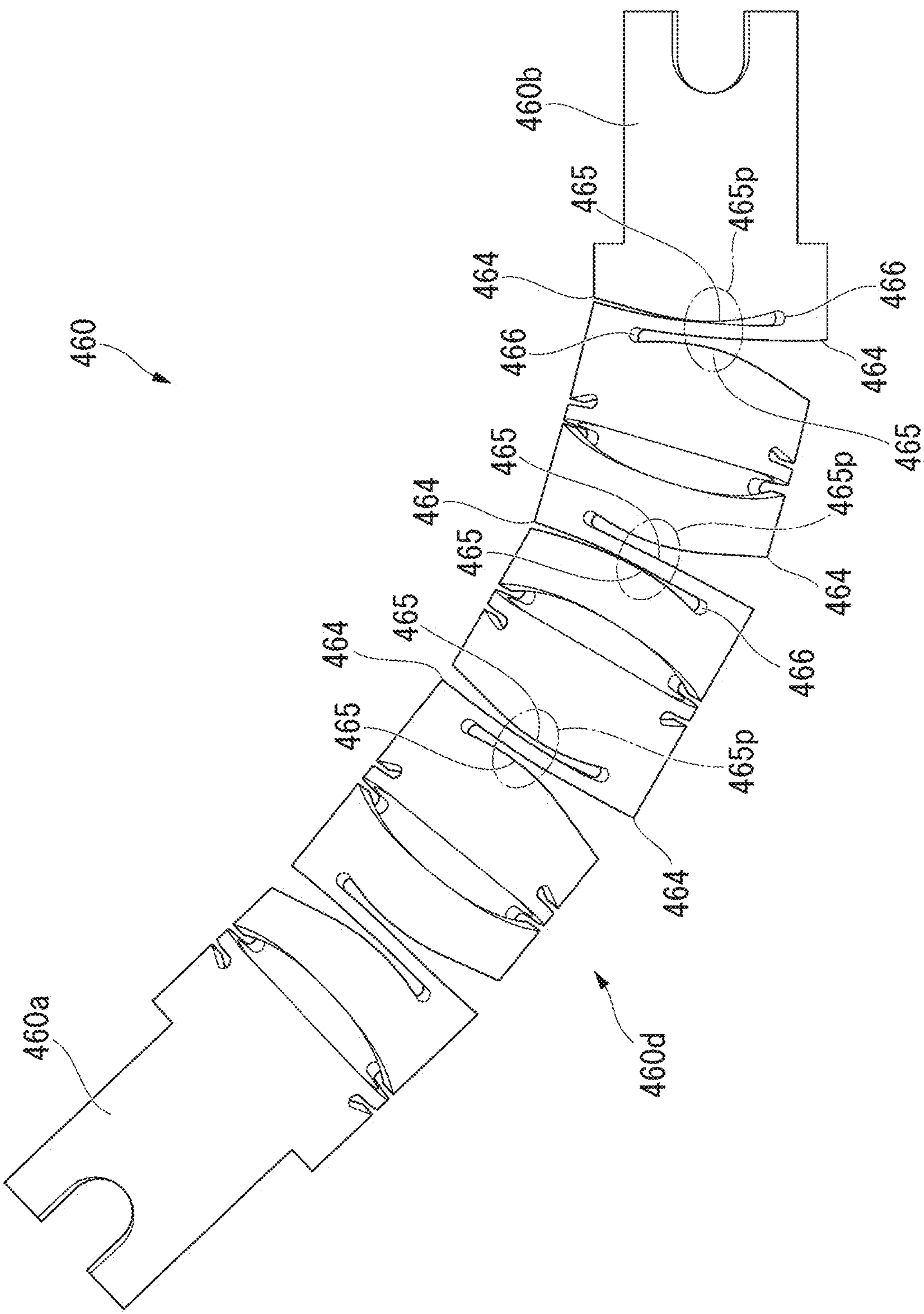


FIG. 63

FIG. 64


460
460b
465p
465
463
462
465
464
465p
465
463
462
465
460d
464
465
463
462
465
465p
464
465
463
462
465
465p
464
460a



460
460a
460b
460d
464
464
464
464
464
464
465
465
465
465
465
465
465p
465p
465p
466
466
466

FIG. 65

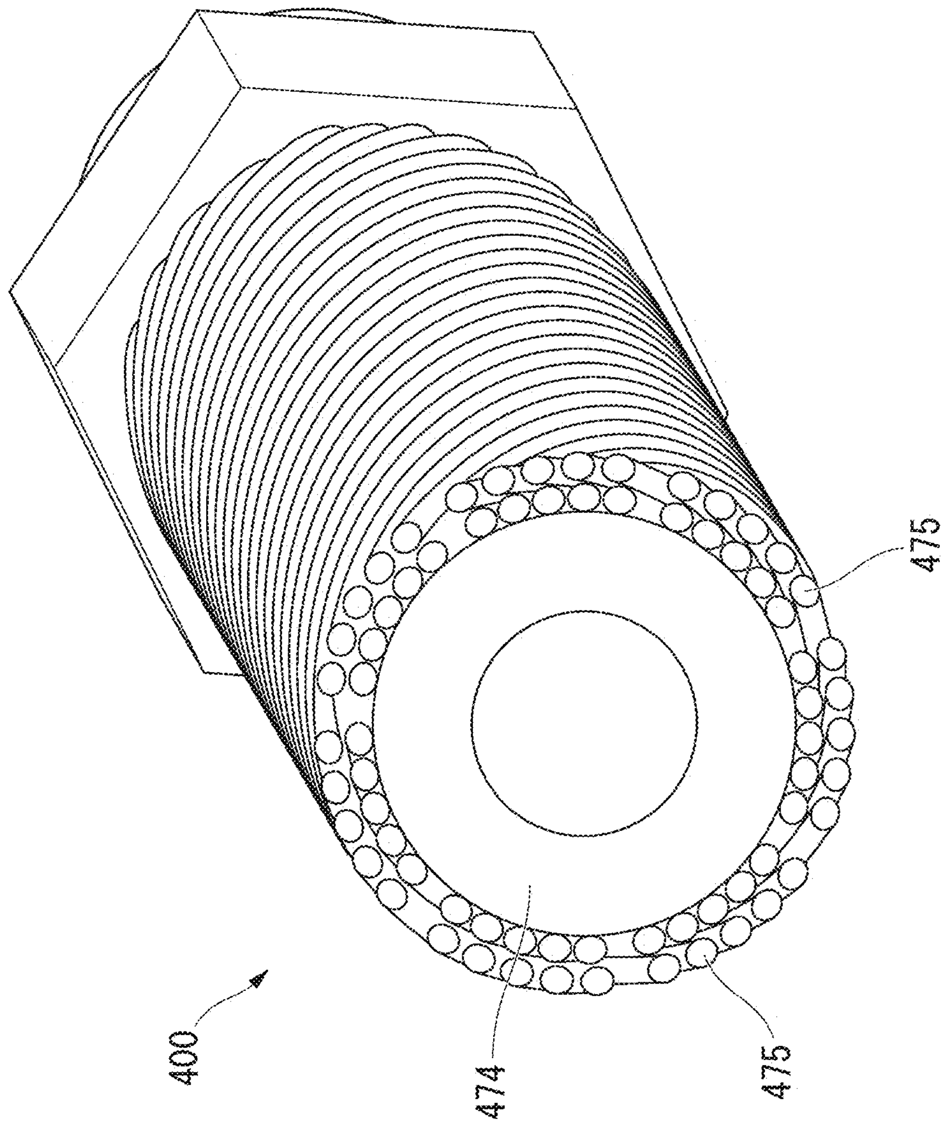


FIG. 66

FIG. 67

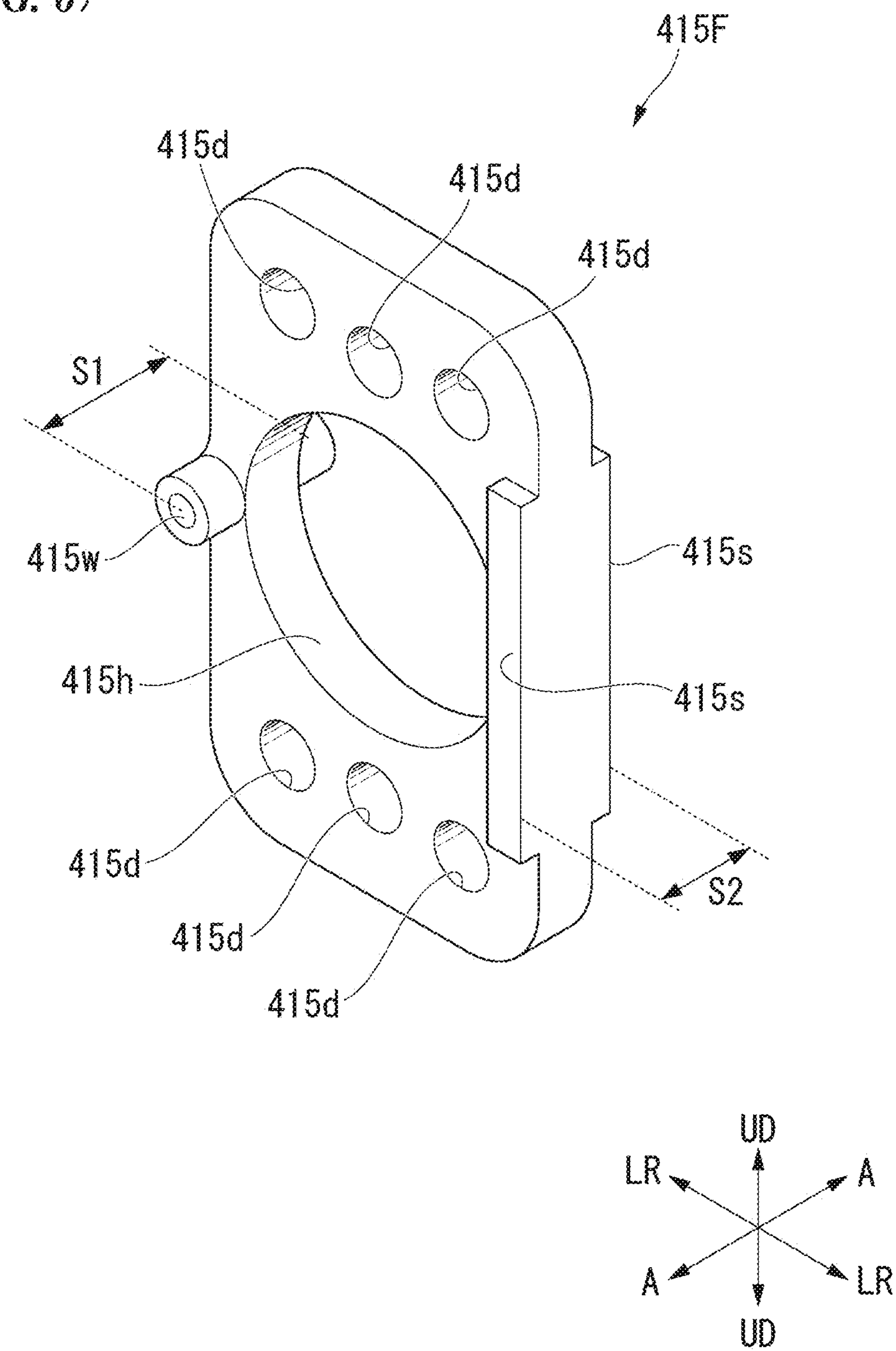

415F
415d
415d
415d
S1
415w
415s
415h
415s
415d
415d
415d
S2
UD
LR
A
A
LR
UD

FIG. 68

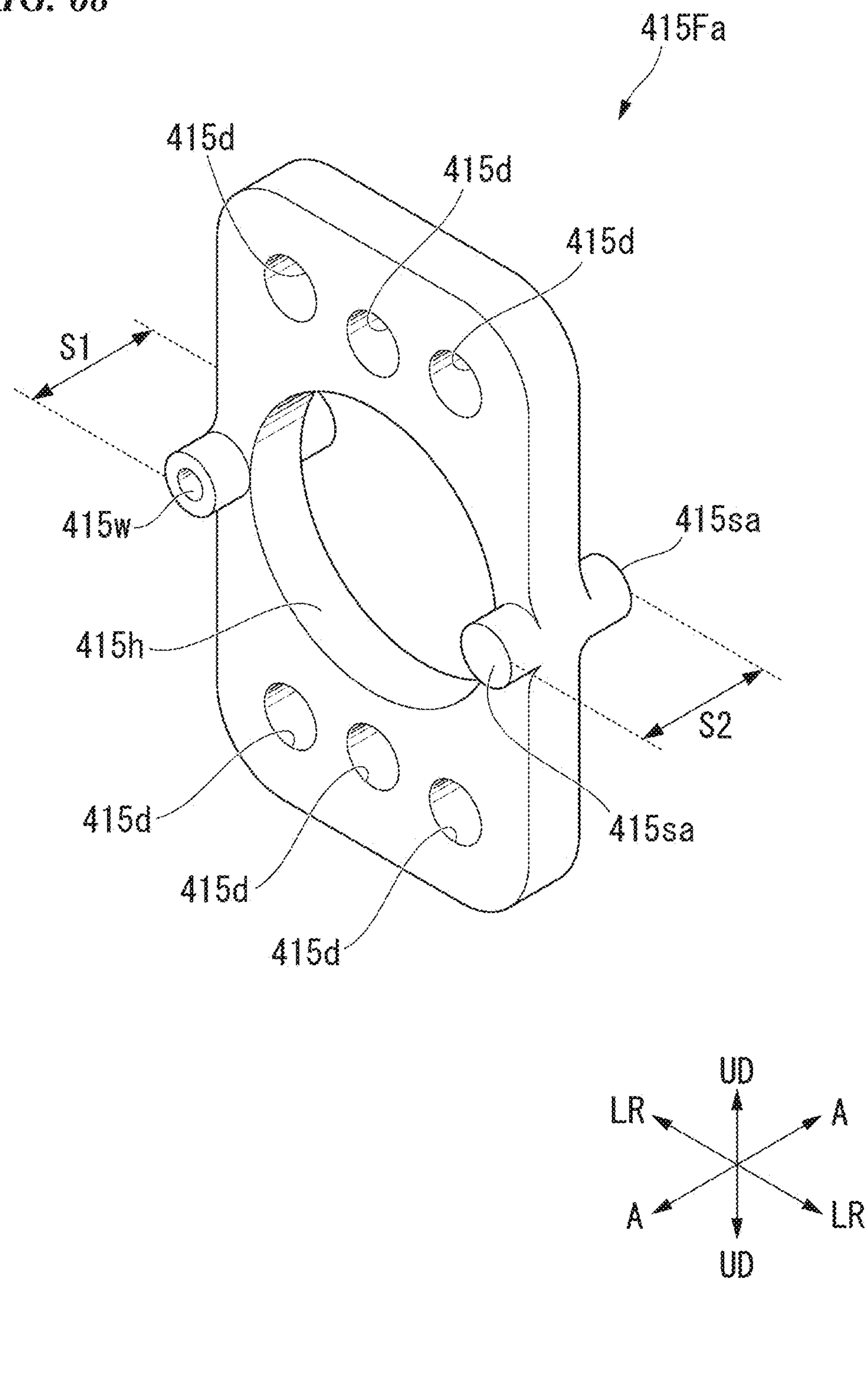

415Fa
415d
415d
415d
S1
415w
415sa
415h
S2
415d
415sa
415d
415d
UD
LR
A
A
LR
UD

FIG. 69

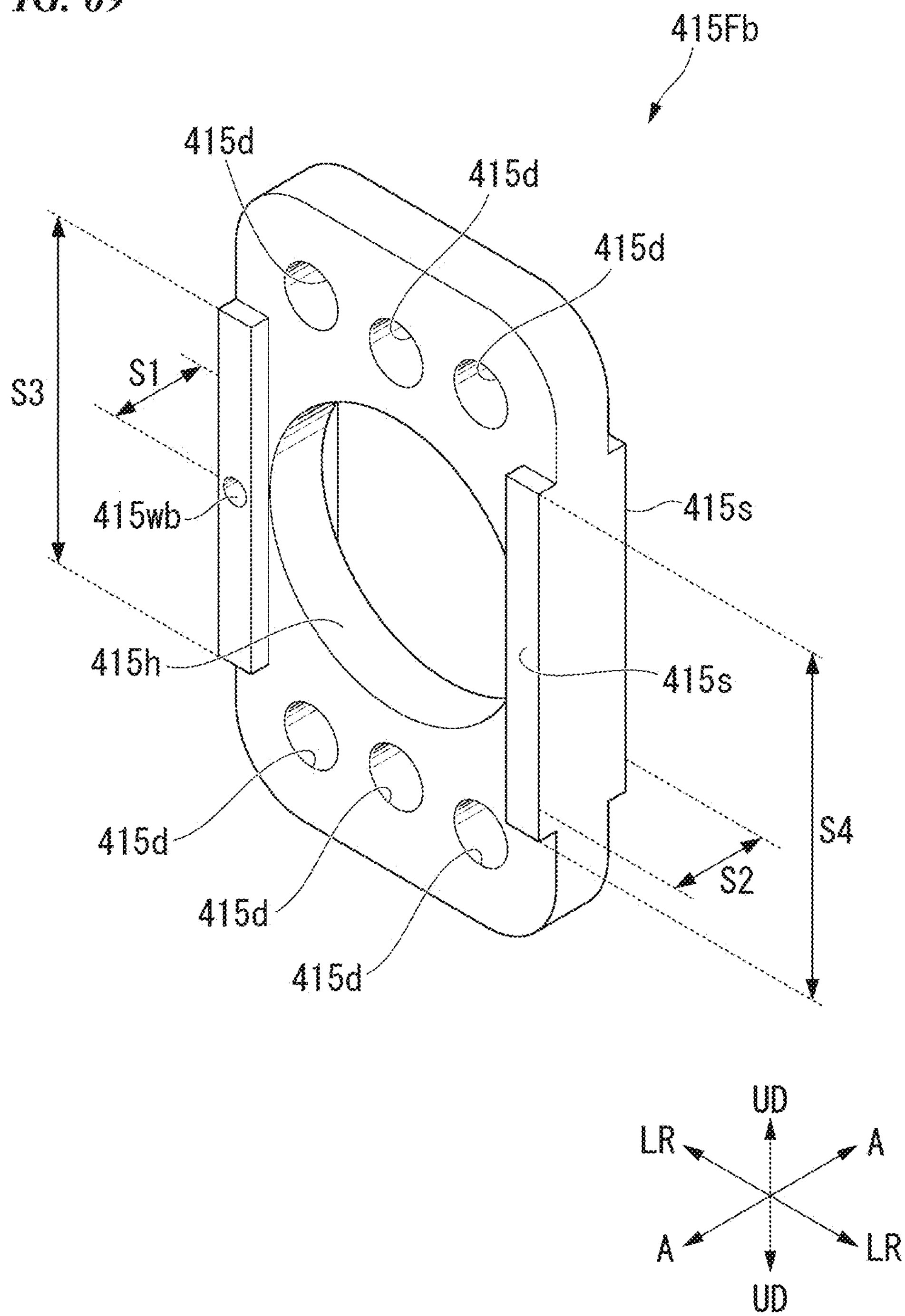

415Fb
415d
415d
415d
S3
S1
415wb
415s
415h
415s
415d
415d
415d
S2
S4
UD
LR
A
A
LR
UD

FIG. 70

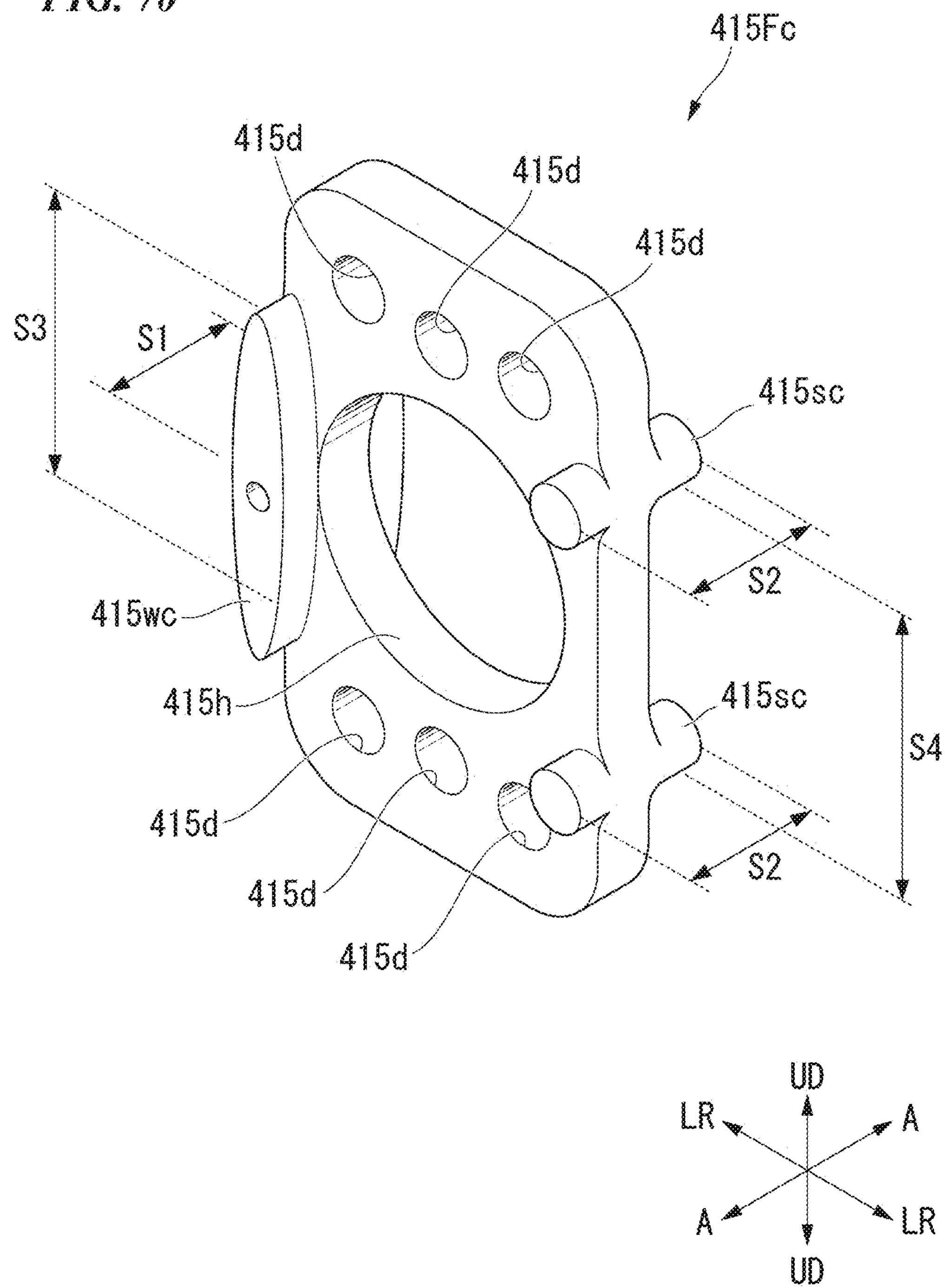

415Fc
415d
415d
415d
S3
S1
415sc
S2
415wc
415h
415sc
S4
415d
415d
415d
S2
UD
LR
A
A
LR
UD

FIG. 71

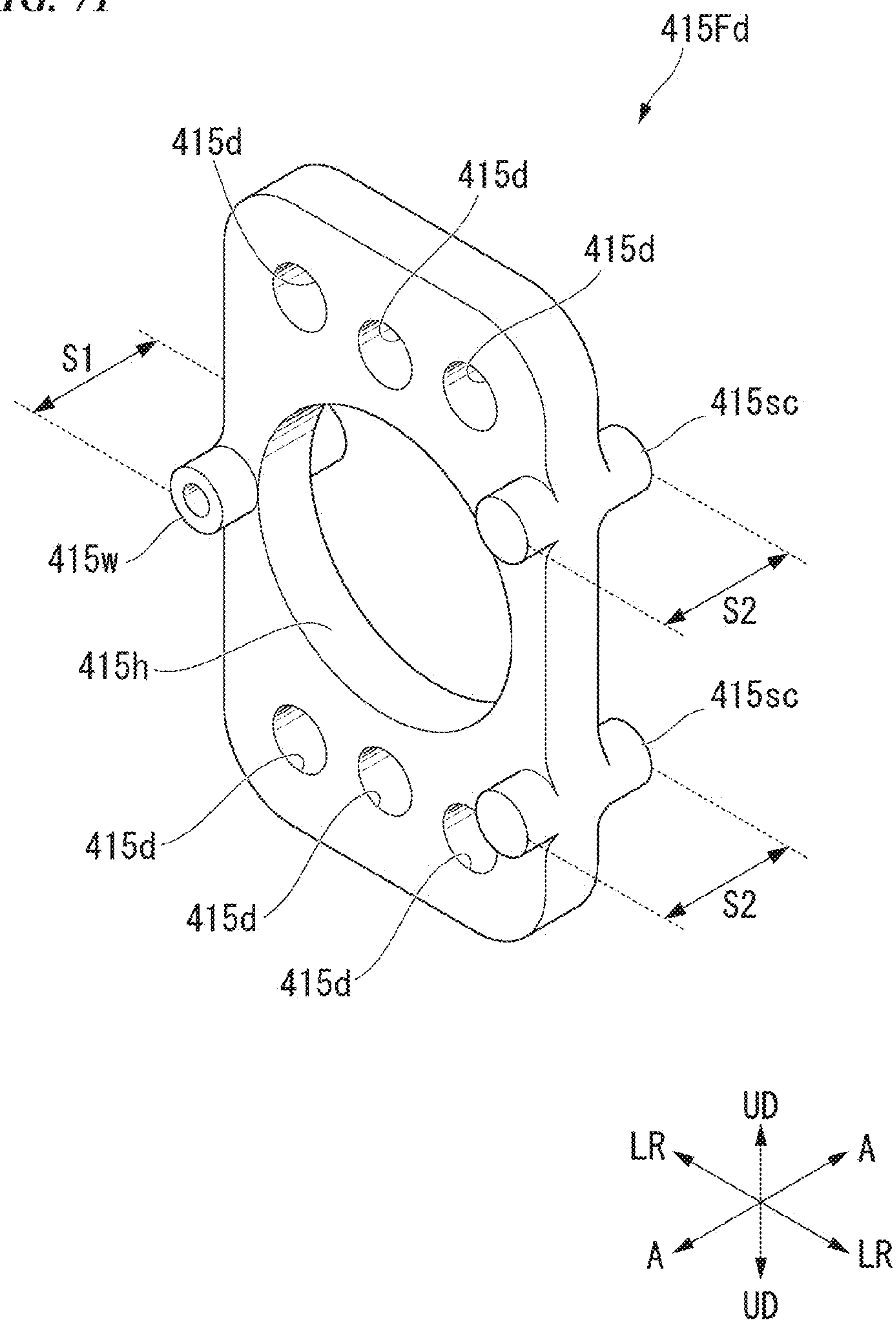
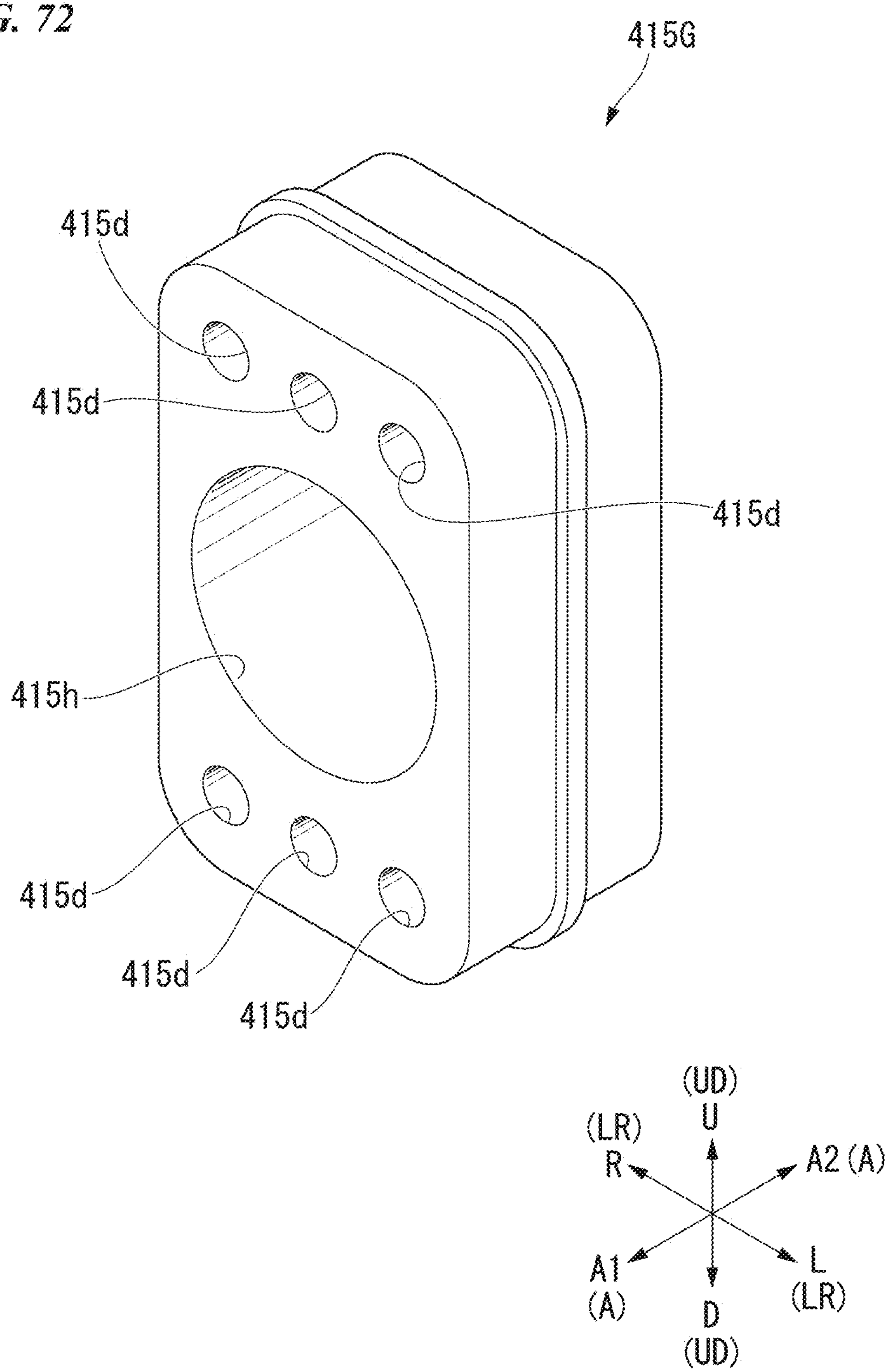

415Fd
415d
415d
415d
S1
415sc
415w
S2
415h
415sc
415d
415d
415d
S2
UD
LR
A
A
LR
UD

FIG. 72


415G
415d
415d
415d
415h
415d
415d
415d
(UD)
U
(LR)
R
A2 (A)
A1
(A)
D
(UD)
L
(LR)

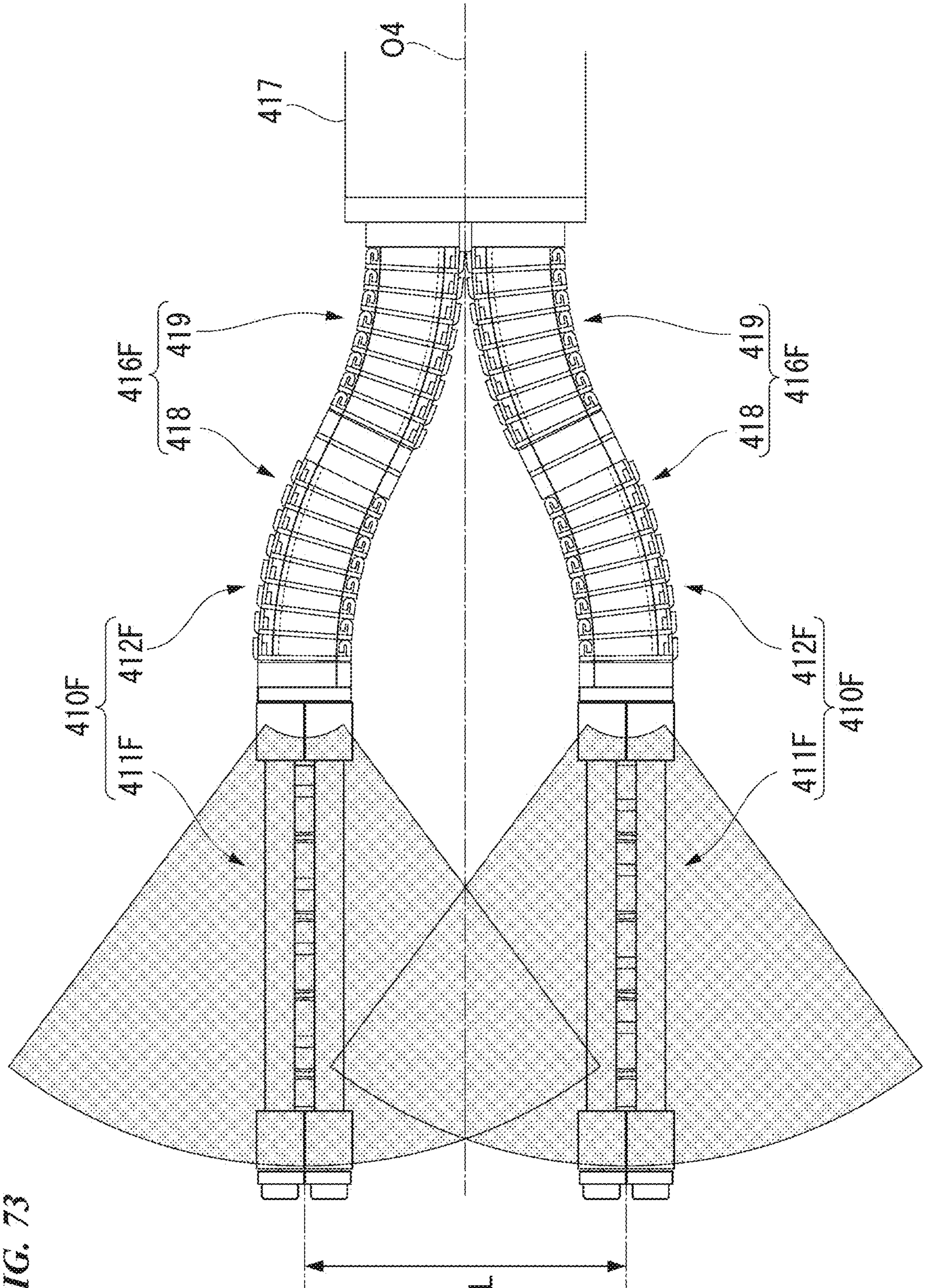


FIG. 73

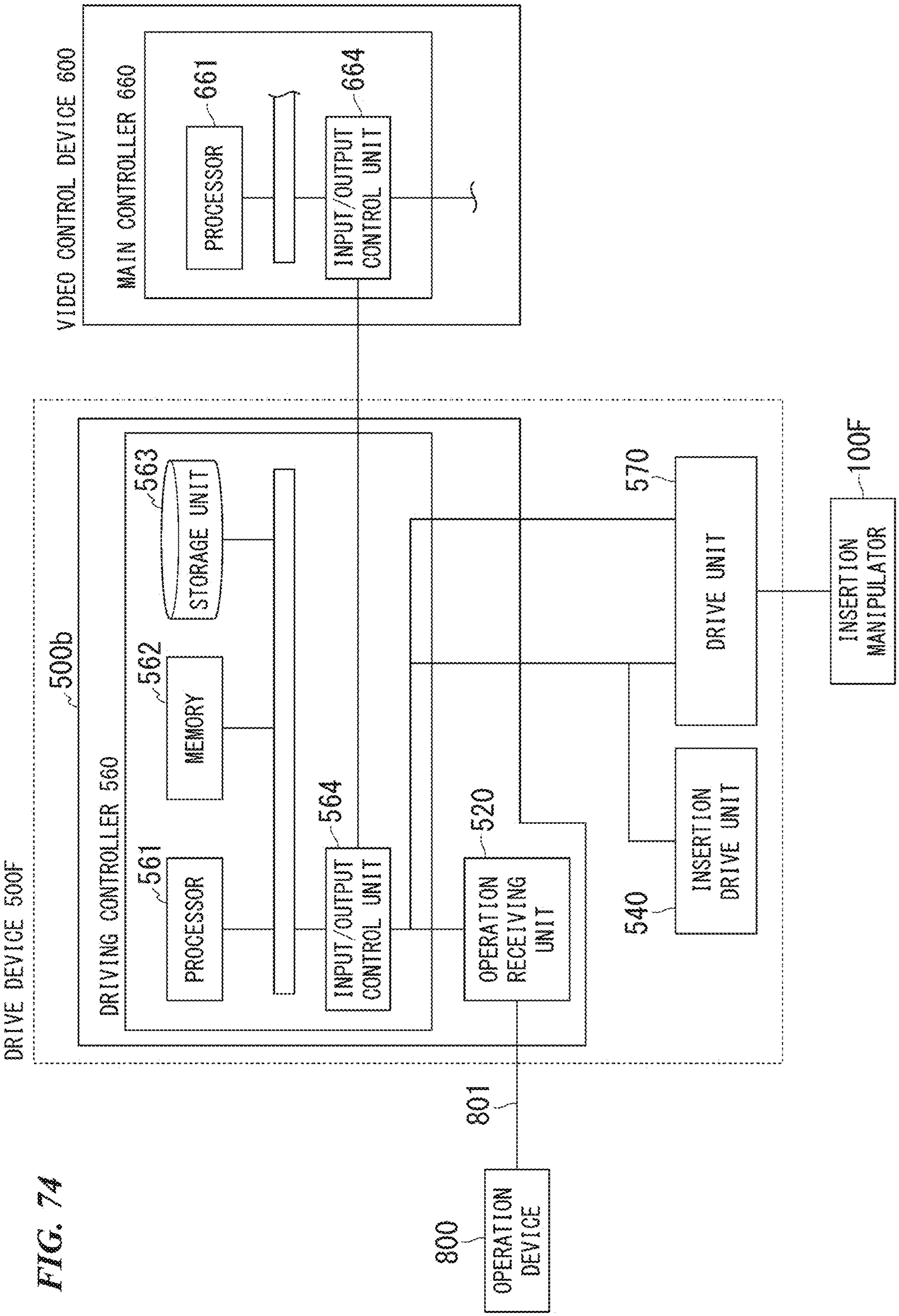
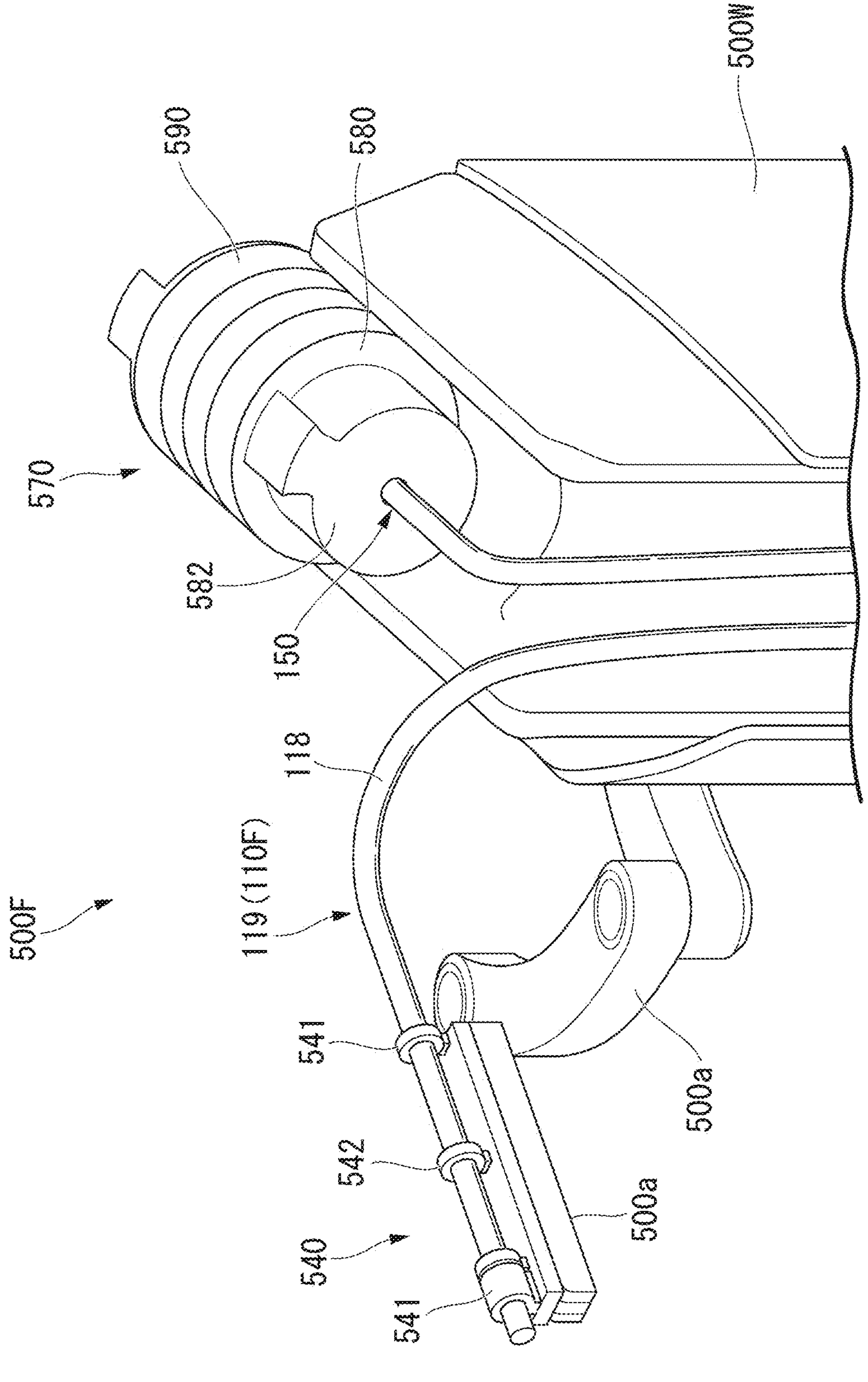

VIDEO CONTROL DEVICE 600
MAIN CONTROLLER 660
PROCESSOR
661
INPUT/OUTPUT CONTROL UNIT
664
DRIVE DEVICE 500F
500b
DRIVING CONTROLLER 560
PROCESSOR
561
MEMORY
562
STORAGE UNIT
563
INPUT/OUTPUT CONTROL UNIT
564
OPERATION RECEIVING UNIT
520
INSERTION DRIVE UNIT
540
DRIVE UNIT
570
INSERTION MANIPULATOR
100F
OPERATION DEVICE
800
801

FIG. 74

FIG. 75


500W
590
580
570
582
150
118
119(110F)
500F
541
542
540
541
500a
500a

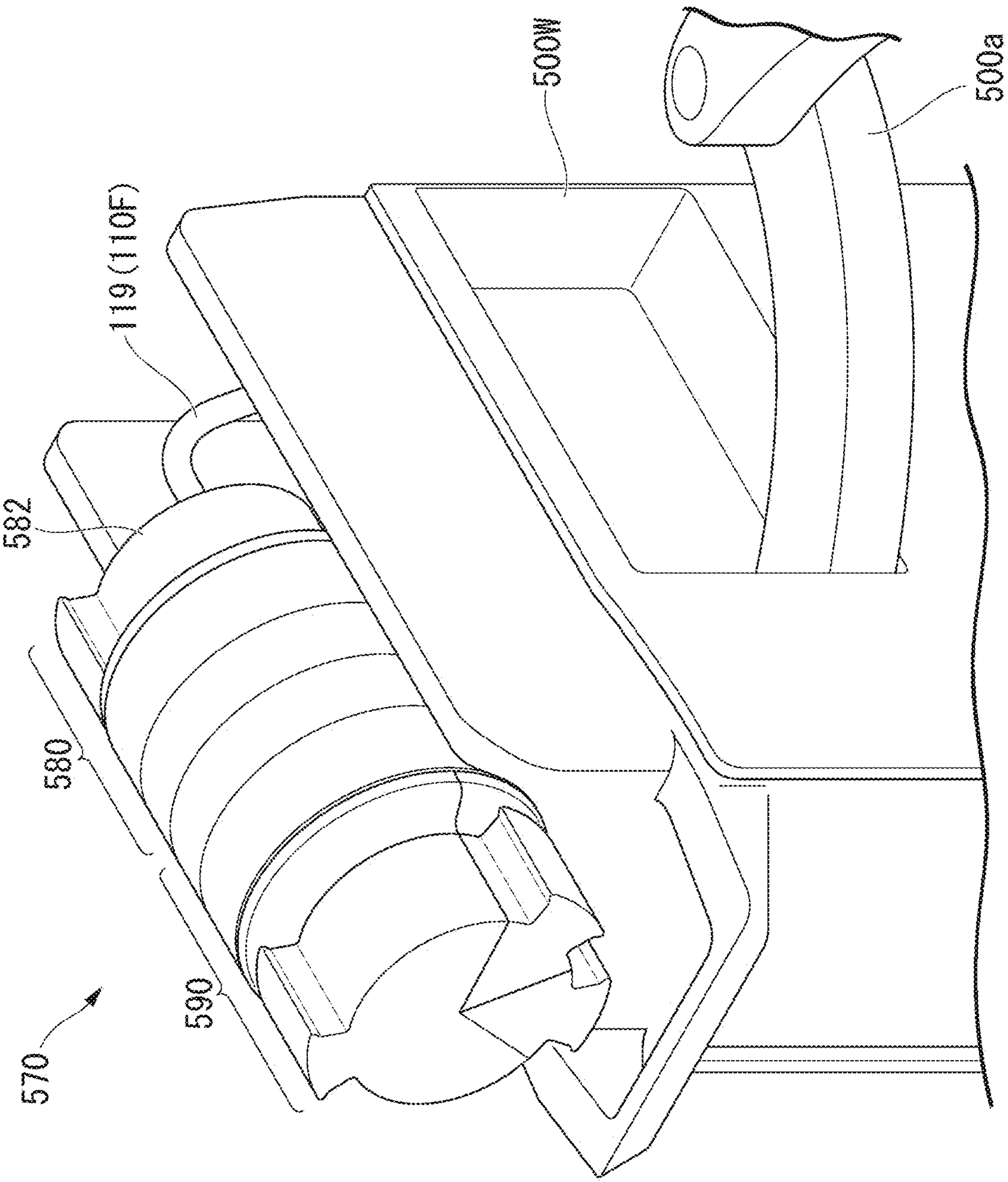


FIG. 76

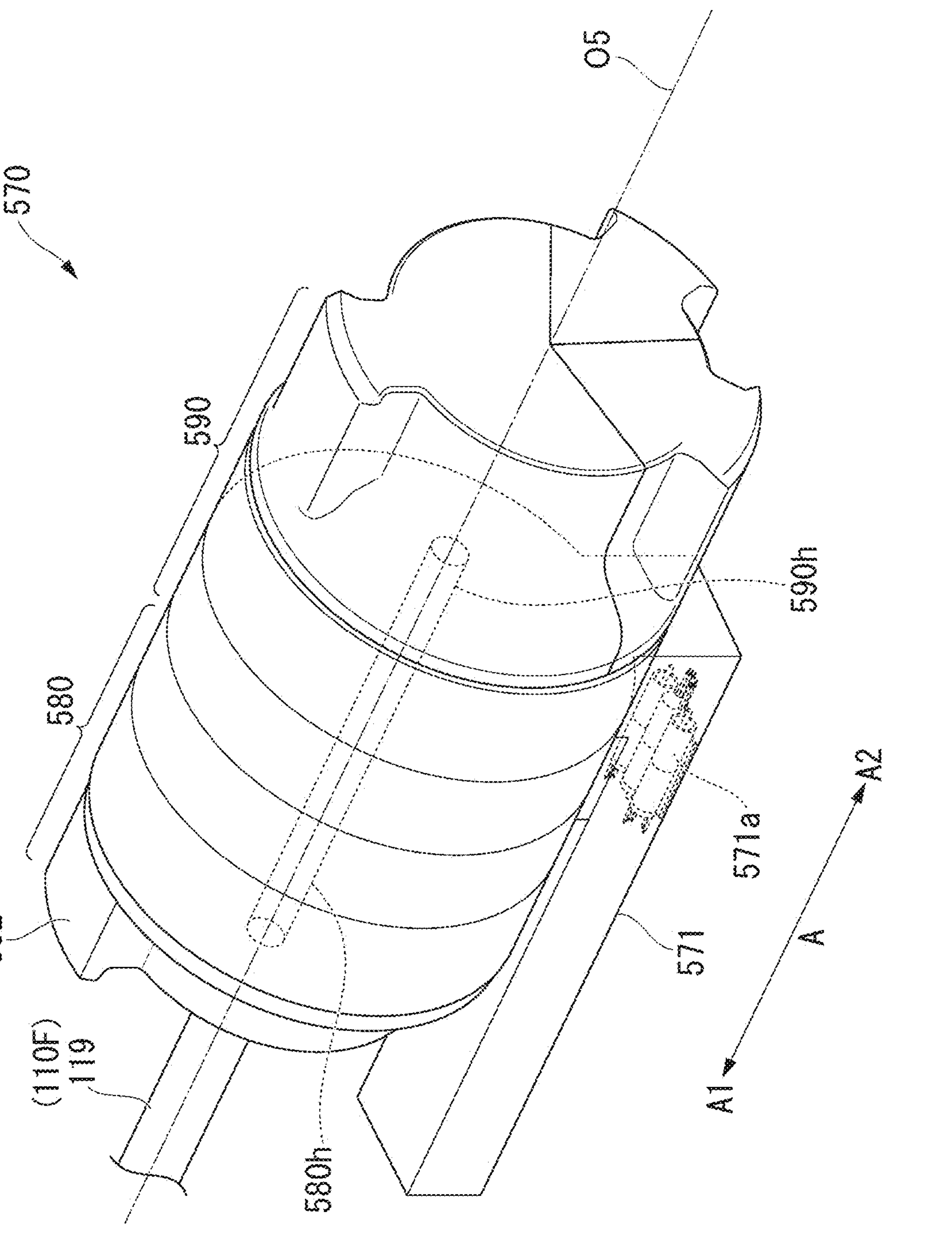


FIG. 77

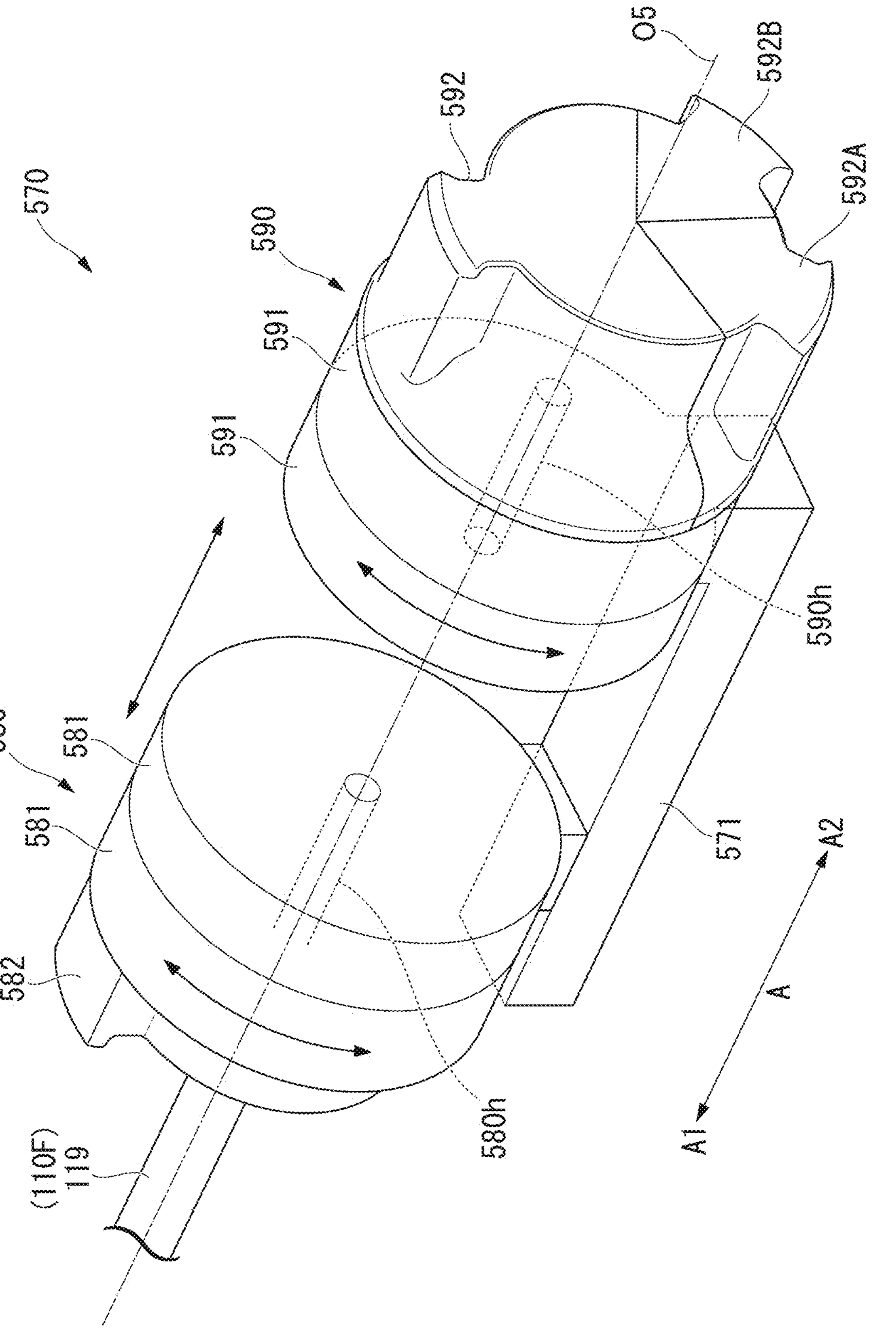
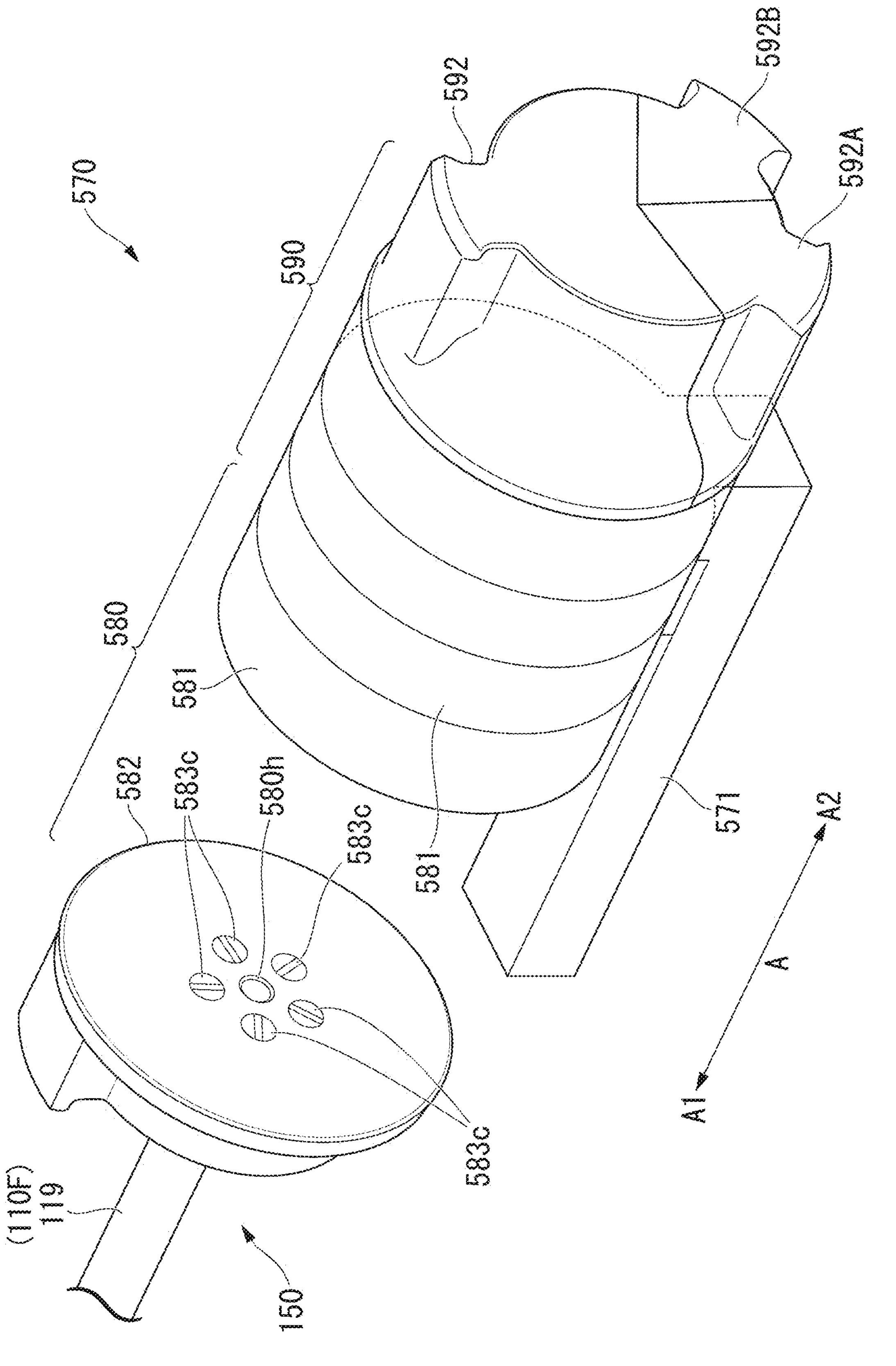

570
590
592
O5
592B
592A
591
591
590h
580
581
581
582
580h
571
A2
A
A1
119
(110F)

FIG. 78

FIG. 79


570
590
592
592B
592A
580
581
581
582
583c
580h
583c
583c
571
A2
A
A1
(110F)
119
150

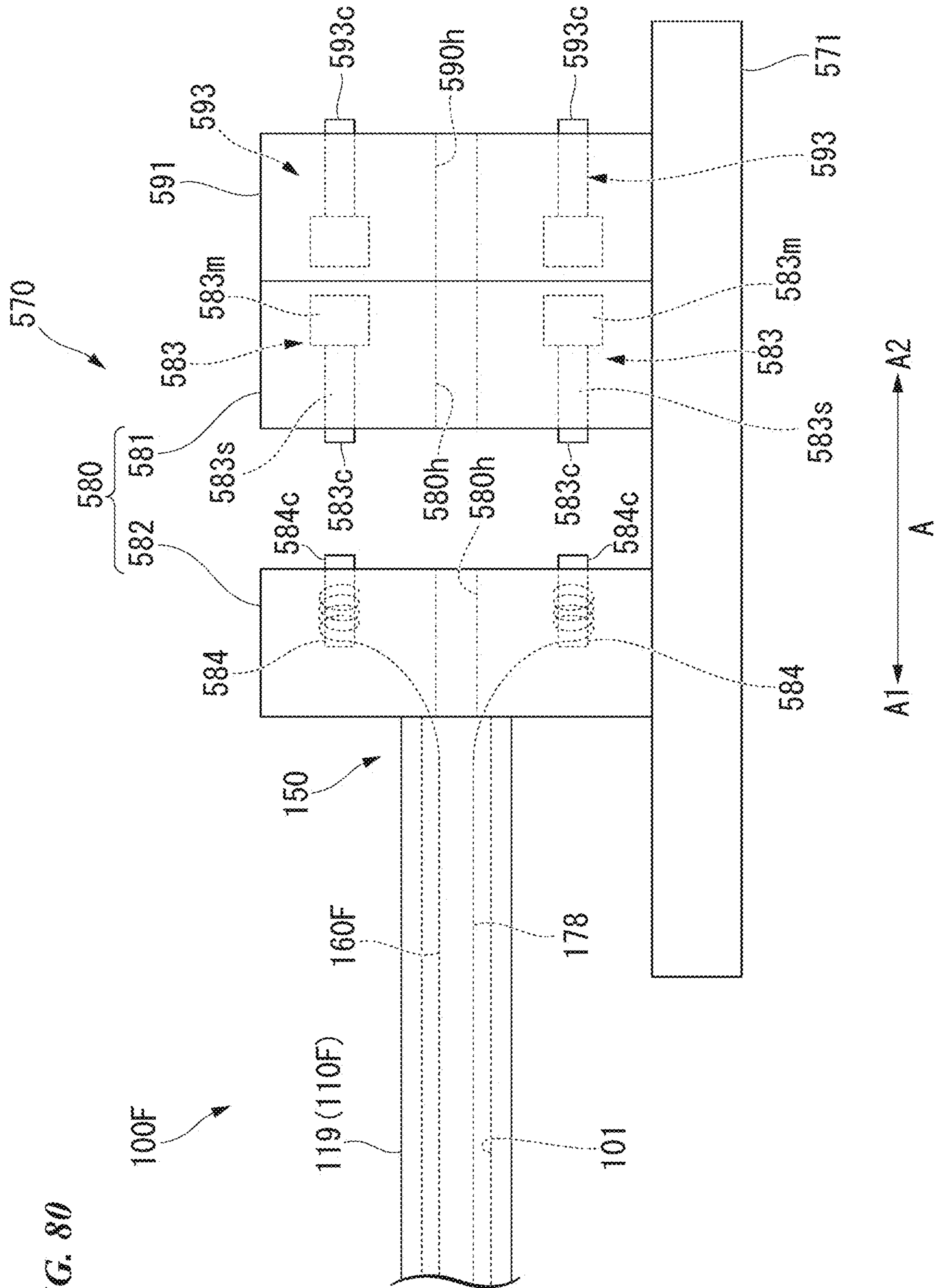

570
580
581
582
583
583m
583s
583c
584
584c
591
593
593c
590h
580h
571
150
160F
178
100F
119(110F)
101
A
A1
A2

FIG. 80

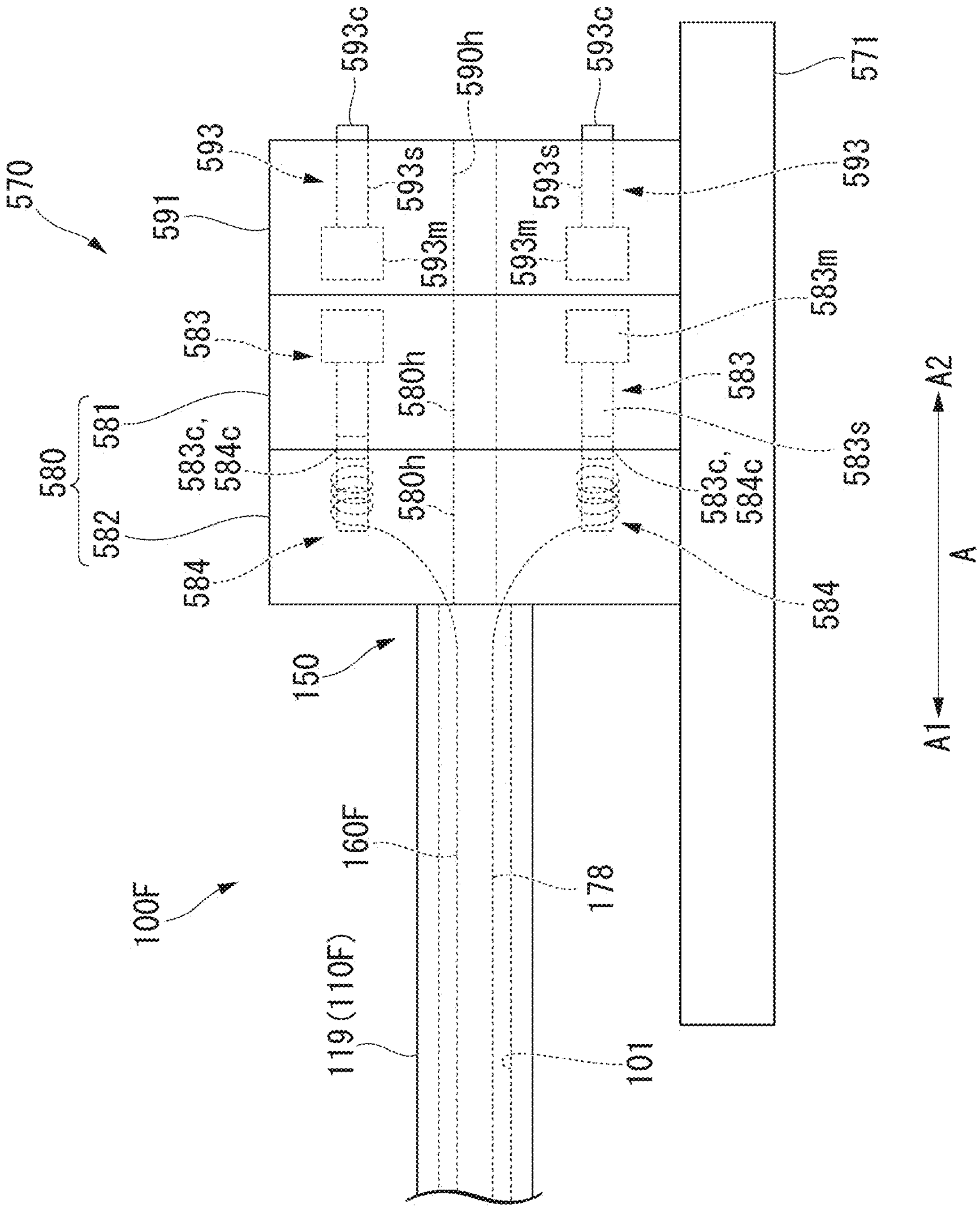

570
591
593
593c
590h
593c
571
593s
593s
593
593m
593m
583
583m
580h
583
581
580
583c, 584c
583c, 584c
583s
582
580h
584
584
A2
A
A1
150
160F
178
100F
119 (110F)
101

FIG. 81

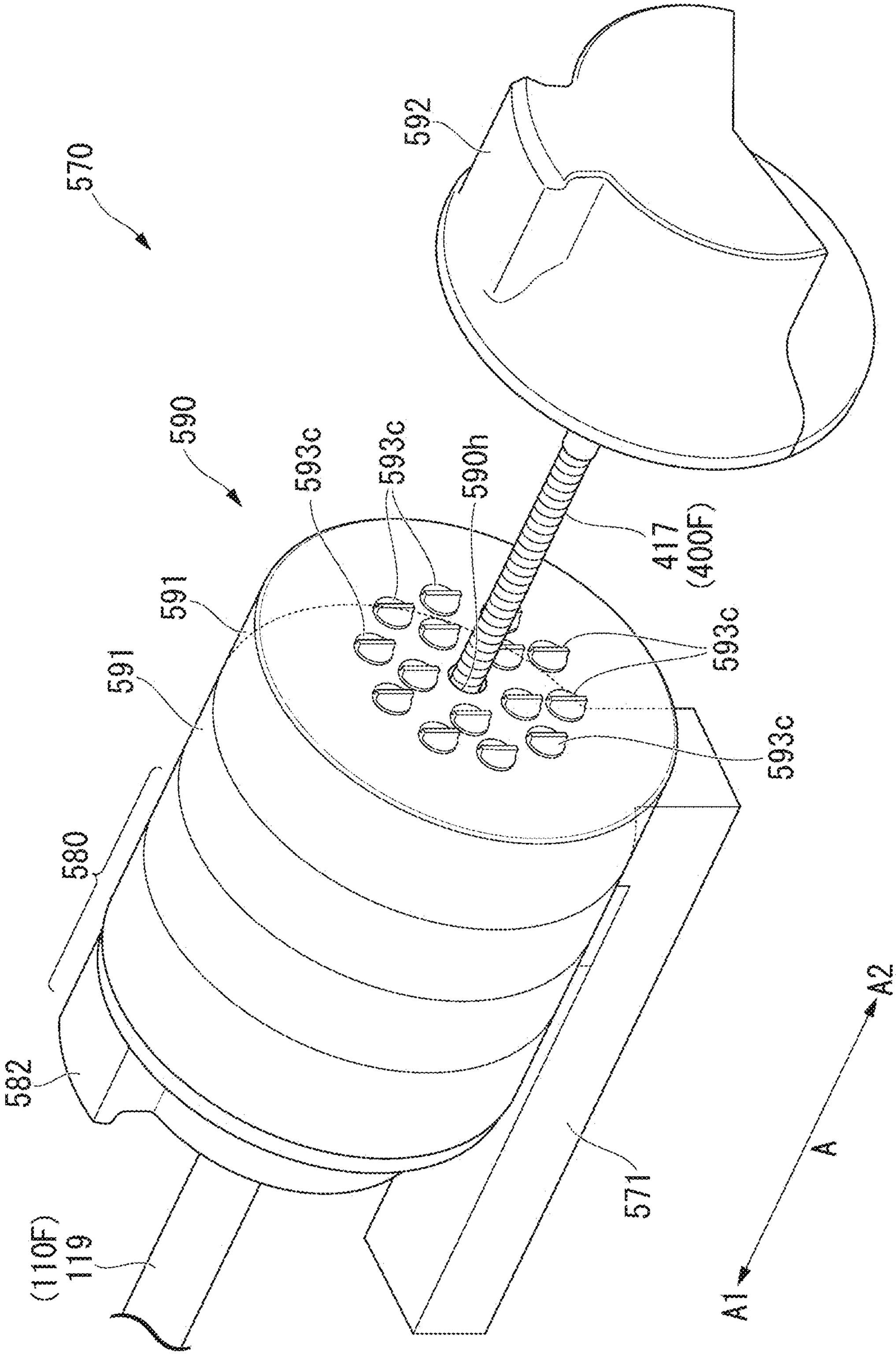
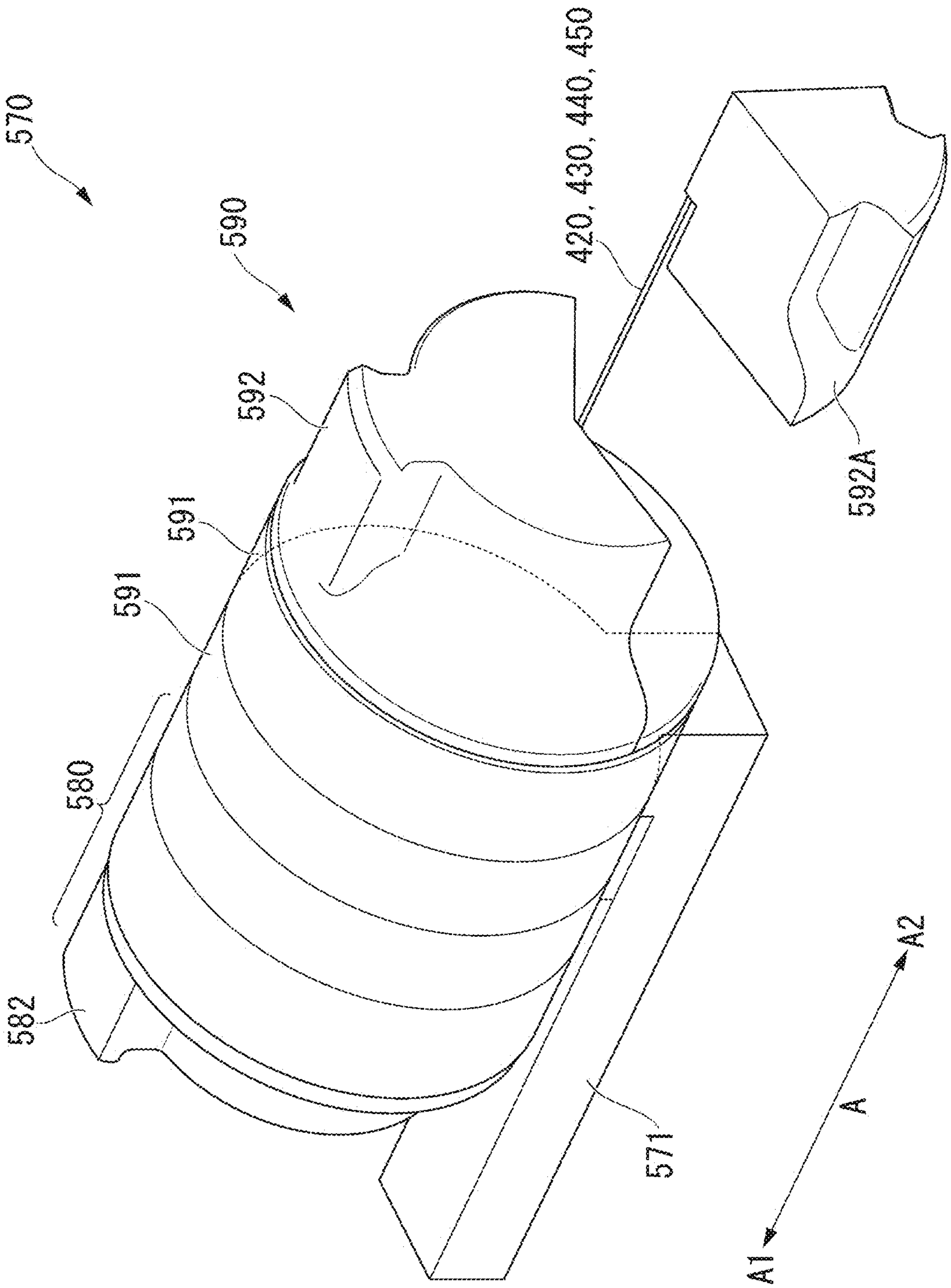

570
592
590
593c
593c
590h
417
(400F)
591
591
593c
593c
580
582
571
(110F)
119
A1
A
A2

FIG. 82

FIG. 83



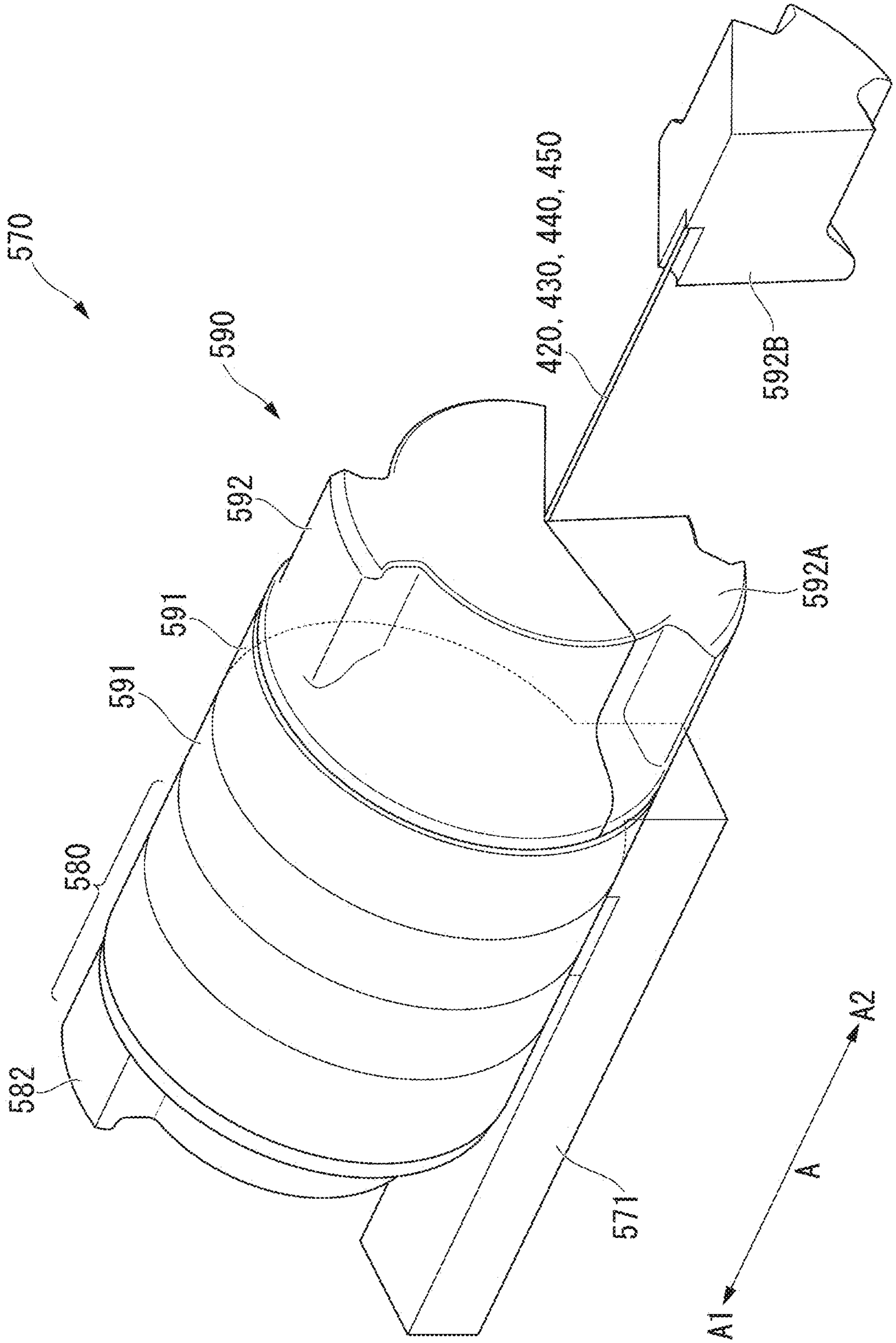
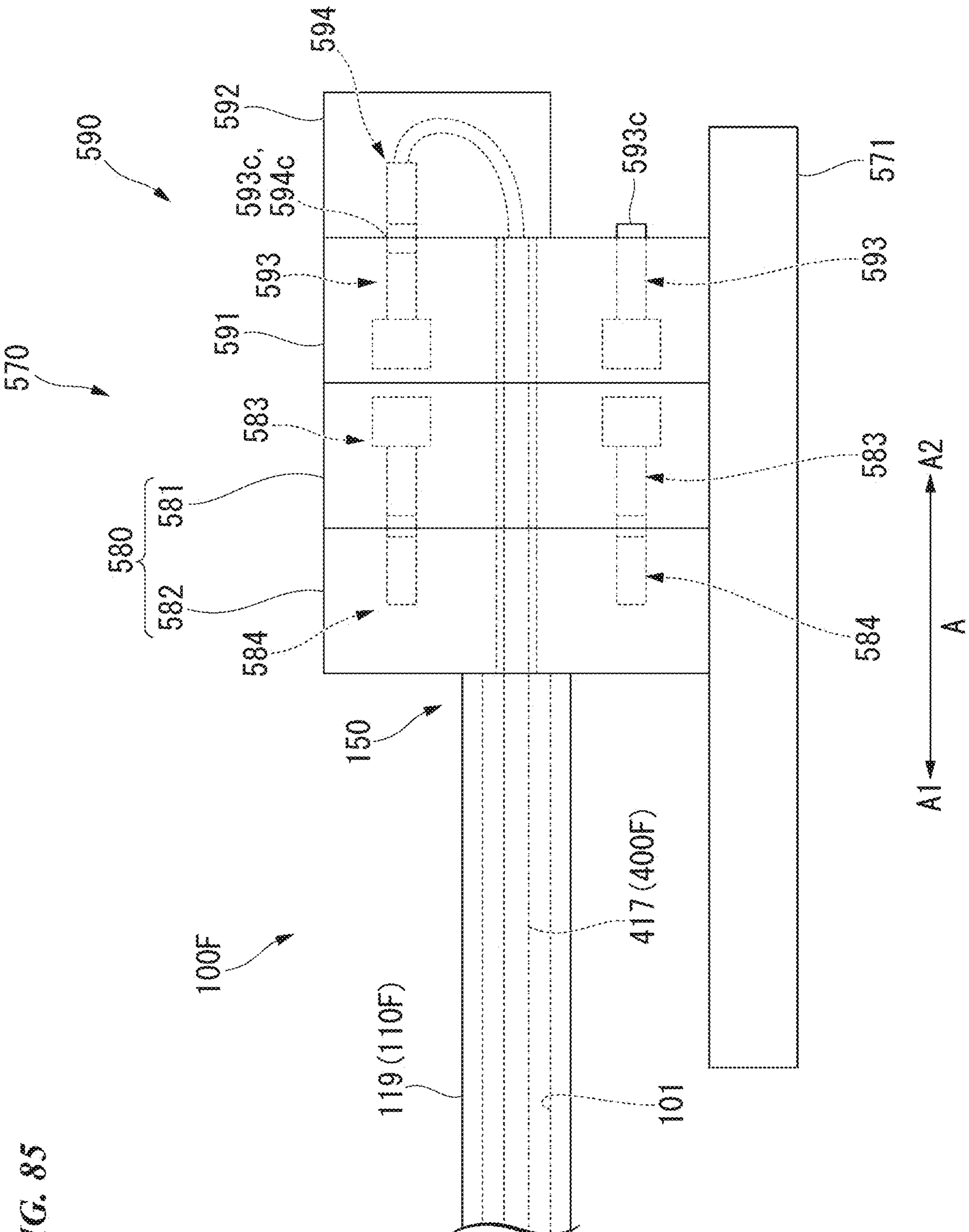
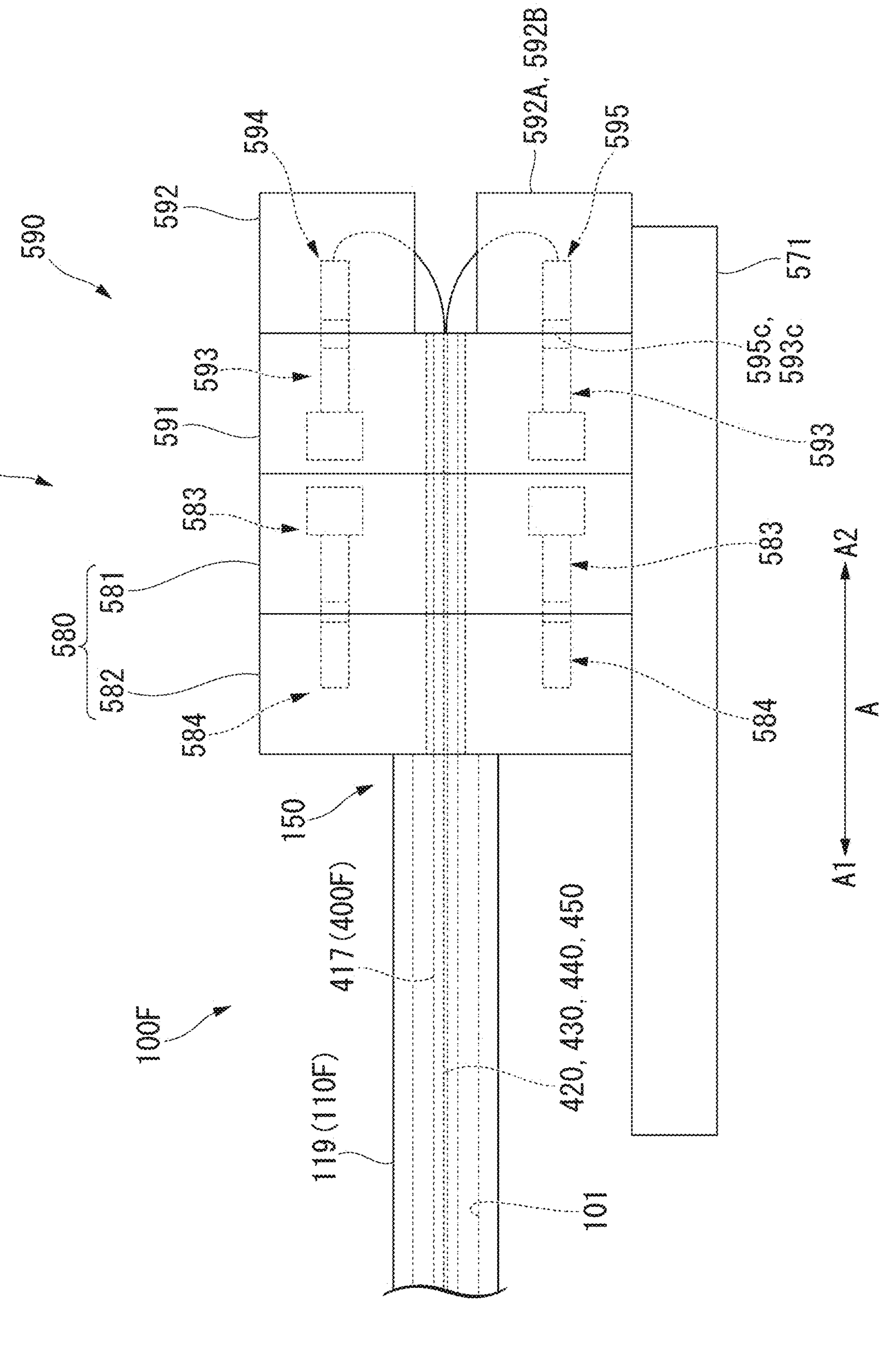


FIG. 84

FIG. 85


590
570
594
592
593c, 594c
593c
571
593
591
583
580
581
582
584
A2
A
A1
150
417 (400F)
100F
119 (110F)
101



590
570
594
592
592A, 592B
595
571
595c,
593c
593
591
593
583
581
580
582
584
583
584
A2
A
A1
150
417 (400F)
420, 430, 440, 450
100F
119 (110F)
101

FIG. 86

FIG. 87

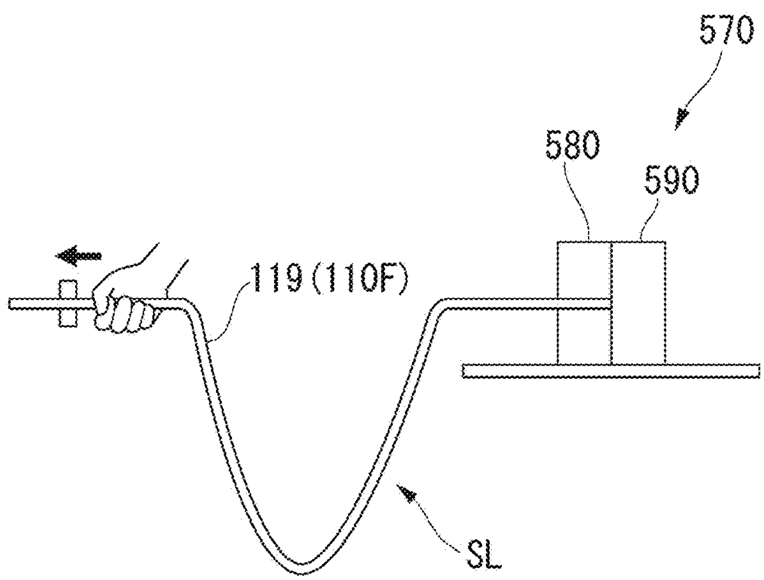


FIG. 88

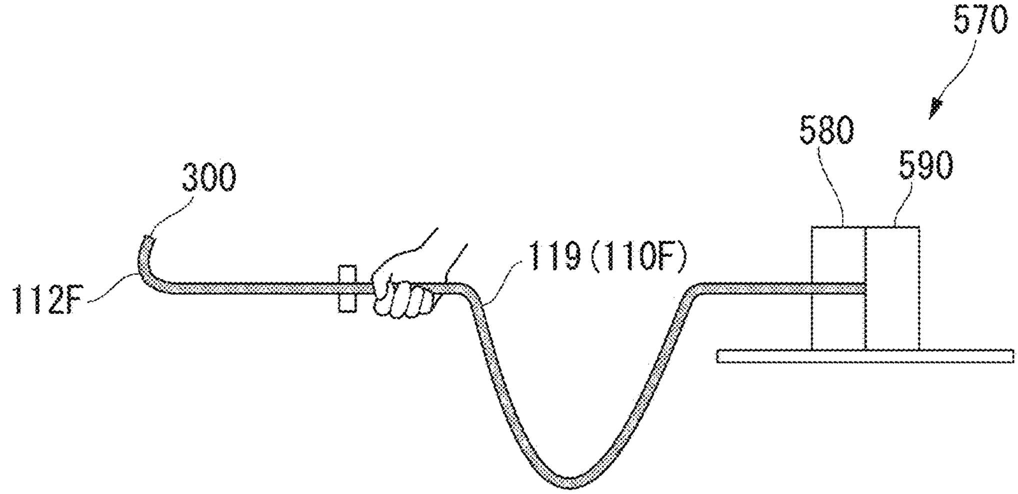


FIG. 89

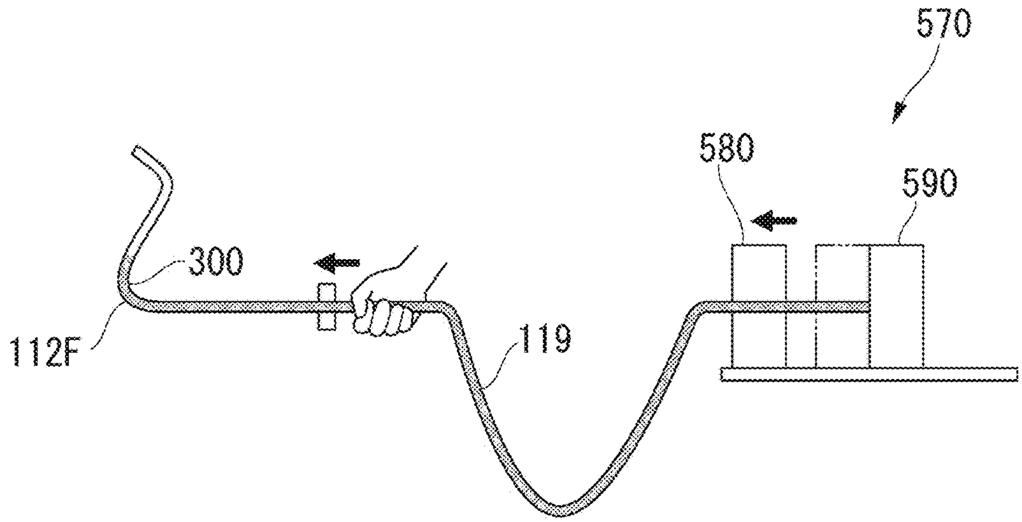


FIG. 90

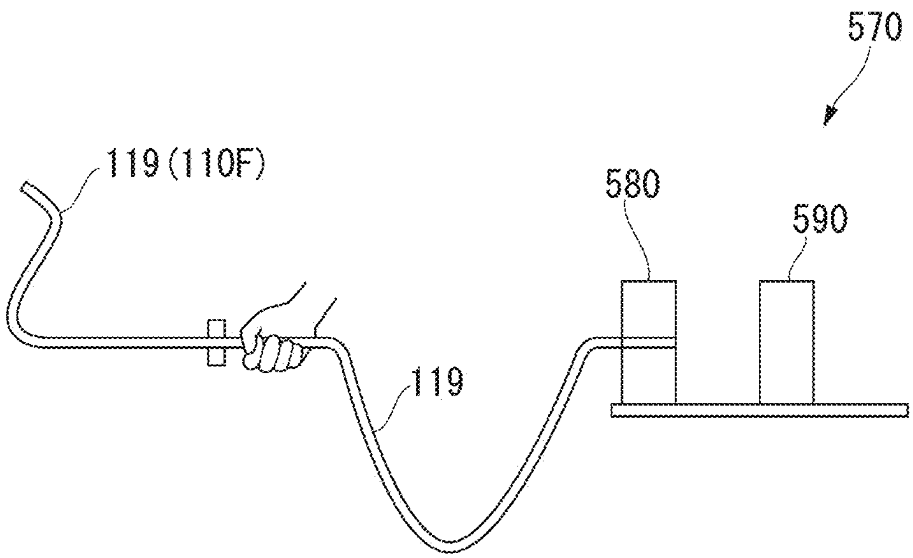


FIG. 91

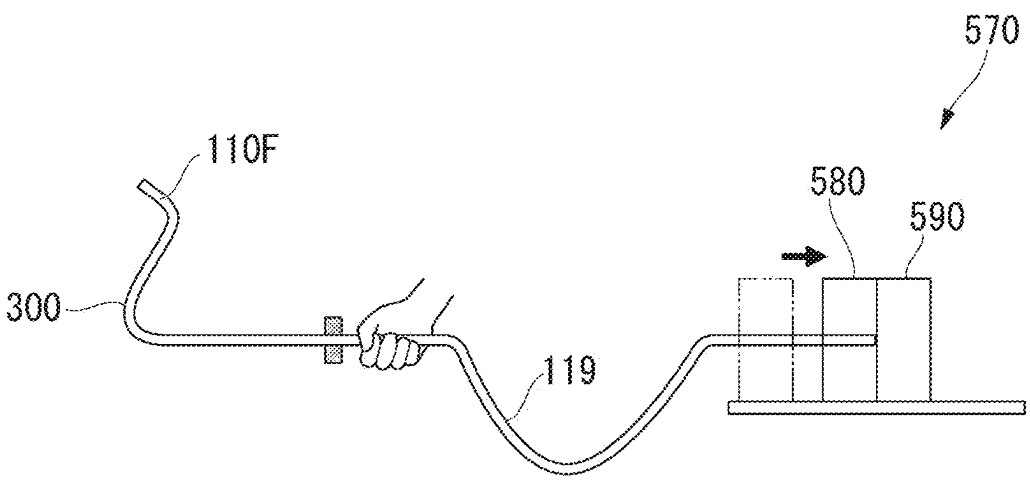
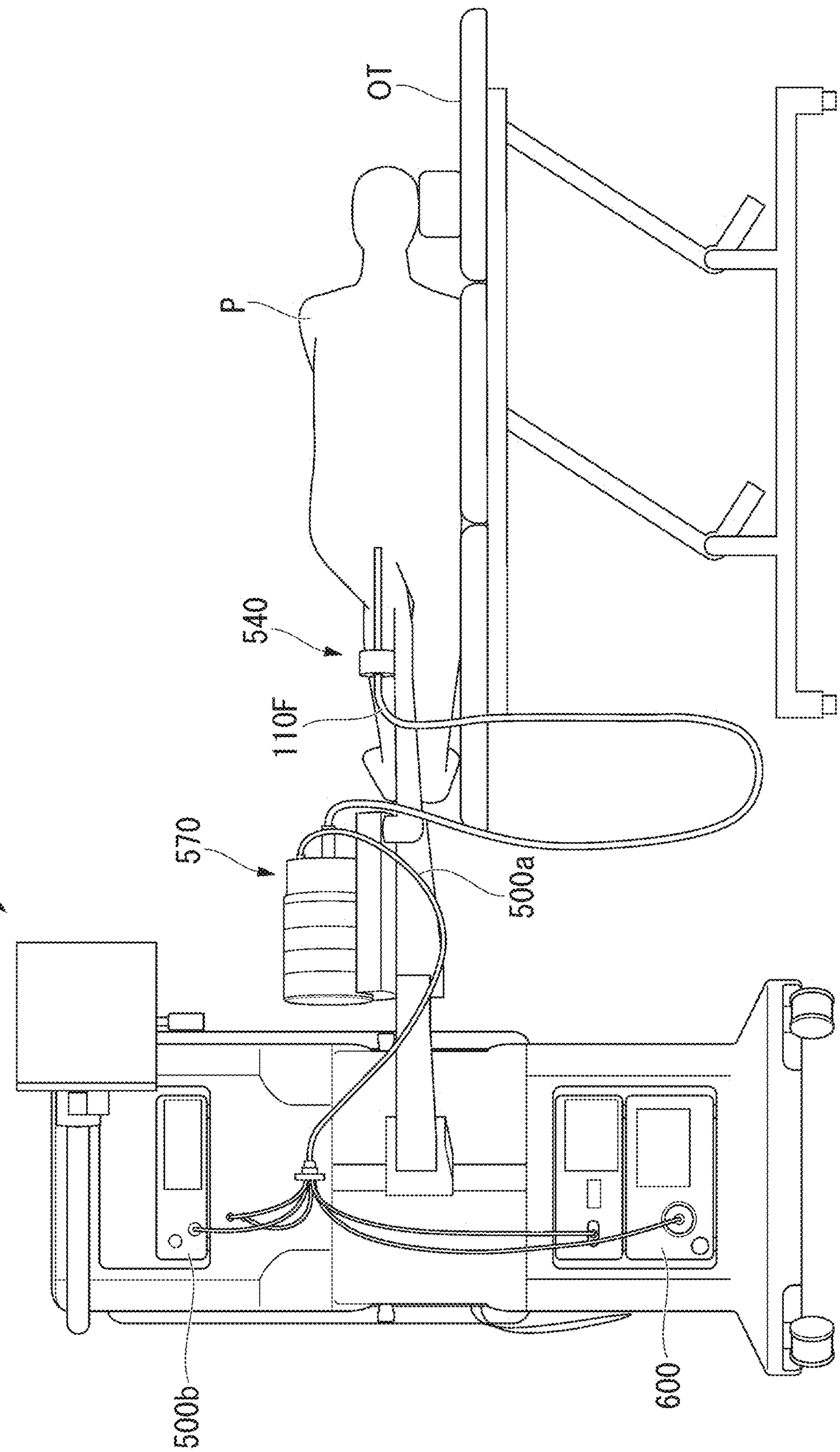


FIG. 92



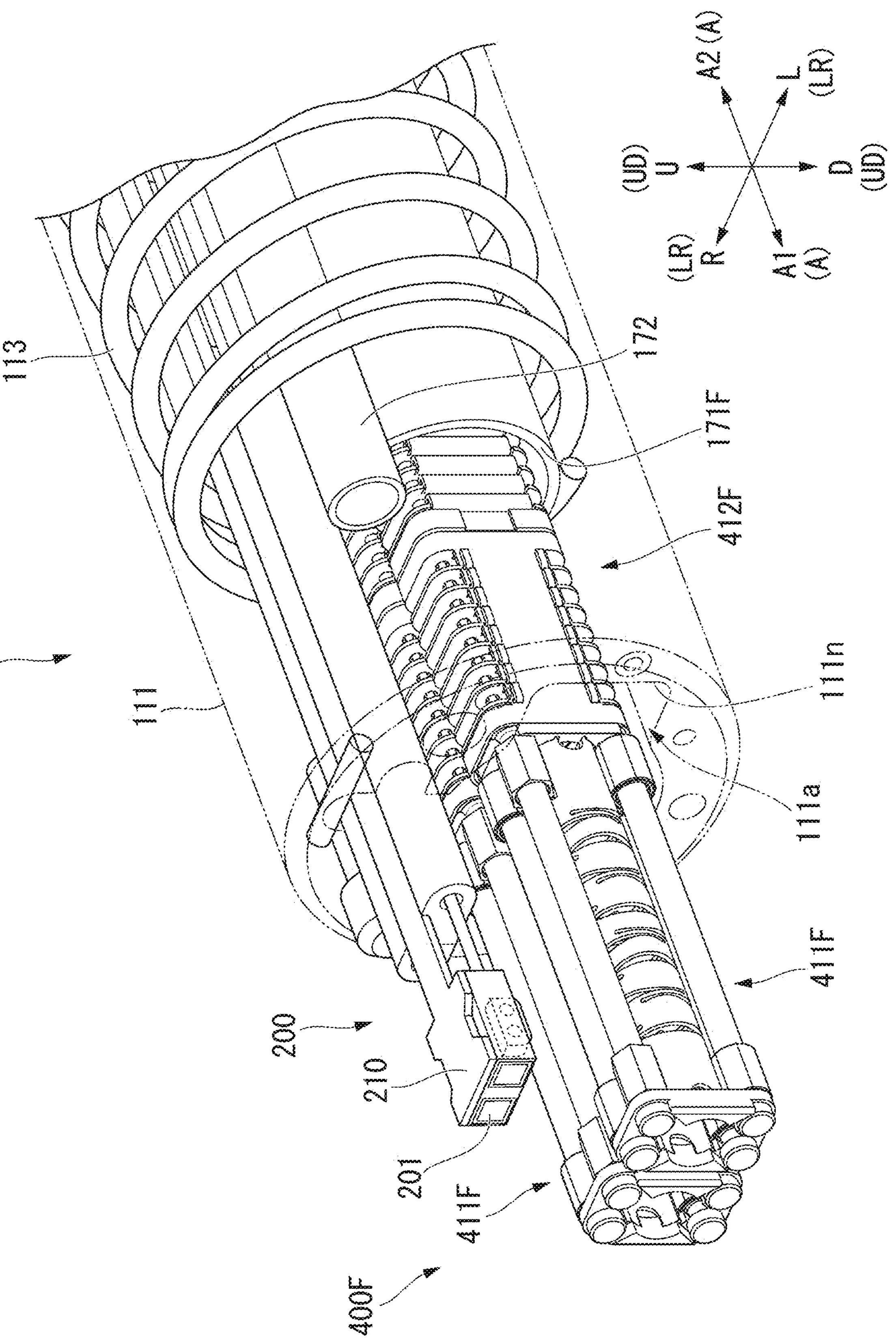
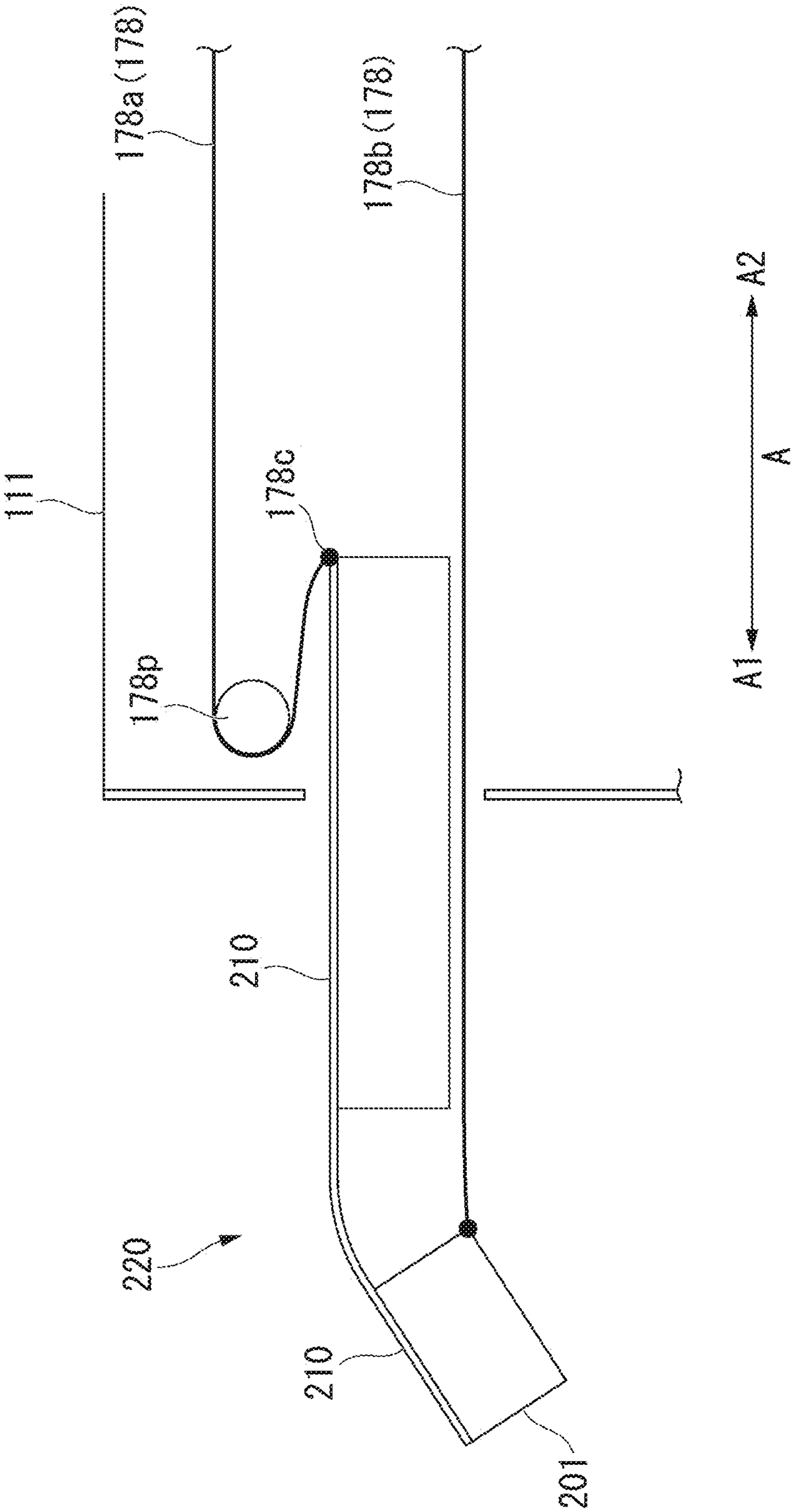

113
172
171F
412F
100F
111
111n
111a
411F
200
210
201
411F
400F
A2 (A)
L (LR)
(UD) U
D (UD)
(LR) R
A1 (A)

FIG. 93

FIG. 94



MEDICAL MANIPULATOR AND MEDICAL MANIPULATOR

[0001] The present disclosure relates to a medical manipulator and a medical manipulator system. The present application claims priority on U.S. Provisional Patent Applications No. 63/562,855 filed on Mar. 8, 2024, No. 63/635,021 filed on Apr. 17, 2024, No. 63/648,937 filed on May 17, 2024, No. 63/664,879 filed on Jun. 27, 2024, No. 63/669,306 filed on Jul. 10, 2024, No. 63/688,972 filed on Aug. 30, 2024, No. 63/691,009 filed on Sep. 5, 2024, No. 63/695,602 filed on Sep. 17, 2024, No. 63/737,339 filed on Dec. 20, 2024, and No. 63/763,398, provisionally filed on Feb. 26, 2025, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

Background

[0002] Conventionally, a medical manipulator system for observation or treatment in a hollow organ such as the alimentary canal has been used. The medical manipulator system can electrically drive an insertion section or the like inserted into a hollow organ. A user can control the operation of the insertion section or the like using an operation device provided outside of an internal part.

[0003] In Patent Document 1, a medical system including an endoscope which is electrically driven is described. In the medical system described in Patent Document 1, since the endoscope is electrically driven, it is possible to reduce fatigue of an operator.

PRIOR ART DOCUMENTS

Patent Documents

[0004] [Patent document 1] PCT International Publication No. WO 2021/145411

SUMMARY

[0005] A medical manipulator according to a first aspect of the present invention includes a manipulator flexible portion extending in a longitudinal direction; and two arms arranged at a distal end of the manipulator flexible portion, and the two arms are arranged in a left-right direction perpendicular to the longitudinal direction, each of the two arms includes a second bending portion, and distal ends of second bending portions of the two arms are spaced apart in the left-right direction by bending the second bending portions.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** A diagram illustrating the entire configuration of an electric endoscope system according to a first embodiment.

[0007] FIG. **2** A diagram illustrating an insertion manipulator of an electric endoscope system inserted into a large intestine.

[0008] FIG. **3** A diagram illustrating a distal end portion of an insertion section of the insertion manipulator.

[0009] FIG. **4** A front view of the distal end portion when seen from the distal end side.

[0010] FIG. **5** A sectional view of the distal end portion.

[0011] FIG. **6** A diagram illustrating a bendable portion of the insertion manipulator.

[0012] FIG. **7** A diagram illustrating joint rings of the bendable portion.

[0013] FIG. **8** A diagram illustrating a first channel tube of the insertion manipulator.

[0014] FIG. **9** A sectional view of a proximal end channel tube of the first channel tube.

[0015] FIG. **10** A functional block diagram of a drive device.

[0016] FIG. **11** A functional block diagram of a video control device.

[0017] FIG. **12** A diagram illustrating a modified example of the bendable portion.

[0018] FIG. **13** A diagram illustrating a first channel tube inserted into the modified example of the bendable portion.

[0019] FIG. **14** A diagram illustrating a modified example of the insertion section.

[0020] FIG. **15** A diagram illustrating a spiral tube.

[0021] FIG. **16** A diagram illustrating a modified example of the distal end portion.

[0022] FIG. **17** A diagram illustrating a hardness-changing device according to a second embodiment.

[0023] FIG. **18** A diagram illustrating the entire configuration of the hardness-changing device.

[0024] FIG. **19** A diagram illustrating a hardness-changing unit.

[0025] FIG. **20** A diagram illustrating the hardness-changing unit.

[0026] FIG. **21** A diagram illustrating an operation of the hardness-changing unit.

[0027] FIG. **22** A diagram illustrating an operation of the hardness-changing unit.

[0028] FIG. **23** A diagram illustrating an operation of the hardness-changing unit.

[0029] FIG. **24** A diagram illustrating an operation of the hardness-changing unit.

[0030] FIG. **25** A diagram illustrating an operation of the hardness-changing unit.

[0031] FIG. **26** A diagram illustrating a modified example of the hardness-changing unit.

[0032] FIG. **27** A diagram illustrating the modified example of the hardness-changing unit.

[0033] FIG. **28** A diagram illustrating another modified example of the hardness-changing unit.

[0034] FIG. **29** A diagram illustrating the modified example of the hardness-changing unit.

[0035] FIG. **30** A diagram illustrating another modified example of the hardness-changing unit.

[0036] FIG. **31** A diagram illustrating the modified example of the hardness-changing unit.

[0037] FIG. **32** A diagram illustrating another modified example of the hardness-changing unit.

[0038] FIG. **33** A diagram illustrating a high-frequency knife in a manipulator tool according to a third embodiment.

[0039] FIG. **34** A diagram illustrating a high-frequency knife for marking.

[0040] FIG. **35** A diagram illustrating a local injection needle.

[0041] FIG. **36** A diagram illustrating the local injection needle that locally injects a liquid.

[0042] FIG. **37** A diagram illustrating the high-frequency knife for incision.

[0043] FIG. **38** A diagram illustrating the high-frequency knife for incision.

[0044] FIG. **39** A diagram illustrating a basket.

[0045] FIG. **40** A diagram illustrating the basket for recovering a target part.
[0046] FIG. **41** A diagram illustrating a manipulator tool according to a fourth embodiment.
[0047] FIG. **42** A sectional view of the manipulator tool.
[0048] FIG. **43** A diagram illustrating a modified example of the manipulator tool.
[0049] FIG. **44** A diagram illustrating an artificial muscle disposed at another position.
[0050] FIG. **45** A diagram illustrating an artificial muscle disposed at another position.
[0051] FIG. **46** A diagram illustrating an artificial muscle disposed at another position.
[0052] FIG. **47** A diagram illustrating an artificial muscle disposed at another position.
[0053] FIG. **48** A diagram illustrating an artificial muscle disposed at another position.
[0054] FIG. **49** A diagram illustrating another modified example of the manipulator tool.
[0055] FIG. **50** An overall view of an electric endoscope system according to a fifth embodiment.
[0056] FIG. **51** A perspective view of the distal end of an insertion manipulator of the electric endoscope system.
[0057] FIG. **52** A cross-sectional view of the insertion section of the insertion manipulator.
[0058] FIG. **53** A view showing the insertion section.
[0059] FIG. **54** A view showing the bending portion of the insertion manipulator.
[0060] FIG. **55** A view showing a ring member.
[0061] FIG. **56** A view showing the ring member as viewed from the longitudinal direction.
[0062] FIG. **57** A view showing a first channel tube.
[0063] FIG. **58** A view showing a treatment manipulator protruding from a first opening.
[0064] FIG. **60** A view showing the treatment manipulator.
[0065] FIG. **61** Same as above.
[0066] FIG. **62** Perspective view of the bending tube.
[0067] FIG. **63** Exploded view of the bending tube expanded in the circumferential direction.
[0068] FIG. **64** View of the bending tube compressed.
[0069] FIG. **65** View of the bending tube bending.
[0070] FIG. **66** View of an artificial muscle.
[0071] FIG. **67** View of a bending piece.
[0072] FIG. **68** View of a modified bending piece.
[0073] FIG. **69** View of another modified bending piece.
[0074] FIG. **70** View of another modified bending piece.
[0075] FIG. **71** View of another modified bending piece.
[0076] FIG. **72** View of a connecting piece.
[0077] FIG. **73** View of a treatment instrument arm with a bending second bending portion.
[0078] FIG. **74** Functional block diagram of a drive device.
[0079] FIG. **75** View of an insertion drive unit.
[0080] FIG. **76** View of a drive unit.
[0081] FIG. **77** View of the operation of the drive unit.
[0082] FIG. **78** Same as above.
[0083] FIG. **79** Exploded view of the first drive unit.
[0084] FIG. **80** Cross-sectional view of the first motor unit with the first adapter separated.
[0085] FIG. **81** Cross-sectional view of the first motor unit with the first adapter attached.
[0086] FIG. **82** Exploded view of the second drive unit.
[0087] FIG. **83** Same as above.
[0088] FIG. **84** Same as above.
[0089] FIG. **85** Cross-sectional view of the second motor unit with the second adapter attached.
[0090] FIG. **86** Cross-sectional view of the second motor unit with the third adapter and the fourth adapter attached.
[0091] FIG. **87** A diagram showing an example of the operation of the drive unit.
[0092] FIG. **88** Same as above.
[0093] FIG. **89** Same as above.
[0094] FIG. **90** Same as above.
[0095] FIG. **91** Same as above.
[0096] FIG. **92** A diagram showing a rack as a modified example of the cart.
[0097] FIG. **93** A diagram showing a scope protruding from the distal end.
[0098] FIG. **94** A diagram explaining the operation of the scope operating wire.

DETAILED DESCRIPTION

First Embodiment

[0099] An electric endoscope system **1000** according to a first embodiment of the present invention will be described below with reference to FIGS. **1** to **13**. FIG. **1** is a diagram illustrating the entire configuration of the electric endoscope system **1000** according to the present embodiment. The electric endoscope system **1000** is an example of a medical manipulator system. A medical manipulator includes an insertion manipulator **100** inserted into a human body and an endoscope, a catheter, a treatment tool, and an endoluminal device which are electrically driven.

[Electric Endoscope System **1000**]

[0100] The electric endoscope system **1000** is a medical system used to observe and treat an internal part of a patient. The electric endoscope system **1000** includes an insertion manipulator **100**, a treatment manipulator **400**, a drive device **500**, a video control device **600**, an operation device **800**, and a display device **900**.
[0101] FIG. **2** is a diagram illustrating an insertion manipulator **100** inserted into a large intestine.
[0102] The insertion manipulator **100** is a device inserted into a lumen of a patient to observe and treat a lesion. The insertion manipulator **100** has excellent insertability and can be inserted into, for example, an ascending colon AC or a cecum CE of a large intestine as illustrated in FIG. **2**. The insertion manipulator **100** is detachably attached to the drive device **500** and the video control device **600**. An internal path **101** is formed in the insertion manipulator **100**. In the following description, a side of the insertion manipulator **100** inserted into a lumen of a patient is referred to as a "distal end side (distal side) A**1**," and a side attached to the drive device **500** is referred to as a "proximal side (proximal side) A**2**."
[0103] The treatment manipulator **400** is, for example, a device inserted into a first channel tube **171** of the insertion manipulator **100**, protrudes from a first opening **111***a*, and is inserted into a lumen of a patient to treat a lesion. An end effector (a treatment portion) that treats a lesion is disposed at the distal end of the treatment manipulator **400**.
[0104] The drive device **500** is detachably connected to the insertion manipulator **100** and the operation device **800**. The drive device **500** drives a motor built thereinto on the basis of an operation input to the operation device **800** to

electrically drive the insertion manipulator **100**. The drive device **500** drives a pump or the like built thereinto on the basis of an operation input to the operation device **800** and causes the insertion manipulator **100** to perform air supply, water supply, and suction.

[0105] The video control device **600** is detachably connected to the insertion manipulator **100** and acquires a captured image from the insertion manipulator **100**. The video control device **600** causes the display device **900** to display the captured image acquired from the insertion manipulator **100** or a GUI image or a CG image to provide information to an operator.

[0106] The drive device **500** and the video control device **600** constitute a control device **700** that controls the electric endoscope system **1000**. The control device **700** may further include peripherals such as a video printer. The drive device **500** and the video control device **600** may be a unified device.

[0107] The operation device **800** is detachably connected to the drive device **500** via an operation cable **801**. The operation device **800** may communicate with the drive device **500** in a wireless manner instead of in a wired manner. An operator S can electrically drive the insertion manipulator **100** by operating the operation device **800**.

[0108] The display device **900** is a device that can display an image such as an LCD. The display device **900** is connected to the video control device **600** via a display cable **901**.

[0109] The operation device **800** and the display device **900** are mounted in a trolley. The trolley on which the operation device **800** and the display device **900** are mounted is also referred to as a "console CON."

[0110] The constituent devices of the electric endoscope system **1000** will be described below.

[Insertion Manipulator **100**]

[0111] FIG. **3** is a diagram illustrating a distal end portion of an insertion section **110**. The insertion manipulator **100** includes an insertion section **110**, an attachment portion **150**, a bending wire **160**, a built-in element **170**, and a scope **200** as illustrated in FIGS. **1** and **3**.

[0112] In the insertion manipulator **100**, an internal path (a lumen) **101** extending in a longitudinal direction (a longitudinal axis direction, an axial direction) A of the insertion manipulator **100** is formed from the distal end of the insertion section **110** to the proximal end of the attachment portion **150**. The bending wire **160** and the built-in element **170** are inserted into the internal path **101**.

[0113] The insertion section **110** is a thin longitudinal member that can be inserted into a lumen. The insertion section **110** includes a distal end portion **111**, a bending portion **112**, and a flexible portion **119**. The distal end portion **111**, the bending portion **112**, and the flexible portion **119** are sequentially connected from the distal end side **A1** to the proximal side **A2**. The insertion section **110** includes an outer sheath **118** which is the outermost part.

[0114] FIG. **4** is a front view of the distal end portion **111** when seen from the distal end side **A1**.

[0115] The distal end portion **111** is formed in a cylindrical shape. The distal end portion **111** includes a first opening **111***a*, a second opening **111***b*, a water supply nozzle **111***d*, an air supply nozzle **111***e*, and a suction nozzle **111***f*. The first opening **111***a*, the second opening **111***b*, the water supply nozzle **111***d*, the air supply nozzle **111***e*, and the suction nozzle **111***f* are formed on a distal end surface of the distal end portion **111**.

[0116] FIG. **5** is a sectional view of the distal end portion **111**.

[0117] The built-in element **170** is inserted into the internal path **101**. The built-in element **170** includes a first channel tube **171**, a second channel tube **172**, an imaging cable **173**, a light guide **174**, a water supply tube **175**, an air supply tube **176**, and a suction tube **177**. In FIG. **5**, two treatment manipulators **400** that are inserted into the first channel tube **171** are also illustrated.

[0118] The first opening **111***a* is an opening that communicates with the first channel tube **171**. The first opening **111***a* is a circular opening in a front view when seen from the distal end side **A1**. The distal end of the treatment manipulators **400** inserted into the first channel tube **171** protrudes from and retract to the first opening **111***a*.

[0119] In the first opening **111***a*, a notch portion **111***n* is formed at both ends in a direction (an LR direction which will be described later) perpendicular to a longitudinal direction A. As illustrated in FIG. **3**, the treatment manipulators **400** protruding from the first opening **111***a* to the distal end side **A1** can be inserted into the notch portions **111***n*. The notch portions **11** In may not penetrate in the direction perpendicular to the longitudinal direction A as long as they are formed on the inner circumferential surface of the first opening **111***a*.

[0120] The second opening **111***b* is an opening that communicates with the second channel tube **172**. The second opening **111***b* is a circular opening in a front view when seen from the distal end side **A1**. The distal end of the treatment manipulator **400** inserted into the second channel tube **172** protrudes from and retracts to the second opening **111***b*. Devices inserted into the second channel tube **172** are not limited to the treatment manipulator **400** and include medical devices such as manual instruments.

[0121] An inner diameter D1 of a part other than the notch portions **111***n* in the first opening **111***a* is larger than an inner diameter D2 of the second opening **111***b*. Specifically, the diameter D1 of the part other than the notch portions **11** In in the first opening **111***a* is three to five times the inner diameter D2 of the second opening **111***b*.

[0122] The water supply nozzle **111***d* is an opening that communicates with the water supply tube **175**. A liquid in a tank installed in the vicinity of the control device **700** is sent out from the water supply nozzle **111***d* via the water supply tube **175**.

[0123] The air supply nozzle **111***e* is an opening that communicates with the air supply tube **176**. A gas in a tank installed in the vicinity of the control device **700** is sent out from the air supply nozzle **111***e* via the air supply tube **176**.

[0124] The suction nozzle **111***f* is an opening that communicates with the suction tube **177**. A tank installed in the vicinity of the control device **700** sucks a gas or a liquid from the suction nozzle **111***f* via the suction tube **177**.

[Scope **200**]

[0125] The scope **200** is a unit that observes a lesion or the like and is attached to the distal end portion **111**. The scope **200** may be bendably attached to protrude from the distal end portion **111** to the distal end side **A1**. The scope **200** includes an imaging unit **201** and an illumination unit **202**.

[0126] The imaging unit (camera) **201** includes a stereo lens and an imaging element such as a CMOS and images an imaging subject. An imaging signal is sent to the video control device **600** via the imaging cable **173**.

[0127] The illumination unit **202** is connected to a light guide **174** that guides illumination light and emits illumination light illuminating an imaging subject.

[0128] In the electric endoscope system **1000**, the entire insertion manipulator **100** may be considered to be an "endoscope." In addition, the scope **200**, the imaging cable **173**, and the light guide **174** may be considered to be an "endoscope."

[0129] FIG. **6** is diagram illustrating the bending portion **112**.

[0130] The bending portion **112** includes a plurality of joint rings (also referred to as bending pieces) **115**, distal end portions **116** connected to the distal ends of the plurality of joint rings **115**, and an outer sheath **118**. The plurality of joint rings **115** are connected in the longitudinal direction A in the outer sheath **118**. The joint ring **115** at the distal end is connected to the distal end portion **111**. In the bending portion **112** illustrated in FIG. **3**, the outer sheath **118** is not illustrated.

[0131] FIG. **7** is a diagram illustrating joint rings **115**.

[0132] A joint ring **115** is a short cylindrical member formed of metal. A plurality of joint rings **115** are connected such that internal spaces of the neighboring joint rings **115** form a continuous space.

[0133] Each joint ring **115** includes a first joint ring **115*a*** on the distal end side and a second joint ring **115*b*** on the proximal side. The first joint ring **115*a*** and the second joint ring **115*b*** are connected by a first rotary pin **115*p*** to be rotatable in a vertical direction (also referred to as an "UD direction") perpendicular to the longitudinal direction A.

[0134] In the neighboring joint rings **115**, the second joint ring **115*b*** of the joint ring **115** on the distal end side and the first joint ring **115*a*** of the joint ring **115** on the proximal side are connected by a second rotary pin **115*q*** to be rotatable in a lateral direction (also referred to as an "LR direction") perpendicular to the longitudinal direction A and the UD direction.

[0135] The first joint rings **115*a*** and the second joint rings **115*b*** are alternately connected by the first rotary pin **115*p*** and the second rotary pin **115*q***, and the bending portion **112** is bendable in a desired direction.

[0136] An upper wire-guide **115*u*** and a lower wire-guide **115*d*** are formed on the inner circumferential surface of the second joint ring **115*b***. The upper wire-guide **115*u*** and the lower wire-guide **115*d*** are disposed at both ends in the UD direction with the center axis O1 in the longitudinal direction A interposed therebetween. A left wire-guide **115*l*** and a right wire-guide **115*r*** are formed in the inner circumferential surface of the first joint ring **115*a***. The left wire-guide **115*l*** and the right wire-guide **115*r*** are disposed at both ends in the LR direction with the center axis O1 in the longitudinal direction A interposed therebetween.

[0137] The upper wire-guide **115*u***, the lower wire-guide **115*d***, the left wire-guide **115*l***, and the right wire-guide **115*r*** are formed such that through-holes into which the bending wire **160** is inserted are parallel to the longitudinal direction A.

[0138] The bending wire **160** is a wire for bending the bending portion **112**. The bending wire **160** extends to the attachment portion **150** via the internal path **101**. The bending wire **160** includes an upper bending wire **161*u***, a lower bending wire **161*d***, a left bending wire **161*l*** (see FIG. **3**), and a right bending wire **161*r***.

[0139] The upper bending wire **161*u*** and the lower bending wire **161*d*** are wires for bending the bending portion **112** in the UD direction. The upper bending wire **161*u*** is inserted into the upper wire-guide **115*u***. The lower bending wire **161*d*** is inserted into the lower wire-guide **115*d***.

[0140] The left bending wire **161*l*** and the right bending wire **161*r*** are wires for bending the bending portion **112** in the LR direction. The left bending wire **161*l*** is inserted into the left wire-guide **115*l***. The right bending wire **161*r*** is inserted into the right wire-guide **115*r***.

[0141] The distal end of the bending wire **160** is fixed to the distal end portion **116** of the bending portion **112**. The bending portion **112** is bendable in a desired direction by pulling or relaxing the bending wires **160** (the upper bending wire **161*u***, the lower bending wire **161*d***, the left bending wire **161*l***, and the right bending wire **161*r***).

[0142] The bending wire **160** and the built-in elements **170** (the first channel tube **171**, the second channel tube **172**, the imaging cable **173**, the light guide **174**, the water supply tube **175**, the air supply tube **176**, and the suction tube **177**) are inserted into the internal path **101** formed in the bending portion **112**. In FIG. **6**, built-in elements **170** other than the first channel tube **171** are not illustrated.

[0143] The flexible portion **119** is a tubular member which is longitudinal and flexible. The flexible portion **119** includes an outer sheath **118** which is the outermost part. The distal end of the flexible portion **119** is connected to the bending portion **112**. The bending wire **160** and the built-in elements **170** (the first channel tube **171**, the second channel tube **172**, the imaging cable **173**, the light guide **174**, the water supply tube **175**, the air supply tube **176**, and the suction tube **177**) are inserted into the internal path **101** formed in the flexible portion **119**.

[0144] The attachment portion **150** is provided at the proximal end of the flexible portion **119** as illustrated in FIG. **1**. The attachment portion **150** is attached to the drive device **500** and the video control device **600**.

[First Channel Tube **171**]

[0145] As illustrated in FIG. **5**, the first channel tube **171** is a tube including a first treatment tool lumen **171*r*** with a large diameter. The second channel tube **172** is a tube including a second treatment tool lumen **172*r***. The inner diameter D1 of the first treatment tool lumen **171*r*** is larger than the inner diameter D2 of the second treatment tool lumen **172*r***. Specifically, the inner diameter D1 of the first treatment tool lumen **171*r*** is three to five time the inner diameter D2 of the second treatment tool lumen **172*r***. As illustrated in FIG. **3**, two treatment manipulators **400** can be inserted into the first treatment tool lumen **171*r***.

[0146] The inner diameter D12 of the first treatment tool lumen **171*r*** is equal to or greater than half the outer diameter of the outer sheath **118**. In an existing endoscope (for example, a nasal endoscope) in which the inner diameter of a treatment tool channel is relatively large, the outer diameter is about 6 mm and the inner diameter of the treatment tool channel is about 2.4 mm (about ⅖ times the outer diameter). In the insertion manipulator **100**, the inner diameter D1 of the first treatment tool lumen **171*r*** with respect to the outer diameter of the outer sheath **118** is larger than that in the existing endoscope. Accordingly, a large treatment manipulator **400** can be inserted into the first treatment tool lumen **171***r*, and a range of a manual operation is widened.

[0147] FIG. **8** is a diagram illustrating the first channel tube **171**. In FIG. **8**, the built-in elements **170** other than the first channel tube **171** are not illustrated.

[0148] The first channel tube **171** includes a proximal end channel tube **171**A disposed in the flexible portion **119** and a distal end channel tube **171**B disposed in the bending portion **112**. The proximal end channel tube **171**A and the distal end channel tube **171**B are connected to form the first treatment tool lumen **171***r*.

[0149] FIG. **9** is a sectional view of the proximal end channel tube **171**A.

[0150] The proximal end channel tube **171**A includes a coil sheath **171***a*, a blade tube **171***b*, a first fixed portion **171***c*, a second fixed portion **171***d*, and a regulating wire **171***e*.

[0151] The coil sheath **171***a* is a flexibly bendable coil sheath having excellent kink-resistance. The coil sheath **171***a* forms the first treatment tool lumen **171***r* with a large diameter into which the treatment manipulator **400** is inserted.

[0152] The blade tube **171***b* is a tube in which metal strands, resin strands, or the like are woven in a blade shape and is disposed outside of the coil sheath **171***a*. It is preferable that the blade tube **171***b* be disposed to be in contact with an outer circumferential surface of the coil sheath **171***a*.

[0153] The proximal end channel tube **171**A is a tube with a double structure of the coil sheath **171***a* and the blade tube **171***b*. Accordingly, the proximal end channel tube **171**A also has excellent torque transmissivity based on the blade tube **171***b* while maintaining excellent kink-resistance based on the coil sheath **171***a*. By fitting the coil sheath **171***a* and the blade tube **171***b*, a frictional force between the coil sheath **171***a* and the blade tube **171***b* is further enhanced, and the torque transmissivity is further enhanced.

[0154] The coil sheath **171***a* is fixed to the blade tube **171***b* at a first fixed portion **171***c* on the distal end side A**1** in the axial direction A and a second fixed portion **171***d* on the proximal side A**2** with respect to the first fixed portion **171***c*.

[0155] The regulating wire **171***e* is connected to the first fixed portion **171***c* and the second fixed portion **171***d* and regulates a distance L**1** in the axial direction A between the first fixed portion **171***c* and the second fixed portion **171***d*. It is preferable that a plurality of regulating wires **171***e* be provided. The regulating wire **171***e* is a wire with high rigidity and is, for example, an NiTi wire.

[0156] Since the distance L**1** in the axial direction A between the first fixed portion **171***c* and the second fixed portion **171***d* is regulated by the regulating wire **171***e*, it is possible to prevent a decrease in torque transmissivity or a decrease in pushability due to expansion and contraction of the coil sheath **171***a* and the blade tube **171***b*.

[0157] The regulating wire **171***e* is provided to move alternately inside and outside of the blade tube **171***b*. Accordingly, the regulating wire **171***e* can appropriately regulate the distance L**1** in the axial direction A between the first fixed portion **171***c* and the second fixed portion **171***d* regardless of a bent shape of the insertion section **110**.

[0158] The coil sheath **171***a* may be fixed to the blade tube **171***b* by the first fixed portion **171***c* and the second fixed portion **171***d* in a state in which the coil sheath is compressed in the axial direction A. In this case, the coil sheath **171***a* has a biasing force for separating the first fixed portion **171***c* and the second fixed portion **171***d* in the axial direction A. A tension in the axial direction A acts on the blade tube **171***b*. As a result, it is possible to prevent a decrease in torque transmissivity or a decrease in pushability due to contraction of the coil sheath **171***a* and the blade tube **171***b*. When this effect is not sufficient, the regulating wire **171***e* is not necessarily required.

[0159] As described above, the proximal end channel tube **171**A includes the first treatment tool lumen **171***r* with a large diameter, but has excellent kink-resistance, excellent torque transmissivity, and excellent pushability. Accordingly, even when the outer sheath **118** which is the outermost part of the flexible portion **119** is thin, the flexible portion **119** can maintain excellent kink-resistance, excellent torque transmissivity, and excellent pushability.

[0160] The distal end channel tube **171**B may have the same configuration as the proximal end channel tube **171**A. However, the bending portion **112** in which the distal end channel tube **171**B is provided includes the joint rings **115** with high rigidity and thus has excellent torque transmissivity and pushability. Accordingly, the distal end channel tube **171**B has only to include the coil sheath **171***a* forming the first treatment tool lumen **171***r* and may not include the blade tube **171***b* and the regulating wire **171***e*.

[0161] Since the outer sheath **118** can be formed of a thin film as described above, it is possible to achieve a decrease in diameter of the insertion manipulator **100**. However, when the first channel tube **171** does not have sufficient kink-resistance, torque transmissivity, or pushability, the outer sheath **118** may be a multi-layered tube including an outer coil sheath into which the first channel tube **171** is inserted and an outer blade tube disposed outside of the outer coil sheath.

[Drive Device **500**]

[0162] FIG. **10** is a functional block diagram illustrating the drive device **500**.

[0163] The drive device **500** includes an endoscope adapter **510**, an operation-receiving unit **520**, an air supply/suction driving unit **530**, a drive unit **550**, and a drive controller **560**.

[0164] The endoscope adapter **510** is an adapter to which the insertion manipulator **100** is detachably connected. The endoscope adapter **510** connects the bending wire **160**, the suction tube **177**, the water supply tube **175**, and the air supply tube **176** to the drive device **500**.

[0165] The operation-receiving unit **520** receives an operation input from the operation device **800** via the operation cable **801**. When the operation device **800** and the drive device **500** communicate in a wireless manner instead of a wired manner, the operation-receiving unit **520** includes a known wireless receiver module.

[0166] The air supply/suction driving unit **530** is connected to the suction tube **177**, the water supply tube **175**, and the air supply tube **176** via the endoscope adapter **510**. The air supply/suction driving unit **530** includes a pump and supplies a liquid to the water supply tube **175**. The air supply/suction driving unit **530** supplies air to the air supply tube **176**. The air supply/suction driving unit **530** sucks air from the suction tube **177**.

[0167] The drive unit (actuator) **550** is connected to the bending wire **160** of the insertion manipulator **100** via the endoscope adapter **510**. The drive unit **550** includes a drive unit and an encoder which are not illustrated. The drive unit pulls or relaxes the bending wire **160** using a pulley or the like. The encoder detects an amount of pulling of the bending wire **160**. The detection result of the encoder is acquired by the drive controller **560** of the drive device **500**.

[0168] The drive unit (actuator) **550** may be connected to the treatment manipulator **400** to drive the treatment manipulator **400**.

[0169] The drive controller **560** controls the drive device **500** as a whole. The drive controller **560** acquires the operation input received by the operation-receiving unit **520**. The drive controller **560** controls the air supply/suction driving unit **530** and the drive unit **550** on the basis of the acquired operation input.

[0170] The drive controller **560** is a program-executable computer including a processor **561**, a memory **562**, a storage unit **563** storing a program and data, and an input/output control unit **564**. The functions of the drive controller **560** are realized by causing the processor **561** to execute a program. At least some functions of the drive controller **560** may be realized by a dedicated logical circuit.

[0171] The drive controller **560** may further include an element in addition to the processor **561**, the memory **562**, the storage unit **563**, and the input/output control unit **564**. For example, the drive controller **560** may further include an image computing unit that performs some or all of image processing and image recognition processing. When the image-computing unit is additionally provided, the drive controller **560** can perform specific image processing or image recognition processing at a high speed. The image-computing unit may be mounted in a separate hardware device connected thereto via a communication line.

[Video Control Device **600**]

[0172] FIG. **11** is a functional block diagram illustrating the video control device **600**.

[0173] The video control device **600** includes an endoscope adapter **610**, an imaging processing unit **620**, a light source unit **630**, and a main controller **660**.

[0174] The endoscope adapter **610** is an adapter to which the insertion manipulator **100** is detachably connected. The endoscope adapter **610** connects the imaging cable **173** and the light guide **174** to the video control device **600**.

[0175] The imaging processing unit **620** is connected to the imaging cable **173**. The imaging processing unit **620** converts an imaging signal acquired from the imaging unit **201** of the scope **200** via the imaging cable **173** and the imaging cable **173** to a captured image.

[0176] The light source unit **630** is connected to the light guide **174**. The light source unit **630** generates illumination light applied to an imaging subject. The illumination light generated by the light source unit **630** is guided to an illumination unit **202** of the scope **200** via the light guide **174**.

[0177] The main controller **600** is a program-executable computer including a processor **661**, a memory **662**, a storage unit **663** that can store a program and data, and an input/output control unit **664**. The functions of the main controller **660** are realized by causing the processor **661** to execute a program. As least some functions of the main controller **660** may be realized by a dedicated logical circuit.

[0178] The main controller **660** can perform image processing on a captured image acquired by the imaging processing unit **620**. The main controller **660** can generate a GUI image or a CG image to provide information to an operator S. The main controller **660** can display the captured image, the GUI image, or the CG image on the display device **900**.

[0179] The main controller **660** is not limited to an integrated hardware device. For example, the main controller **660** may be constituted by separating a part thereof as a separate hardware device and connecting the separated hardware device thereto via a communication line. For example, the main controller **660** may be a cloud system that connects the separated storage units **663** via a communication line.

[0180] The main controller **660** may further include an element in addition to the processor **661**, the memory **662**, the storage unit **663**, and the input/output control unit **664**. For example, the main controller **660** may further include an image-computing unit that performs some or all of image processing or image recognition processing performed by the processor **661**. Since the image-computing unit is further provided, the main controller **660** can perform specific image processing or image recognition processing at a high speed. The image-computing unit may be mounted in a separate hardware device connected thereto via a communication line.

[Operation of Electric Endoscope System **1000**]

[0181] Operations of the electric endoscope system **1000** according to the present embodiment will be described below. Specifically, manual operations of observing and treating a lesion formed in a wall of a large intestine using the electric endoscope system **1000** will be described.

[0182] An operator S inserts the insertion section **110** of the insertion manipulator **100** into the large intestine via an anal passage of a patient from its distal end. The operator S operates the operation device **800** while observing the captured image displayed on the display device **900** such that the insertion section **110** is moved to cause the distal end portion **111** to approach a lesion. The operator S operates the operation device **800** to bend the bending portion **112** according to necessity.

[0183] Since the insertion manipulator **100** has excellent kink-resistance, excellent torque transmissivity, and excellent pushability, the operator S can easily operate the insertion manipulator **100**.

[0184] The operator S inserts the treatment manipulator **400** into the first channel tube **171** or the second channel tube **172**. The operator S operates the operation device **800** while observing the captured image displayed on the display device **900** such that the treatment manipulator **400** is operated to treat the lesion.

[0185] Since the insertion manipulator **100** includes the first treatment tool lumen **171***r* with a large diameter, the operator S can appropriately perform treatment using the large treatment manipulator **400**.

[0186] After treating the lesion, the operator S removes the insertion manipulator **100** and the treatment manipulator **400** and ends the manual operation.

[0187] With the electric endoscope system **1000** according to the present embodiment, it is possible to more efficiently perform observation or treatment. Since the insertion manipulator **100** has excellent kink-resistance, excellent torque transmissivity, and excellent pushability, the operator S can easily operate the insertion manipulator **100**. Since the insertion manipulator **100** includes the first treatment tool lumen **171***r* with a large diameter, the operator S can appropriately perform treatment using the large treatment manipulator **400**.

[0188] While the first embodiment of the present invention has been described above in detail with reference to the drawings, a specific configuration is not limited to this embodiment and includes changes in design without departing from the gist of the present invention. Elements described in the aforementioned embodiment and modified examples can be appropriately combined into a configuration.

Modified Example 1-1

[0189] FIG. **12** is a diagram illustrating a bending portion **112**A which is a modified example of the bending portion **112**. In FIG. **12**, the built-in elements **170** other than the first channel tube **171** are not illustrated.

[0190] The bending portion **112** does not include a plurality of joint rings **115** but includes a plurality of ring members **115**A. The plurality of ring members **115**A are arranged in the axial direction A. The built-in elements **170** including the first channel tube **171** are inserted into the plurality of ring members **115**A. The bending wire **160** is inserted into a wire-guide **115***g* formed in the ring members **115**A.

[0191] FIG. **13** is a diagram illustrating the first channel tube **171** inserted into the bending portion **112**A.

[0192] The bending portion **112**A has lower torque transmissivity or lower pushability in comparison with the bending portion **112** in which the joint rings **115** are connected. Accordingly, a distal end portion of the first channel tube **171** into which the bending portion **112**A is inserted is preferably a tube with a double structure of the coil sheath **171***a* and the blade tube **171***b*. The distal end portion of the first channel tube **171** into which the bending portion **112**A is inserted preferably includes a regulating wire **171***e* for regulating a distance between the ends.

[0193] FIG. **14** is a diagram illustrating an insertion section **110**A which is a modified example of the insertion section **110**.

[0194] The insertion section **110**A includes an expanding and contracting portion **117** in addition to a distal end portion **111**, a bending portion **112**, and a flexible portion **119**. The expanding and contracting portion **117** is a member that connects the distal end portion **116** of the bending portion **112** to the distal end portion **111** and is driven by the drive unit (actuator) **550** to expand and contract in the longitudinal direction A. An outer circumferential portion of the expanding and contracting portion **117** is formed in a bellows shape. The operator S can accurately control a position of the distal end portion **111** in the longitudinal direction A with expansion and contraction of the expanding and contracting portion **117**.

[0195] FIG. **15** is a diagram illustrating a spiral tube **180**.

[0196] The operator may insert the insertion section **110** into a spiral tube **180** which is separate from the insertion section **110**. The spiral tube **180** is a tube in which a fin **181** is wound in a spiral shape on an outer circumference thereof. The spiral tube **180** is driven by the drive unit (actuator) **550** to be rotatable about a rotation axis extending in the longitudinal direction A. The operator can cause the spiral tube **180** and the insertion section **110** to move forward and rearward in the lumen by rotating the spiral tube **180**. For example, the operator can easily insert the distal end portion **111** of the insertion section **110** to a deep region of a large intestine.

[0197] FIG. **16** is a diagram illustrating a distal end portion **111**A which is a modified example of the distal end portion **111**.

[0198] The distal end portion **111**A includes a cutout portion **111***g* in which a part of a cylindrical body is cut out. The cutout portion **111***g* extends in the longitudinal direction A. A first opening **111***a* of the distal end portion **111**A is provided on the proximal side A**2** of the cutout portion **111***g*. The first opening **111***a* is open to the distal end side A**1**. The scope **200** is attached to the distal end of the distal end portion **111**A. The distal end portion **111**A includes a treatment camera **111***c* in addition to the scope **200** which is an insertion camera. The treatment camera **111***c* is provided on the distal end side A**1** of the cutout portion **111***g*. The first opening **111***a* and the treatment camera **111***c* are provided to face each other. The treatment camera **111***c* can image a situation in which the treatment manipulator **400** protruding from the first opening **111***a* treats a lesion. Since the direction which the treatment camera **111***c* faces and the direction in which the treatment manipulator **400** protrudes are opposite, the operator can perform treatment while viewing the distal end of the treatment manipulator **400**.

Second Embodiment

[0199] A hardness-changing device **300** according to a second embodiment of the present disclosure will be described below with reference to FIGS. **17** to **25**. In the following description, the same constituents as in the above description will be referred to by the same reference signs, and repeated description thereof will be omitted.

[0200] FIG. **17** is a diagram of a hardness-changing device **300** inserted into the bending portion **112**.

[0201] The hardness-changing device (rigidizer, introducer) **300** is a device including a hardness-changing portion **310** into which the first treatment tool lumen **171***r* can be inserted. In FIG. **17**, the hardness-changing device **300** is inserted into the first treatment tool lumen **171***r*. In FIG. **17**, the first channel tube **171** and the second channel tube **172** are not illustrated.

[0202] FIG. **18** is a diagram illustrating the entire configuration of the hardness-changing device **300**.

[0203] The hardness-changing device **300** includes a hardness-changing portion **310**, an insertion section **320**, and a drive unit **340**.

[0204] FIGS. **19** and **20** are diagrams illustrating the hardness-changing portion **310**.

[0205] The hardness-changing portion **310** is a long member which can be inserted into the first treatment tool lumen **171***r* and in which hardness changes by performing a predetermined operation thereon. The hardness-changing portion **310** includes a plurality of spinal portions **311** and a wire **312**. Each spinal portion **311** is formed in a bowl shape. The plurality of spinal portions **311** are connected in a longitudinal axis direction while overlapping each other. The wire **312** is inserted into the plurality of spinal portions **311** and is fixed to a spinal portion at the distal end. When the wire **312** is pulled to the proximal side as illustrated in FIG. **20**, neighboring spinal portions **311** come into close contact, and thus a frictional force is enhanced and the shape of the hardness-changing portion **310** is fixed. By loosening the wire **312**, the shape of the hardness-changing portion **310** changes.

[0206] The insertion section **320** is a long member that can be inserted into the first treatment tool lumen **171***r* and has flexibility. The insertion section **320** is provided on the proximal side of the hardness-changing portion **310**.

[0207] The drive unit **340** is provided on the proximal side of the insertion section **320**. The drive unit **340** causes the insertion section **320** to move forward and rearward in the longitudinal axis direction and causes the insertion section **320** to rotate about the longitudinal axis. The drive unit **340** may be incorporated into the drive device **500**.

[0208] FIGS. **21** to **25** are diagrams illustrating operations of the hardness-changing portion **310**.

[0209] As illustrated in FIG. **21**, the operator S inserts the distal end of the insertion section **110** of the insertion manipulator **100** into a large intestine of a patient via an anal passage. As illustrated in FIG. **22**, the operator S disposes the bending portion **112** in a part of the large intestine which is greatly bent. As illustrated in FIG. **23**, the operator S causes the hardness-changing portion **310** with a variable shape to move forward and locates the hardness-changing portion **310** in the bending portion **112**. The operator S fixes the shape of the hardness-changing portion **310** inserted into the bending portion **112**. As illustrated in FIG. **24**, the operator S causes the insertion section **110** to move forward. The bending portion **112** moves forward along the hardness-changing portion **310** with a fixed shape. The insertion manipulator **100** can smoothly pass through the part of the large intestine which is greatly bent. When the hardness-changing portion **310** is not necessary, the operator S opens the wire of the hardness-changing portion **310** and removes the hardness-changing portion **310** from the bending portion **112** in a state in which it is softened as illustrated in FIG. **25**.

[0210] With the hardness-changing device **300** according to the present embodiment, it is possible to more efficiently perform observation or treatment. The operator S can smoothly insert the insertion manipulator **100** to a target position using the hardness-changing device **300**.

[0211] While the second embodiment of the present invention has been described above in detail with reference to the drawings, a specific configuration is not limited to this embodiment and includes changes in design without departing from the gist of the present invention. Elements described in the aforementioned embodiment and modified examples can be appropriately combined into a configuration.

Modified Example 2-1

[0212] FIGS. **26** and **27** are diagrams illustrating a hardness-changing portion **310**A which is a modified example of the hardness-changing portion **310**. The hardness-changing portion **310**A includes a tube **313** and a plurality of wire members **314** which are inserted into the tube **313**. Each wire member **314** is a metal strand or a resin strand. As illustrated in FIG. **27**, by making the internal space of the tube **313** have a negative pressure through suction and brining the plurality of wire members **314** into contact with each other, relative movement of the wire members **314** is curbed due to a frictional force between the plurality of wire members **314** and between the wire member **314** and the tube. Accordingly, the shape of the hardness-changing portion **310**A is fixed. Since the frictional force is increased by roughening the surfaces of the wire members **314** to increase contact resistance, it is possible to further enhance hardness of the hardness-changing portion **310**A with a fixed shape.

Modified Example 2-2

[0213] FIGS. **28** and **29** are diagrams illustrating a hardness-changing portion **310**B which is a modified example of the hardness-changing portion **310**. The hardness-changing portion **310**B includes a plurality of spinal portions **311** and a tube **313**. The tube **313** is inserted into the plurality of spinal portions **311**. As illustrated in FIG. **29**, by making the internal space of the tube **313** have a positive pressure through suction such that the tube **313** expands, a frictional force between the neighboring spinal portions **311** is increased, and the shape of the hardness-changing portion **310**B is fixed. Since the hardness-changing portion **310**B makes the internal space of the tube **313** have a positive pressure, it is possible to easily increase the frictional force between the plurality of spinal portions **311** and to more strength the hardness-changing portion **310**B in comparison with a mode in which the internal space of the tube **313** is made to have a negative pressure. In the mode in which the internal space of the tube **313** is made to have a negative pressure, the pressure has only to be reduced by only the atmospheric pressure. However, when the internal space of the tube **313** is made to have a positive pressure, such limitation is removed, and thus it is possible to apply a larger pressure.

Modified Example 2-3

[0214] FIGS. **30** and **31** are diagrams illustrating a hardness-changing portion **310**C which is a modified example of the hardness-changing portion **310**. The hardness-changing portion **310**C includes an outer tube **315**, an inner tube **316**, and a plurality of cables **317**. The inner tube **316** is inserted into the inner space of the outer tube **315**. The plurality of cables **317** are inserted into a space V interposed between the outer tube **315** and the inner tube **316**. As illustrated in FIG. **31**, by making the space V have a negative pressure through suction, the plurality of cables **317** come into contact with each other, and the shape of the hardness-changing portion **310**C is fixed. The space V may be made to have a negative pressure using a hydraulic pressure.

Modified Example 2-4

[0215] FIG. **32** is a diagram illustrating a hardness-changing portion **310**D which is a modified example of the hardness-changing portion **310**. The hardness-changing portion **310**D includes a plurality of spinal portions **311** and a shape memory alloy wire **318**. In the hardness-changing portion **310**D, when the shape memory alloy wire **318** contracts with supply of electric power, the spinal portions **311** come into close contact with each other similarly to the hardness-changing portion **310**, and thus the hardness-changing portion **310**D can switch between a state in which the shape is fixed and a state in which the shape is variable.

Third Embodiment

[0216] A treatment manipulator **400** according to a third embodiment of the present disclosure will be described below with reference to FIGS. **33** to **40**. In the following description, the same constituents as in the above description will be referred to by the same reference signs, and repeated description thereof will be omitted.

[0217] FIG. **33** is a diagram illustrating a high-frequency knife **430**.

[0218] The treatment manipulator **400** includes a bendable treatment tool arm **410** and treatment tools (a forceps **420**, a high-frequency knife **430**, a local injection needle **440**, and a basket **450**) which are inserted into the treatment tool arm **410**.

[0219] The treatment tool arm **410** is a hollow long member. Similar to the bending portion **112** of the insertion manipulator **100**, the treatment tool arm **410** includes a plurality of joint rings (also referred to as bending pieces) **415** and is driven with a wire or the like to be bendable in the vertical direction and the lateral direction.

[0220] The treatment tools (such as the forceps **420**, the high-frequency knife **430**, the local injection needle **440**, and the basket **450**) can be inserted into the internal space (a lumen or a channel) of the treatment tool arm **410**. The treatment tools are manually bendable, but may not have an active bending function.

[0221] FIG. **34** is a diagram illustrating the high-frequency knife **430** for putting a marking M.

[0222] An operator S inserts the high-frequency knife **430** into the treatment tool arm **410** and causes the distal end of the high-frequency knife **430** to protrude from the distal end of the treatment tool arm **410**. The operator S locates the distal end of the high-frequency knife **430** at a desired position by bending the treatment tool arm **410**. The operator S cauterize biological tissue with the distal end portion of the high-frequency knife **430** and puts a marking M on the biological tissue.

[0223] Since the first treatment tool lumen **171***r* has a large diameter, two treatment tool arms **410** can be inserted thereinto. For example, as illustrated in FIG. **34**, the operator S can insert the high-frequency knife **430** into one treatment tool arm **410**, insert the forceps **420** into the other treatment tool arm **410**, and treat a target part using two treatment tools.

[0224] FIG. **35** is a diagram illustrating a local injection needle **440**.

[0225] The operator S removes the high-frequency knife **430** from the treatment tool arm **410**. Then, the operator S inserts the local injection needle **440** into the treatment tool arm **410** and causes the distal end of the local injection needle **440** to protrude from the distal end of the treatment tool arm **410**.

[0226] FIG. **36** is a diagram illustrating the local injection needle **440** which locally injects a liquid.

[0227] The operator S bends the treatment tool arm **410** and locates the distal end of the local injection needle **440** at a desired position. The operator S pierces the biological tissue with the distal end of the local injection needle **440** and locally injects a liquid.

[0228] FIGS. **37** and **38** are diagrams illustrating a high-frequency knife **430** for cutting.

[0229] The operator S removes the local injection needle **440** from the treatment tool arm **410**. Then, the operator S inserts the high-frequency knife **430** into the treatment tool arm **410**. The operator S pulls biological tissue using the forceps **420**. The operator S cuts a target part of the biological tissue with the high-frequency knife **430** while checking the marking M.

[0230] FIG. **39** is a diagram illustrating a basket **450**.

[0231] The operator S removes the two treatment tool arms **410** along with the treatment tools. Then, the operator S inserts a large-diameter treatment tool arm **410**A into the first treatment tool lumen **171***r*. The operator S inserts the basket **450** into the large-diameter treatment tool arm **410**A. The basket **450** has a larger outer diameter than that of the high-frequency knife **430** or the local injection needle **440**. The large-diameter treatment tool arm **410**A has a larger inner diameter of a hollow portion than that of the treatment tool arm **410** and enables the basket **450** to be inserted thereinto. Only a large basket **450** may be directly inserted into the first treatment tool lumen **171***r* without using the large-diameter treatment tool arm **410**A.

[0232] FIG. **40** is a diagram illustrating the basket **450** for recovering a target part T.

[0233] The operator S recovers a target part T which has been cut and removed using the basket **450**. The operator S can efficiently recover the target part T using the large basket **450**. Since the first treatment tool lumen **171***r* has a large diameter, the large basket **450** larger than a basket used in the related art can be inserted thereinto. Since the first treatment tool lumen **171***r* has a large diameter, a larger amount of target part T or tissue than that in the related art can be easily recovered at a time without being caught by the first treatment tool lumen **171***r*. Since the first treatment tool lumen **171***r* has a large diameter, larger treatment tools than those in the related art in addition to the basket can be inserted thereinto.

[0234] With the treatment manipulator **400** according to the present embodiment, it is possible to more efficiently perform observation or treatment. Since the insertion manipulator **100** has the first treatment tool lumen **171***r* with a large diameter, two treatment manipulators **400** can be simultaneously inserted and used. Since the treatment manipulator **400** with a large diameter can be inserted into the first treatment tool lumen **171***r*, a multi-functional and high-performance large treatment manipulator **400** can be used.

[0235] While the third embodiment of the present invention has been described above in detail with reference to the drawings, a specific configuration is not limited to this embodiment and includes changes in design without departing from the gist of the present invention. Elements described in the aforementioned embodiment and modified examples can be appropriately combined into a configuration.

Fourth Embodiment

[0236] A treatment manipulator **400**B according to a fourth embodiment of the present disclosure will be described below with reference to FIGS. **41** to **42**. In the following description, the same constituents as in the above description will be referred to by the same reference signs, and repeated description thereof will be omitted.

[0237] FIG. **41** is a diagram illustrating a treatment manipulator **400**B.

[0238] Similar to the treatment manipulator **400** according to the aforementioned embodiment, the treatment manipulator **400**B is a device inserted into the first channel tube **171** or the like of the insertion manipulator **100** and is inserted into a lumen of a patient to treat a lesion. The treatment manipulator **400**B is driven by the drive unit (actuator) **550** or the like. The treatment manipulator **400**B is different from the treatment manipulator **400** in that two treatment tool arms **410** are arranged on a distal end surface of one manipulator flexible portion **417**. The treatment manipulator **400**B can be more easily inserted in comparison with the treatment manipulator **400**.

[0239] The treatment manipulator **400**B includes a treatment tool (such as a forceps **420**, a high-frequency knife **430**, a local injection needle **440**, or a basket **450**), a treatment tool arm **410**, an artificial muscle **470**, and a wire **480**.

[0240] The treatment tool arm **410** is a hollow long member. The treatment tool arm **410** includes a first bending portion **411** provided on the distal end side A**1** and a second bending portion **412** provided on the proximal side A**2**. In the usage state of the treatment manipulator **400**B illustrated in FIG. **41**, the first bending portion **411** is located on the distal end side A**1** with respect to the distal end surface of the distal end portion **111**. A part of the second bending portion **412** protrudes from the distal end portion **111** to the distal end side A**1**.

[0241] The first bending portion **411** includes a plurality of joint rings (also referred to as wrist joints) **415** similarly to the bending portion **112** of the insertion manipulator **100** and can be driven by the artificial muscle **470** such that it is bendable in the vertical direction and the lateral direction.

[0242] As illustrated in FIG. **41**, the second bending portion **412** includes a shoulder joint **416** that greatly bends the treatment tool arm **410** in the lateral direction (LR direction). The shoulder joint **416** bends the second bending portion **412** in an S-shape when seen in the vertical direction (UD direction). As illustrated in FIG. **41**, the distal ends of the second bending portions **412** of two treatment tool arms **410** are bent and separated from each other in the lateral direction. The distance between the center axes at the distal ends of the second bending portions **412** of the two treatment tool arms **410** is defined as a "distance (shoulder width) L."

[0243] When a target part is treated using two treatment tool arms **410**, the operator S can secure sufficient space for treatment and easily treat the target part by appropriately separating the positions of the proximal ends of the two first bending portions **411**. Therefore, the operator S greatly bends the second bending portions **412** such that the distance L is appropriately increased. At least a part of the two treatment manipulators **400** can be inserted into notch portions **111***n* and spread outward in the lateral direction. Accordingly, the distance L may be larger than the outer diameter of the distal end portion **111** of the insertion manipulator **100**.

[0244] The artificial muscle **470** constitutes at least a part of a mechanism for bending a treatment tool. The artificial muscle **470** is, for example, a Mckibben artificial muscle. The artificial muscle **470** is attached to the first bending portion **411** and bends the first bending portion **411**. In order to bend the first bending portion **411** in the vertical direction and the lateral direction, a plurality of artificial muscles **470** are attached to the first bending portion **411**.

[0245] In the usage state of the treatment manipulator **400**B illustrated in FIG. **41**, the first bending portion **411** is disposed on the distal end side A**1** with respect to the distal end surface of the distal end portion **111**. Accordingly, the artificial muscle **470** is also disposed on the distal end side A**1** with respect to the distal end surface of the distal end portion **111**.

[0246] FIG. **42** is a sectional view of the treatment manipulator **400**B.

[0247] A tube **471** for supplying a fluid for operating the artificial muscle **470** is attached to the artificial muscle **470**. The artificial muscle **470** is contracted with the fluid supplied from the tube **471**.

[0248] The wire **480** is attached to the second bending portion **412**. By pulling the second bending portion **412** from the proximal side A**2**, the second bending portion **412** is bent.

[0249] The first bending portion **411** is bendably driven by the artificial muscle **470**. The artificial muscle **470** can more precisely bend the first bending portion **411** in comparison with the wire **480**. Accordingly, the artificial muscle **470** is suitable for bendable driving of the first bending portion **411** in which precise operations for treatment are necessary. The artificial muscle **470** is attached to the first bending portion **411** and directly bends the first bending portion **411**. Accordingly, it is possible to accurately bend the first bending portion **411** without being affected by the shape on the proximal side of the first bending portion **411**.

[0250] The second bending portion **412** is bendably driven by the wire **480**. The second bending portion **412** does not require a very accurate operation as long as it can greatly bend the shoulder joint **416**. Accordingly, it is possible to bend the second bending portion **412** even using the wire **480**. Through driving using the wire **480**, it is possible to easily secure a longer stroke in comparison with the artificial muscle **470**. Accordingly, the wire **480** is suitable for bendable driving of the second bending portion **412** in which a substantial bending operation is necessary.

[0251] With the treatment manipulator **400**B according to the present embodiment, it is possible to more efficiently perform observation or treatment. Since the first bending portion **411** bendably driven by the artificial muscle **470** can precisely bend the treatment tool, the operator S can more efficiently perform observation or treatment.

[0252] While the fourth embodiment of the present invention has been described above in detail with reference to the drawings, a specific configuration is not limited to this embodiment and includes changes in design without departing from the gist of the present invention. Elements described in the aforementioned embodiment and modified examples can be appropriately combined into a configuration.

Modified Example 4-1

[0253] FIG. **43** is a diagram illustrating a treatment manipulator **400**C which is a modified example of the treatment manipulator **400**B. The treatment manipulator **400**C includes a treatment tool and an artificial muscle **470** and does not include a treatment tool arm **410**. The artificial muscle **470** is attached to the treatment tool using a ring member **472**. A member which is bent with a compression force, for example, a tube or a coil, can be employed instead of the plurality of joint rings **415** illustrated in FIG. **41**, and the artificial muscle **470** can be mounted around the member. Instead of a configuration in which the artificial muscle **470** is disposed outside of the joint rings **415**, a member which is bent with a compression force, for example, a tube or a coil, may be disposed outside of the artificial muscle **470**.

Modified Example 4-2

[0254] FIG. **44** is a diagram illustrating an artificial muscle **470** disposed at a different position.

[0255] The artificial muscle **470** may be provided on the proximal side A**2** with respect to the second bending portion **412**. The first bending portion **411** and the artificial muscle **470** are connected by a distal end wire **473**. The artificial muscle **470** bends the first bending portion **411** by causing the distal end wire **473** to move forward and rearward. The artificial muscle **470** illustrated in FIG. **44** is disposed in the distal end portion **111** on the distal end side A**1** with respect to the bending portion **112**. The artificial muscle **470** is disposed in the distal end portion **111** which is not bent. Accordingly, even when the bending portion **120** is bent, the artificial muscle **470** is not bent.

Modified Example 4-3

[0256] FIG. **45** is a diagram illustrating an artificial muscle **470** disposed at a different position.

[0257] The artificial muscle **470** illustrated in FIG. **45** is disposed in the bending portion **112**. The artificial muscle **470** is disposed in a non-bending area E which is not bent and which is interposed between a first rotary pin **115***p* and a second rotary pin **115***q*. Accordingly, even when the bending portion **120** is bent, the artificial muscle **470** is not bent.

Modified Example 4-4

[0258] FIG. **46** is a diagram illustrating an artificial muscle **470** which is disposed at a different position.

[0259] The artificial muscle **470** illustrated in FIG. **46** is disposed in a flexible portion **119** on the proximal side A**2** with respect to the bending portion **112**. The artificial muscle **470** is disposed in the flexible portion **119** which is not greatly bent. Accordingly, even when the bending portion **120** is bent, the artificial muscle **470** is not bent.

Modified Example 4-5

[0260] FIG. **47** is a diagram illustrating an artificial muscle **470** which is disposed at a different position.

[0261] The artificial muscle **470** illustrated in FIG. **47** is disposed in the flexible portion **119** on the proximal side A**2** with respect to the bending portion **112** and is located at a position separated by a distance L**3** from the proximal end with respect to the bending portion **112**. The distance L**3** is longer than a moving distance of the treatment manipulator **400**B at the time of treatment. Accordingly, even when the treatment manipulator **400**B is intended to move forward and rearward at the time of treatment, the artificial muscle **470** is not inserted into the bending portion **112**. Accordingly, even when the bending portion **120** is bent, the artificial muscle **470** is not bent.

Modified Example 4-6

[0262] FIG. **48** is a diagram illustrating an artificial muscle **470** which is disposed at a different position.

[0263] As illustrated in FIG. **48**, a plurality of artificial muscles **470** may be arranged in the axial direction A. It is preferable that some artificial muscles **470** be disposed in a non-bending area. It is not necessary to arrange the plurality of artificial muscles **470** in the radial direction and it is possible to decrease the diameter of the treatment manipulator **400**B.

Modified Example 4-7

[0264] FIG. **49** is a diagram illustrating a treatment manipulator **400**D which is a modified example of the treatment manipulator **400**B. The treatment manipulator **400**D does not include a wire **480** and bends all bending portions using the artificial muscle **470**. The operator S can precisely bend all the bending portions.

Fifth Embodiment

[0265] An electric endoscope system **1000**F according to a fifth embodiment of the present disclosure will be described below with reference to FIGS. **50** to **60**. In the following description, the same constituents as in the above description will be referred to by the same reference signs, and repeated description thereof will be omitted.

[0266] FIG. **50** is an overall view of the electric endoscope system **1000**F according to this embodiment.

[0267] The electric endoscope system **1000**F is an example of a medical manipulator system. The medical manipulator includes an insertion manipulator **100**F inserted into the body, an electric endoscope, a catheter, a treatment tool, an endoluminal device, and the like.

[Electric Endoscope System **1000**F]

[0268] The electric endoscope system **1000**F is a medical system for observing and treating the inside of the body of a patient P lying on an operating table OT. The electric endoscope system **1000**F includes an insertion manipulator **100**F, a treatment manipulator **400**F (see FIG. **58**), a drive device **500**F, an image control device **600**, an operation device **800**, and a display device **900**. The drive device **500**F and the image control device **600** constitute a control device **700**F that controls the electric endoscope system **1000**F.

[Insertion Manipulator **100**F]

[0269] FIG. **51** is a perspective view of the distal end of the insertion manipulator **100**F. FIG. **52** is a cross-sectional view of the insertion section **110**F. The insertion manipulator **100**F includes an insertion section **110**F, an attachment portion **150**, a bending wire **160**F, a built-in element **170**F, and a scope **200**.

[0270] Inside the insertion manipulator **100**F, an internal passage (lumen) **101** is formed, which extends from the distal end of the insertion section **110**F to the proximal end of the attachment portion **150** along the longitudinal direction (longitudinal axis direction, axial direction) A of the insertion manipulator **100**F. The bending wire **160**F and the built-in element **170**F are inserted into the internal passage **101**.

[0271] The insertion section **110**F is a long, thin member that can be inserted into a lumen. The insertion section **110**F includes a distal end portion **111**, a bending portion **112**F, and a flexible portion **119**. The distal end portion **111**, the bending portion **112**F, and the flexible portion **119** are connected in order from the distal end side A**1** toward the proximal side A**2**. The insertion section **110**F includes an outer sheath **118**F, which is the outermost part.

[0272] Similar to the first embodiment, the distal end portion **111** is provided at the distal end of the insertion section **110**F. The scope **200** is attached so as to be able to protrude from the distal end portion **111** to the distal end side A**1** and bend. The scope **200** is operated by a scope operation wire **178**. The notch portion **111***n* of the distal end portion **111** illustrated in FIG. **51** is formed on the inner circumferential surface of the first opening **111***a* and does not penetrate in a direction perpendicular to the longitudinal direction A.

[0273] As shown in FIG. **52**, the built-in element **170**F passes through the internal path **101**. The built-in element **170** has a first channel tube **171**F, a second channel tube **172**, an imaging cable **173**, a light guide **174**, a water supply tube **175**, an air supply tube **176**, a suction tube **177**, and a scope operation wire **178**.

[0274] FIG. **53** is a diagram showing the insertion section **110**F.

[0275] The insertion section **110**F includes a spring **113**. The spring **113** is formed by winding a linear metal member in a spiral shape. The spring **113** is, for example, a flat wire spring or a round wire spring. By using a flat wire spring instead of a round wire spring formed from a round wire metal member, the diameter of the insertion section **110**F is made thinner. On the other hand, by using a round wire spring, it is expected that the insertion section **110**F will be easier to bend. In addition, the bending wire **160**F and the built-in element **170**F are omitted in FIG. **53**.

[0276] The spring **113** is arranged from the distal end to the proximal end of the insertion section **110**F. Therefore, the insertion section **110**F has good torque transmission.

[0277] The spring **113** is arranged in the internal path **101** of the insertion section **110**F in a preloaded state with a compressive force of, for example, 20 N to 25 N. Since the spring **113** is preloaded, it has good kink resistance and is not easily buckled.

[0278] The outer sheath **118**F is a cylindrical member arranged on the outside of the spring **113**. The outer sheath **118**F includes a braided tube **118***b* and a coating **118***c* arranged on both the inner and outer sides of the braided tube **118***b*. The coating **118***c* arranged on the inner side is also called the inner coating. The coating **118***c* arranged on the outer periphery side is also called the outer coating. The coating **118***c* may be either the inner coating or the outer coating.

[0279] The braided tube **118***b* is a tube in which metal wires, resin wires, etc. are woven into a braid shape. The braided tube **118***b* is made of, for example, aramid or UHMW-PE (Ultra High Molecular Weight Polyethylene).

[0280] The coating **118***c* is sandwiched between the inner and outer periphery sides of the braided tube **118***b* to seal (encapsulate) the braided tube **118***b*. The coating **118***c* is attached to the braided tube **118***b* by coating, co-extruded, over-casting, or over-molding. The coating **118***c* is made of, for example, a low-friction, low-durometer resin.

[0281] In the outer sheath **118**F, the braided tube **118***b* is sealed (encapsulated) by the coating **118***c*. Therefore, the braided tube **118***b* has good rigidity without increasing the diameter. Because the coating **118***c* has low friction and a low durometer, the outer sheath **118**F has suitable elasticity and can balance durability and ease of bending.

[0282] The outer surface of the outer sheath **118**F is provided with a marking **118***m* that can be observed under X-ray fluoroscopy. By checking the marking **118***m*, the surgeon S can easily grasp the insertion position and insertion distance of the insertion section **110**F.

[0283] FIG. **54** is a diagram showing the bending portion **112**F.

[0284] The bending portion **112**F includes a spring **113**, multiple ring members **115**F, and an outer sheath **118**F, which is the outermost part. The bending wire **160**F and the built-in element **170**F are inserted through the internal passage **101** formed in the bending portion **112**F.

[0285] FIG. **55** is a diagram showing the ring member **115**F.

[0286] Multiple ring members **115**F are arranged in the internal passage **101** formed in the bending portion **112**F. The multiple ring members **115**F are arranged in the axial direction A. The multiple ring members **115**F are not connected, and adjacent ring members **115**F are arranged at a distance in the axial direction A. The built-in element **170**F including the first channel tube **171**F is inserted through the multiple ring members **115**A. The bending wire **160**F is inserted through the wire-guide **115**Fg formed in the ring member **115**F.

[0287] The ring member **115**F includes a slit **115***s* formed therein into which the spring **113** is fitted. The ring member **115**F is attached to the spring **113** by fitting the spring **113** into the slit **115***s*.

[0288] FIG. **56** is a diagram showing the ring member **115**F as viewed from the longitudinal direction A.

[0289] Three wire-guides **115**Fg are provided on the inner peripheral surface of the ring member **115**F. The three wire-guides **115**Fg are arranged at equal intervals along the circumferential direction C. The three wire-guides **115**Fg are arranged at intervals of 120 degrees with respect to the central axis O**1** in the longitudinal direction A.

[0290] The bending wire **160**F is a wire that bends the bending portion **112**F. The bending wire **160**F passes through the internal path **101** and extends to the attachment portion **150**. The bending wire **160**F has a first bending wire **161**F, a second bending wire **162**F, and a third bending wire **163**F. The number of bending wires **160**F is three, not four. That is, the bending wire **160**F is composed of three wires, a first bending wire **161**F, a second bending wire **162**F, and a third bending wire **163**F.

[0291] The direction perpendicular to the longitudinal axis of the first bending wire **161**F inserted through the wire-guide **115**Fg and the central axis O**1** of the longitudinal direction A is the up-down direction (also called the "UD direction"). The direction perpendicular to the longitudinal direction A and the up-down direction is the left-right direction (also called the "LR direction"). Although the number of bending wires **160**F is three, the bending wires **160**F can bend the bending portion **112**F in all directions, including the up-down direction and the left-right direction.

[0292] The bending wire **160**F may be inserted through a coil sheath **160***c*, as shown in FIG. **55**. However, the distal end of the bending wire **160**F that passes through the multiple ring members **115**F does not have to pass through the coil sheath **160***c*. Since there is no coil sheath **160***c* in the bending portion **112**F where the ring members **115**F are arranged, the coil sheath does not suppress compressive deformation due to wire traction, and the outer sheath **118**F can be smoothly bent.

[0293] FIG. **57** is a diagram showing the first channel tube **171**F.

[0294] The first channel tube **171**F is a tube having a large-diameter first treatment tool lumen **171***r*. The inner diameter D**1** of the first treatment tool lumen **171***r* is larger than the inner diameter D**2** of the second treatment tool lumen **172***r*. Specifically, the inner diameter D**1** of the first treatment tool lumen **171***r* can be three to five times the inner diameter D**2** of the second treatment tool lumen **172***r*. As shown in FIG. **3**, two treatment manipulators **400** can be inserted into the first treatment tool lumen **171***r*.

[0295] The inner diameter D**1** of the first treatment tool lumen **171***r* can be set to ½ or more of the outer diameter of the outer sheath **118**. Even in an existing endoscope (e.g., a transnasal endoscope) with a relatively large inner diameter of the treatment tool channel, the inner diameter of the treatment tool channel is about 2.4 mm (about ⅖ of the outer diameter) compared to an outer diameter of about 6 mm. Compared to an existing endoscope, the insertion manipulator **100**F has a larger inner diameter D**1** of the first treatment tool lumen **171***r* compared to the outer diameter of the outer sheath **118**. Therefore, a large treatment manipulator **400**F can be inserted into the first treatment tool lumen **171***r*, expanding the range of procedures.

[0296] The first channel tube **171**F includes a blade tube **171***g* and a coating **171***h* arranged on both the inner and outer circumferential sides of the blade tube **171***g*. The coating **171***h* arranged on the outer periphery is also called the outer coating. The coating **171***h* may be either an inner coating or an outer coating.

[0297] The braided tube **171***g* is a tube in which metal wires or resin wires are woven into a braid shape. The braided tube **171***g* is made of, for example, aramid or UHMW-PE (Ultra-High-Molecular-Weight Polyethylene).

[0298] The coating **171***h* is sandwiched between the inner and outer circumferential sides of the braided tube **171***g*, sealing (encapsulating) the braided tube **171***g*. The coating **171***h* is attached to the braided tube **171***g* by coating, co-extrusion, over-casting, or over-molding. The coating **171***h* is made of a low-friction, low-durometer resin or the like.

[0299] The braided tube **171***g* of the first channel tube **171**F is sealed (encapsulated) by the coating **171***h*. Therefore, the first channel tube **171**F has good rigidity without increasing the diameter. Because the coating **171***h* has low friction and a low durometer, the first channel tube **171**F has suitable elasticity and can achieve both durability and ease of bending.

[Treatment Manipulator **400**F]

[0300] FIG. **58** shows the treatment manipulator **400**F protruding from the first opening **111***a*. The treatment manipulator **400**F is a device that protrudes from the first opening **111***a* by inserting, for example, the first channel tube **171**F of the insertion manipulator **100**F, and is inserted into a patient's lumen to treat the affected area. An end effector (treatment portion) for treating the affected area may be disposed at the distal end of the treatment manipulator **400**F, and a treatment tool equipped with an end effector may be insertable into a channel provided in the treatment manipulator **400**F. The treatment manipulator **400**B is driven by a drive unit (actuator) **550** or the like.

[0301] FIGS. **59**, **60**, and **61** show the treatment manipulator **400**F.

[0302] The treatment manipulator **400**F includes a manipulator flexible portion **417**, a bendable treatment instrument arm **410**F, and treatment instruments (forceps **420**, high-frequency knife **430**, local injection needle **440**, basket **450**, etc.) that are inserted through the treatment instrument arm **410**F. The covering member that covers the treatment instrument arm **410**F is omitted from FIG. **59** and other figures.

[0303] The manipulator flexible portion **417** is a long member that extends in the longitudinal direction A and can be inserted into the first channel tube **171**F, etc. Two treatment instrument arms **410**F are provided on the distal end surface **417***a* of the manipulator flexible portion **417**. The two treatment instrument arms **410**F extend from the distal end surface **417***a* toward the distal end side A**1**.

[0304] The two treatment instrument arms **410**F protrude from the first opening **111***a* of the distal end **111** of the insertion manipulator **100**F while being aligned in the left-right direction (LR direction). At least a part of the two treatment instrument arms **410**F can be inserted through the notch portion **111***n* and can be spread outward in the left-right direction. Therefore, by spreading the two treatment instrument arms **410**F in the left-right direction, the surgeon can easily treat the affected area placed near the scope **200** with the treatment instrument.

[0305] By moving or rotating the insertion manipulator **100**F with the two treatment instrument arms **410**F in contact with the notch portion **111***n*, the two treatment instrument arms **410**F move or rotate following the movement with their relative positions to the scope **200** fixed. Therefore, even when the insertion manipulator **100**F is moved or rotated, the surgeon can easily grasp the position of the treatment instrument inserted through the treatment instrument arm **410**F. If the notch portion **111***n* is a groove into which the treatment arm **410**F fits, the relative position of the treatment arm **410**F with respect to the scope **200** is fixed, and the position of the treatment arm **410**F imaged in the field of view does not change even if the insertion manipulator **100**F is moved or rotated.

[0306] The treatment arm **410**F is a hollow, long member and has an insertion path (channel) through which the treatment tool is inserted. The treatment arm **410**F includes a first bending portion **411**F provided on the distal end side A**1** and a second bending portion **412**F provided on the proximal end side A**2**. In the usage state of the treatment manipulator **400**F as shown in FIG. **60**, the first bending portion **411**F can be disposed on the distal end side A**1** from the distal end surface of the distal end portion **111**. A part of the second bending portion **412**F can protrude from the distal end portion **111** to the distal end side A**1**.

[0307] The first bending portion **411**F includes a distal end **411***a*, a proximal end **411***b*, a bendable bending tube **460**, and four artificial muscles **470**. The first bending portion **411**F can be bent in the up-down and left-right directions by being driven by the artificial muscles **470**. The bending tube **460** and the artificial muscles **470** extend along the longitudinal direction A and are arranged side by side. The distal end **411***a*, the proximal end **411***b*, and the artificial muscles **470** are omitted from the illustration in FIG. **61**.

[0308] FIG. **62** is a perspective view of the bending tube **460**.

[0309] The bending tube **460** is a bendable tubular member. The internal space of the bending tube **460** is an insertion path (channel) through which a treatment tool is inserted. The bending tube **460** includes a distal end **460***a* connected to the distal end **411***a*, a proximal end **460***b* connected to the proximal end **411***b*, and a bending tube main body **460***d* sandwiched between the distal end **460***a* and the proximal end **460***b*.

[0310] The bending tube main body **460***d* is a tubular member formed of metal, resin, or the like, and has a plurality of grooves **461** formed along the circumferential direction C. The length of the grooves **461** in the circumferential direction C may be 70% to 80% of the entire circumference of the bending tube main body **460***d*. The grooves **461** penetrate the bending tube main body **460***d* in the radial direction. The bending tube main body **460***d* is bending by the grooves **461** widening and narrowing.

[0311] FIG. **63** is an expanded view of the bending tube **460** expanded in the circumferential direction C.

[0312] The groove (cut) **461** has a first groove **462** and a second groove **463**. The first groove **462** and the second groove **463** are both grooves along the circumferential direction C. The first grooves **462** and the second grooves **463** are alternately arranged along the longitudinal direction A. A set of adjacent first grooves **462** and second grooves **463** is also referred to as a "pair of grooves **461***p*". In the pair of adjacent grooves **461***p*, the pair of grooves **461***p* on the distal end side A**1** is positioned at a position rotated 90 degrees with respect to one side C**1** of the circumferential direction C compared to the pair of grooves **461***p* on the proximal side A**2**.

[0313] The first groove **462** and the second groove **463** are symmetrical with respect to a plane perpendicular to the longitudinal direction A, and are positioned at a position rotated 180 degrees with respect to the circumferential direction C. The first groove **462** is a groove that bulges toward the distal end side A**1**. The second groove **463** is a groove that bulges toward the proximal end side A**2**.

[0314] The first groove **462** and the second groove **463** may have a first gap **464** in the center in the circumferential direction C, a second gap **465** on both sides of the first gap **464** in the circumferential direction C, and a third gap **466** on the outside of the second gap **465**. The length G**1** in the longitudinal direction A of the first gap **464** may be longer than the length G**2** in the longitudinal direction A of the second gap **465** and the length G**3** in the longitudinal direction A of the third gap **466**. In addition, the length G**2** in the longitudinal direction A of the second gap **465** may be shorter than the length

[0315] G**1** in the longitudinal direction A of the first gap **464** and the length G**3** in the longitudinal direction A of the third gap **466** (G**1**>G**3**>G**2**).

[0316] In a pair of grooves **461***p* (first groove **462** and second groove **463**), the position of the second gap **465** of the first groove **462** and the position of the second gap **465** of the second groove **463** in the circumferential direction C may be substantially the same. In a pair of grooves **461***p* (first groove **462** and second groove **463**), the second gap **465** of the first groove **462** and the second gap **465** of the second groove **463** may be arranged adjacent to each other along the longitudinal direction A. The adjacent second gaps **465** in a pair of grooves **461***p* are also referred to as a "pair of second gaps **465***p*." The pair of second gaps **465***p* of a pair of grooves **461***p* and the first gaps **464** of the grooves **461** of the adjacent pair of grooves **461***p* may be arranged alternately in the longitudinal direction A.

[0317] FIG. **64** shows a bending tube **460** compressed at the start of treatment. As shown in FIG. **64**, the bending tube **460** starts treatment in a state where it is compressed in the longitudinal direction A until both ends of the second gap **465** in the longitudinal direction A almost touch. At this time, both ends of the first gap **464** in the longitudinal direction A do not come into contact. This state is called the "initial state." By starting treatment from a state where the surrounding artificial muscle **470** is moderately compressed, it is possible to further compress the artificial muscle **470** during treatment, and it is also possible to further extend the artificial muscle **470** during treatment. The bending tube **460** does not need to be compressed at the time of attachment to the first bending portion **411**F, and both ends of the first gap **464** and the second gap **465** may not come into contact.

[0318] FIG. **65** is a diagram showing a bending tube **460**.

[0319] The pair of second gaps **465***p* function as hinges to smoothly bend the bending tube **460**. The first gap **464** arranged on the outer side of the bending widens, and the first gap **464** arranged on the inner side of the bending narrows. Since the first gap **464** and the pair of second gaps **465***p* are evenly arranged in the circumferential direction C, the bending tube **460** can be bending in all directions including up, down, left, and right. When the groove **461** has the third gap **466** at both ends in the circumferential direction C, the bending tube **460** has the effect of dispersing stress when bending

[0320] The bending tube **460** is a tubular member consisting of a combination of thin struts and thick struts. The thin struts are, for example, the portions sandwiched between the pair of second gaps **465***p*. The thick struts are, for example, the portions sandwiched between the pair of second gaps **465***p* and the first gap **464**. The thin struts function as springs in the compression direction of the artificial muscle **470**. The thick struts function as joints when bending.

[0321] FIG. **66** is a diagram showing the artificial muscle **470**.

[0322] The artificial muscle **470** is, for example, a Mckibben type artificial muscle, as in the fourth embodiment. A tube **471** is attached to the artificial muscle **470**, which supplies a fluid for operating the artificial muscle **470**. The artificial muscle **470** contracts with the fluid supplied from the tube **471**. The artificial muscle **470** illustrated in FIG. **66** includes a silicone tube **474** and a braided tube **475** arranged on the outside of the silicone tube **474**. Fluid is supplied to the silicone tube **474** from the tube **471**. The artificial muscle **470** may have a silicone coating on the outer periphery of the braided tube **475**.

[0323] The four artificial muscles **470** are arranged around the bending tube **460** along the longitudinal direction of the bending tube **460**. The distal end of the bending tube **460** and the distal end of the artificial muscle **470** are connected to the distal end portion **411***a*. The proximal end of the bending tube **460** and the proximal end of the artificial muscle **470** are connected to the proximal portion **411***b*. When the artificial muscle **470** bends, the bending tube **460** passively bends. The number of artificial muscles **470** is not limited to four. For example, as shown in FIG. **43**, the number of artificial muscles **470** may be three.

[0324] The distal end **411***a* and the proximal end **411***b* may be formed in a rectangular shape extending in the up-down direction when viewed from the longitudinal direction A. Two artificial muscles **470** may be arranged on the upper side (U side) of the bending tube **460**, and two artificial muscles **470** may be arranged on the lower side (D side) of the bending tube **460**. Therefore, the dimension in the left-right direction of the first bending portion **411**F may be shorter than the dimension in the up-down direction. Therefore, it is easy to arrange the two first bending portions **411**F side by side in the left-right direction. Furthermore, the bending tube **460** is less likely to interfere with the artificial muscles **470** when bending in the left-right direction compared to when bending in the up-down direction. Therefore, the movable range in the left-right direction of the two first bending portions **411**F can be widened.

[0325] The artificial muscle **470** may be silicone-coated. In this case, even if there is a bias or variation in the structure of the blade tube **475**, the silicone coating absorbs the variation in deformation, realizing uniform deformation of the entire artificial muscle **470**. In addition, even if the artificial muscle **470** repeatedly contracts and relaxes, the deviation of the blade tube **475** can be suppressed and the restoration to the original shape is improved, so the reproducibility and stability of the operation of the artificial muscle **470** is improved. Therefore, the operator can easily control the bending operation of the artificial muscle **470** in the up, down, left, and right directions.

[0326] The operation of the artificial muscle **470** is divided into a "nonlinear region" in which the amount of deformation relative to the amount of pressurization of the fluid is nonlinear, and a "linear region" in which the amount of change relative to the amount of pressurization of the fluid is linear or can be handled as approximately linear in terms of the control of the manipulator. When the fluid is pressurized from a non-pressurized state, the operation of the artificial muscle **470** transitions to the linear region via the nonlinear region. Therefore, the artificial muscle **470** is in the "initial state" when the fluid is compressed by pressurizing it until it transitions to the linear region, the responsiveness of the artificial muscle **470** improves, and the controllability of the first bending portion **411**F improves.

[0327] The first bending portion **411**F is driven by the artificial muscle **470**, not by a wire. Therefore, there is no need to insert a wire into the internal space of the bending tube **460**, and sufficient space can be secured for inserting the treatment tool. In addition, there is no need to provide a path for inserting the wire in the bending tube **460**, and the structure of the bending tube **460** can be simplified. In addition, since wire breakage does not occur, maintenance is easy.

[0328] As shown in FIG. **60**, the second bending portion **412**F has a shoulder joint **416**F that greatly bends the treatment tool arm **410**F in the left-right direction (LR direction), and a wire **480**. The shoulder joint **416**F bends the second bending portion **412**F into an S-shape when viewed from the up-down direction (UD direction). As shown in FIG. **59**, the distal ends of the second bending portions **412**F of the two treatment instrument arms **410**F are bent in the left-right direction and move away from each other.

[0329] As shown in FIG. **61**, the wire **480** is attached to the second bending portion **412**F. The second bending portion **412**F is bent by pulling the second bending portion **412**F from the proximal end side A**2**. Two wires **480** are attached to one second bending portion **412**F.

[0330] As shown in FIG. **61**, the shoulder joint **416**F has a first shoulder joint **418** provided on the distal end side A**1** and a second shoulder joint **419** provided on the proximal end side A**2**.

[0331] The first shoulder joint **418** is connected to the distal end of the second shoulder joint **419** and extends to the distal end side A**1**. The first shoulder joints **418** of the two treatment instrument arms **410**F are bending in a direction approaching each other (inward) in the left-right direction (LR direction).

[0332] The second shoulder joint **419** is connected to the distal end surface **417***a* of the manipulator flexible portion **417** and extends to the distal end side A**1**. The second shoulder joints **419** of the two treatment instrument arms **410**F are bending in a direction moving away from each other (outward) in the left-right direction (LR direction). The first shoulder joint **418** and the second shoulder joint **419** are bending in opposite directions in the left-right direction, so that the second bending portion **412**F is bending in an S-shape.

[0333] As shown in FIG. **61**, the shoulder joint **416**F has a bending piece **415**F and a connecting piece **415**G. The first shoulder joint **418** and the second shoulder joint **419** have connecting pieces **415**G arranged at the distal end and proximal end, and multiple bending pieces **415**F are arranged between the two connecting pieces **415**G. When the wire **480** is pulled, adjacent bending pieces **415**F come into contact with each other.

[0334] FIG. **67** is a diagram showing the bending piece **415**F.

[0335] The bending piece **415**F may be formed in a substantially rectangular parallelepiped shape, or may be in a rectangular shape extending in the up-down direction as viewed from the longitudinal direction A. The bending piece **415**F has an insertion hole **415***h* that penetrates in the longitudinal direction A at the center. The insertion hole **415***h* is an insertion path (channel) for a treatment tool. The bending piece **415**F has a wire hole **415***w* at one end in the left-right direction, and a spacer **415***s* at the other end in the left-right direction. The wire hole **415***w* is formed along the longitudinal direction A, and is a wire-guide through which the wire **480** passes. The spacer **415***s* protrudes on both sides in the longitudinal direction A. On the upper and lower sides of the insertion hole **415***h*, there are provided connection holes **415***d* through which the connection members **415***c* (see FIG. **59**) that connect the bending pieces **415**F and the connecting pieces **415**G pass.

[0336] When adjacent bending pieces **415**F come into contact, the wire holes **415***w* come into contact with each other, and the spacers **415***s* come into contact with each other. The length S**1** of the wire hole **415***w* in the longitudinal direction A is longer than the length S**2** of the spacer **415***s* in the longitudinal direction A (S**1**>S**2**). Therefore, when the wire **480** is pulled, the adjacent bending pieces **415**F come into contact with each other and bend in the left-right direction.

[0337] FIG. **68** is a diagram showing bending piece **415**Fa, which is a modified example of bending piece **415**F.

[0338] The bending piece **415**Fa includes a spacer **415***sa*, which is a modified example of spacer **415***s*. The spacer **415***sa* is formed in a cylindrical shape. The spacer **415***sa* is provided in the center in the up-down direction (UD direction).

[0339] FIG. **69** is a diagram showing a bending piece **415**Fb, which is a modified example of bending piece **415**F.

[0340] The bending piece **415**Fb includes a wire hole **415***wb*, which is a modified example of wire hole **415***w*. The wire hole **415***wb* is formed in a rectangular shape extending in the up-down direction (UD direction) when viewed from longitudinal direction A. In the wire hole **415***wb*, the hole through which wire **480** is inserted is provided in the center in the up-down direction. Since the wire hole **415***wb* extends in the vertical direction, it is possible to prevent the bending piece **415**Fb from wobbling in the vertical direction.

[0341] The vertical length S**3** of the wire hole **415***wb* is preferably shorter than the vertical length S**4** of the spacer **415***s* (S3<S4). The vertical length **3** of the wire hole **415***wb* on the force point side of the wire **480** is short, and the vertical length S**4** of the spacer **415***s* on the side farther from the force point side of the wire **480** is long, so that the bending operation is stable.

[0342] FIG. **70** is a diagram showing a bending piece **415**Fc, which is a modified example of the bending piece **415**F.

[0343] The bending piece **415**Fc has a wire hole **415***wc*, which is a modified example of the wire hole **415***w*. The wire hole **415***wc* is formed in an elliptical shape extending in the vertical direction (UD direction) when viewed from the longitudinal direction A. The hole through which the wire **480** passes in the wire hole **415***wc* is provided in the center in the vertical direction. Because the wire hole **415***wc* extends in the vertical direction, it is possible to prevent the bending piece **415**Fc from wobbling in the vertical direction.

[0344] The bending piece **415**Fc includes a spacer **415***sc*, which is a modified example of the spacer **415***s*. The spacer **415***sc* is formed in a cylindrical shape and is provided on both sides in the vertical direction.

[0345] It is desirable that the vertical length S3 of the wire hole **415***wc* is shorter than the vertical length S4 of the spacer **415***sc* (S3<S4). The bending operation is stabilized by having a short vertical length **3** of the wire hole **415***wc* on the force point side of the wire **480** and a long vertical length S4 of the spacer **415***sc* on the side farther from the force point side of the wire **480**.

[0346] FIG. **71** is a diagram showing a bending piece **415**Fd, which is a modified example of bending piece **415**F.

[0347] The bending piece **415**Fd includes spacer **415***sc*, which is a modified example of spacer **415***s*. The spacer **415***sc* is formed in a cylindrical shape and is provided on both the top and bottom sides.

[0348] FIG. **72** is a diagram showing a connecting piece **415**G.

[0349] The connecting piece **415**G may be formed in a substantially rectangular parallelepiped shape, or may be in a rectangular shape extending in the top and bottom direction when viewed from the longitudinal direction A. The bending piece **415**F includes an insertion hole **415***h* that penetrates in the longitudinal direction A at the center. The insertion hole **415***h* is an insertion path (channel) for a treatment tool. At the upper and lower sides of the insertion hole **415***h*, there are provided connection holes **415***d* through which the connection members **415***c* (see FIG. **59**) that connect the bending piece **415**F and the connecting piece **415**G are inserted.

[0350] As shown in FIG. **61**, the bending pieces **415**F are provided so that the wire holes **415***w* are located on the outside in the left-right direction in the first shoulder joints **418** of the two treatment instrument arms **410**F. The length S1 in the longitudinal direction A of the wire hole **415***w* located on the outside is longer than the length S2 in the longitudinal direction A of the spacer **415***s* located on the inside (S1>S2). Therefore, when the wire **480** is pulled, the first shoulder joints **418** of the two treatment instrument arms **410**F are bent in a direction approaching each other (inside) in the left-right direction (LR direction).

[0351] As shown in FIG. **61**, a bending pieces **415**F are provided in the second shoulder joints **419** of the two treatment instrument arms **410**F so that the wire hole **415***w* is located on the inside in the left-right direction. The length S1 in the longitudinal direction A of the wire hole **415***w* located on the inside is longer than the length S2 in the longitudinal direction A of the spacer **415***s* located on the outside (S1>S2). Therefore, when the wire **480** is pulled, each of the second shoulder joints **419** of the two treatment instrument arms **410**F bends in a direction away from each other (outside) in the left-right direction (LR direction).

[0352] FIG. **73** shows a treatment instrument arm **410**F with a bent second bending portion **412**F.

[0353] As the first shoulder joint **418** and the second shoulder joint **419** bend in opposite directions in the left-right direction, the second bending portion **412**F bends in an S-shape. Therefore, the distance (shoulder width) L between the central axes at the distal ends of the second bending portions **412**F of the two treatment instrument arms **410**F can be widened. When treating a target site using two treatment instrument arms **410**F, by appropriately separating the positions of the proximal ends of the two first bending portions **411**F, the surgeon S can secure sufficient space for treatment and easily treat the target site. In addition, the configuration in which the artificial muscle **470** can be driven while protruding from the insertion manipulator **100**F has the advantage that the artificial muscle **470** can be driven without being limited by the outer diameter of the insertion manipulator **100**F. The configuration in which the distal end portion **111** has the notch portion **111***n* has the advantage that the treatment tool protruding from the distal end of the treatment manipulator **400**F can maintain a positional relationship that does not interfere with the treatment tool protruding from the second opening **111***b*, and the advantage that a positional relationship that does not obstruct the field of view can be maintained. Furthermore, by arranging the artificial muscle **470** at the distal end, it has the advantage that it is less susceptible to changes in shape in the entire length of the treatment manipulator **400**F and the insertion manipulator **100**F. Furthermore, by selecting a fluid-driven artificial muscle **470**, the degree of freedom in arranging the ducts inside the treatment manipulator **400**F is high.

[0354] The first bending portion **411**F is driven to bend by the artificial muscle **470**, so the bending angle may be smaller than other bending portions driven by wires or the like. However, as shown in FIG. **73**, when the second bending portion **412**F is bent into an S-shape, the two first bending portions **411**F can be arranged approximately parallel along the longitudinal direction A while being spaced apart in the left-right direction. Therefore, even if the treatment manipulator **400**F includes the first bending portion **411**F with a relatively small bending angle, it is easy to treat the affected area arranged on the central axis O**4** of the treatment manipulator **400**F. Furthermore, since the bending angle is relatively small but delicate driving is possible, it is also suitable for treatment in deep areas such as the right colon.

[0355] In the treatment manipulator **400**F, the treatment tool does not have a bending mechanism, but the treatment tool arm **410**F through which the treatment tool is inserted has the bending mechanism. Since the treatment tool advances and retracts relative to the bending treatment tool arm **410**F, the treatment manipulator **400**F has a wider approachable range than other treatment tools with bending mechanisms. Even the treatment manipulator **400**F with the first bending portion **411**F, which has a small bending angle, can approach a wider range. In addition, since the treatment tool of the treatment manipulator **400**F does not have a bending mechanism, the mechanism can be simplified and maintenance is easy.

[Driver **500**F]

[0356] FIG. **74** is a functional block diagram of the driver **500**F.

[0357] The driver **500**F includes an operation receiver **520**, a drive controller **560**, an insertion drive unit **540**, and a drive unit **570**. The operation-receiving unit **520** and the drive controller **560** are mounted on the drive device main body **500***b*. The insertion drive unit **540** and the drive unit **570** are devices separated from the drive device main body **500***b*. The drive controller **560** controls the insertion drive unit **540** and the drive unit **570**. The insertion drive unit **540** and the drive unit **570** are mounted on the cart **500**W.

[0358] Compared to the drive device **500** of the first embodiment, the drive device **500**F is incorporated into a "drive unit **570**" which is a device separated from the air supply and a suction drive unit **530** and a drive unit **550**.

[Insertion Drive Unit **540**]

[0359] FIG. **75** is a diagram showing the insertion drive unit **540**.

[0360] The insertion drive unit **540** assists the operation of inserting the insertion manipulator **100**F into the body of the patient P. The insertion drive unit **540** is mounted on an arm **500***a* that extends deformably from the cart **500**W. The surgeon S can operate the arm **500***a* to place the insertion drive unit **540** in a position that allows the insertion manipulator **100**F to be easily inserted into the patient P's body. In addition, when the position of the insertion manipulator **100**F is to be readjusted (repositioned) during treatment, the insertion drive unit **540** can be used to readjust the position, which has the effect of preventing buckling of the insertion manipulator **100**F.

[0361] The insertion drive unit **540** has two rings **541** through which the flexible portion **119** of the insertion manipulator **100**F is inserted, and a drive unit **542** disposed between the two rings **541**.

[0362] The drive unit **542** can be fixed by sandwiching the flexible portion **119**. The drive unit **542** can move between the two rings **541**, and can move the flexible portion **119**, which is fixed by clamping it, forward and backward. The drive unit **542** can rotate in the circumferential direction, and can rotate the flexible portion **119**, which is fixed by clamping it. The insertion drive unit **540** is not limited to the mode in which the drive unit **542** clamps and fixes the flexible portion **119**. For example, the insertion drive unit **540** may be equipped with rollers, and the rollers may be pressed against the flexible portion **119** to fix it, and the flexible portion **119** may be moved forward and backward and rotated by the rotation of the rollers.

[Drive Unit **570**]

[0363] FIG. **76** is a diagram showing the drive unit **570**.

[0364] The drive unit **570** is a device that drives the insertion manipulator **100**F and the treatment manipulator **400**F. The drive unit **570** is mounted on the top of the cart **500**W. The base unit **571** and the drive unit **570** have a first drive unit **580** and a second drive unit **590**.

[0365] FIGS. **77** and **78** are diagrams showing the operation of the drive unit **570**.

[0366] The drive unit **570** has a generally cylindrical shape extending in the longitudinal direction A. The first drive unit **580** is disposed on the distal end side **A1**, and the second drive unit **590** is disposed on the proximal side **A2**. The base unit **571** supports the first drive unit **580** and the second drive unit **590** so that they can be driven. The base unit **571** has a motor **571***a* that drives the first drive unit **580** and the second drive unit **590**.

[0367] The first drive unit **580** is a device that drives the insertion manipulator **100**F. The first drive unit **580** is supported by the base unit **571** so as to be movable back and forth in the longitudinal direction A and rotatable about a central axis O**5** along the longitudinal direction A. The first drive unit **580** has a first insertion passage **580***h* that passes through in the longitudinal direction A, through which the treatment manipulator **400**F or the hardness-changing device **300** can be inserted.

[0368] FIG. **79** is an exploded view of the first drive unit **580**.

[0369] The first drive unit **580** has a first motor unit **581** on the proximal end side A**2** and a first adaptor **582** on the distal end side A**1**.

[0370] FIG. **80** is a cross-sectional view of the first motor unit **581** from which the first adaptor **582** has been separated.

[0371] The first motor unit **581** is a unit to which the first adaptor **582** can be attached from the distal end side A**1**. The first motor unit **581** drives the first drive unit **584** of the attached first adaptor **582**. The first motor unit **581** includes a first drive unit **583**.

[0372] The first drive unit (first actuator) **583** includes a first motor **583***m*, a first shaft **583***s* driven by the first motor **583***m*, and a first coupled portion **583***c* connected to the first shaft **583***s*. The first coupled portion **583***c* is exposed at the distal end side A**1** of the first motor unit **581**.

[0373] The first adaptor **582** is an adaptor that can be attached to and detached from the distal end side A**1** of the first motor unit **581**. The distal end side A**1** of the first adaptor **582** is connected to the attachment portion **150** provided at the proximal end of the flexible portion **119** of the insertion manipulator **100**F. The internal path **101** of the insertion manipulator **100**F and the first insertion passage **580***h* are in communication with each other. The first adaptor **582** has a first drive unit **584**.

[0374] The first drive unit (driving force transmission part) **584** is a member to which a driving force for driving the bending wire **160**F and the scope operation wire **178** is input. The first drive unit **584** is, for example, a rotating drum that rotates about a central axis along the longitudinal direction A. The first drive unit **584** has a first coupling portion **584***c*. The first coupling portion **584***c* is exposed to the proximal end side A**2** of the first adapter **582**. The first drive unit **584** is not limited to a rotating drum. For example, the first drive unit **584** may be a ball screw. Alternatively, an artificial muscle may be used for the bending portion **112**F, and the first drive unit **584** may be a cylinder. The cylinder may be a hydraulic cylinder, a water pressure cylinder, or a pneumatic cylinder.

[0375] FIG. **81** is a cross-sectional view of the first motor unit **581** to which the first adapter **582** is attached.

[0376] When the first adapter **582** is attached to the first motor unit **581**, the first coupled portion **583***c* and the first coupling portion **584***c* are coupled. As a result, the rotation of the first shaft **583***s* by the first motor **583***m* is transmitted to the first drive unit **584**. This allows the first drive unit **583** to drive the bending wire **160**F and the scope operation wire **178**.

[0377] The second drive unit **590** is a device that drives the treatment manipulator **400**F and the hardness-changing device **300**. The second drive unit **590** is supported by the base unit **571** so as to be rotatable about the central axis O**5**. The second drive unit **590** has a second insertion passage **590***h* that passes through in the longitudinal direction A, through which the treatment manipulator **400**F and the hardness-changing device **300** can be inserted.

[0378] FIGS. **82** to **84** are exploded views of the second drive unit **590**. The second drive unit **590** has a second motor unit **591** on the distal end side A**1**, and a second adaptor **592**, a third adaptor **592**A, and a fourth adaptor **592**B on the proximal side A**2**.

[0379] When the first drive unit **580** moves to the proximal side A**2**, the first drive unit **580** comes into contact with the second drive unit **590**. As shown in FIG. **77**, when the first drive unit **580** and the second drive unit **590** come into contact, the first insertion passage **580***h* and the second insertion passage **590***h* are connected.

[0380] As shown in FIG. **82** to FIG. **84**, the second motor unit **591** is a unit to which the second adaptor **592**, the third adaptor **592**A, and the fourth adaptor **592**B can be attached from the proximal side A**2**. The second motor unit **591** drives the second drive units **594** of the attached second adaptor **592**, the third adaptor **592**A, and the fourth adaptor **592**B. The second motor unit **591** has a second drive unit **593**.

[0381] As shown in FIG. **81**, the second drive unit (second actuator) **593** has a second motor **593***m*, a second shaft **593***s* driven by the second motor **593***m*, and a second coupled portion **593***c* connected to the second shaft **593***s*. The second coupled portion **593***c* is exposed on the proximal end side A**2** of the second motor unit **591**.

[0382] FIG. **85** is a cross-sectional view of the second motor unit **591** to which the second adaptor **592** is attached.

[0383] The second adaptor **592** is an adaptor that can be attached and detached to the proximal end side A**2** of the second motor unit **591**. The treatment manipulator **400**F and the hardness-changing device **300** are connected to the distal end side A**1** of the second adaptor **592**. The treatment manipulator **400**F and the hardness varying device **300** are inserted into the internal path **101** of the insertion manipulator **100**F via the first insertion passage **580***h* and the second insertion passage **590***h*. The second adapter **592** has a second drive unit **594**.

[0384] The second drive unit (driving force transmission part) **594** is a member to which a driving force is input to drive the tube **471** and wire **480** of the treatment manipulator **400**F and the wire **312** of the hardness varying device **300**. The second drive unit **594** is, for example, a rotating drum, a cylinder, or a ball screw. The second drive unit **594** has a second coupling portion **594***c*. The second coupling portion **594***c* is exposed to the distal end side A**1** of the second adapter **592**.

[0385] When the second adaptor **592** is attached to the second motor unit **591**, the second coupled portion **593***c* and the second coupling portion **594***c* are coupled. As a result, the rotation of the second shaft **593***s* by the second motor **593***m* is transmitted to the second drive unit **594**. This allows the second drive unit **593** to drive the tube **471**, the wire **480**, the wire **312**, etc.

[0386] FIG. **86** is a cross-sectional view of the second motor unit **591** to which the third adaptor **592**A and the fourth adaptor **592**B are attached.

[0387] The third adaptor **592**A and the fourth adaptor **592**B are adaptors that can be attached and detached to the proximal end side A**2** of the second motor unit **591**. The distal end side A**1** of the third adaptor **592**A and the fourth adaptor **592**B is connected to a treatment tool (forceps **420**, high-frequency knife **430**, local injection needle **440**, basket **450**, etc.). The treatment tool is inserted through the treatment tool arm **410**F via the manipulator flexible portion **417** of the treatment manipulator **400**F. The third adaptor **592**A and the fourth adaptor **592**B have a third drive unit **595**.

[0388] The third drive unit (driving force transmission part) **595** is a member to which a driving force for driving a treatment tool (forceps **420**, high-frequency knife **430**, local injection needle **440**, basket **450**, etc.) is input. The third drive unit **595** is, for example, a rotating drum. The third drive unit **595** has a third coupling portion **595***c*. The third coupling portion **595***c* is exposed at the distal end side A**1** of the third adaptor **592**A and the fourth adaptor **592**B.

[0389] When the third adaptor **592**A and the fourth adaptor **592**B are attached to the second motor unit **591**, the second coupled portion **593***c* and the third coupling portion **595***c* are coupled. As a result, the rotation of the second shaft **593***s* by the second motor **593***m* is transmitted to the third drive unit **595**. This allows the second drive unit **593** to drive the treatment tools (forceps **420**, high-frequency knife **430**, local injection needle **440**, basket **450**, etc.).

[0390] The second motor unit **591** has a plurality of second drive units **593** that are assigned to the second adaptor **592**, the third adaptor **592**A, and the fourth adaptor **592**B. The number of second drive units **593** to which the second adaptor **592**, the third adaptor **592**A, and the fourth adaptor **592**B are connected is determined by the number of drive systems for driving the treatment tools, etc.

[0391] The second adaptor **592**, the third adaptor **592**A, and the fourth adaptor **592**B each have an identifier such as an RFID. The drive controller **560** can recognize the type of adaptor attached to the second motor unit **591**.

[0392] FIGS. **87** to **91** are diagrams showing an example of the operation of the drive unit **570**. The surgeon S inserts the insertion section **110**F of the insertion manipulator **100**F into the patient's large intestine. The hardness-changing device **300** is connected to the second adapter **592** of the second drive unit **590**.

[0393] As shown in FIG. **87**, the surgeon S inserts the insertion section **110**F of the insertion manipulator **100**F from the distal end into the large intestine from the patient's anus. At this time, the surgeon S provides slack SL in the flexible portion **119** between the part holding the flexible portion **119** and the drive unit **570**. The advancement and retraction of the insertion section **110**F may be performed by the surgeon S's hand or by the insertion drive unit **540**.

[0394] As shown in FIG. **88**, the surgeon S places the bending portion **112**F at a portion that curves significantly in the large intestine. The surgeon S advances the hardness-changing portion **310**, whose shape is variable, to place the hardness-changing portion **310** at the bending portion **112**F. The surgeon S fixes the shape of the hardness-changing portion **310** that passes through the bending portion **112**F.

[0395] As shown in FIG. **89**, the surgeon S advances the first drive unit **580** toward the distal end side A**1** to advance the insertion section **110**F. The bending portion **112**F advances along the hardness-changing portion **310**, the shape of which is fixed. The insertion manipulator **100**F can smoothly pass through the large bending portion in the large intestine.

[0396] As shown in FIG. **90**, the surgeon S releases the wire of the hardness-changing portion **310** to soften the hardness-changing portion **310**.

[0397] As shown in FIG. **91**, the surgeon S holds and fixes the flexible portion **119**. Fixing of the insertion section **110**F may be performed by the surgeon S's hand or by the insertion drive unit **540**. The surgeon S retracts the first drive unit **580** to the proximal end side A**2**. This reduces the slack SL of the flexible portion **119**, and the hardness-changing portion **310** advances.

[0398] As described above, by moving the first drive unit **580** back and forth while operating the insertion section **110**F, the insertion manipulator **100**F can be advanced. Since it is not necessary to keep advancing the first drive unit **580** to advance the insertion manipulator **100**F, the movable range of the first drive unit **580** can be limited. The cart **500**W on which the drive unit **570** is mounted can be made smaller.

[0399] FIG. **92** shows a rack **500**R, which is a modified example of the cart **500**W.

[0400] The drive unit **570** is mounted on an arm **500***a* that extends deformably from the rack **500**R. The drive unit main body **500***b* and the image control device **600** can be mounted on the rack **500**R.

[Scope **200**]

[0401] FIG. **93** shows the scope **200** protruding from the distal end **111**.

[0402] The scope (camera unit) **200** is attached so that it can protrude from the distal end **111** to the distal end side A**1** and can be bent. The scope **200** is operated by the scope operation wire **178**. The scope (camera unit) **200** includes an imaging unit (camera) **201** and a support portion **220** that supports the imaging unit **201**. The imaging unit **201** is provided on the distal end side A**1** of the support portion **220**.

[0403] The support portion **220** of the scope **200** has a leaf spring **210** on the upper side (U side). The leaf spring **210** is formed in a flat plate shape extending in the longitudinal direction A. When the scope operation wire **178** is not driven, the leaf spring allows the scope **200** to maintain a linear shape extending in the longitudinal direction A.

[0404] FIG. **94** is a diagram explaining the operation of the scope operation wire **178**. The scope operation wire **178** includes a first scope operation wire **178***a* and a second scope operation wire **178***b*.

[0405] The first scope operation wire **178***a* is connected to the proximal end side A**2** of the scope **200** via a pulley **178***p*. The pulley **178***p* is disposed on the distal end side A**1** of the connection part **178***c* where the first scope operation wire **178***a* is connected to the scope **200**. The pulley **178***p* reverses the forward and backward direction of the first scope operation wire **178***a* in the longitudinal direction A.

[0406] The distal end of the second scope operation wire **178***b* is connected to the lower side (D side) of the imaging unit (camera) **201** provided on the distal end side A**1** of the scope **200**.

[0407] The scope **200** transitions between a first state in which the scope **200** advances relative to the distal end **111**, a second state in which the scope **200** retracts relative to the distal end **111**, and a third state in which the scope **200** bends downward.

[0408] When the first scope manipulation wire **178***a* is pulled, the first scope manipulation wire **178***a* pulls the connection part **178***c* toward the distal end side A**1**, and the scope **200** advances relative to the distal end **111** (first state).

[0409] When the second scope manipulation wire **178***b* is pulled, the scope **200** retracts relative to the distal end **111** (second state).

[0410] When the first scope operation wire **178***a* and the second scope operation wire **178***b* are pulled, the force pulling the second scope operation wire **178***b* becomes greater than the holding force of the leaf spring, so that the leaf spring **210** bends downward and the scope **200** bends downward (third state).

[0411] The two scope operation wires **178** allow the forward movement, backward movement, and bending of the scope **200** to be controlled, and the distal end **111** can be simplified and made thinner.

[0412] The electric endoscope system **1000**F according to this embodiment allows observation and treatment to be performed more efficiently.

[0413] The fifth embodiment of the present invention has been described above in detail with reference to the drawings, but the specific configuration is not limited to this embodiment and design changes within the scope of the present invention are also included. In addition, the components shown in the above-mentioned embodiments and modifications can be configured in any suitable combination.

INDUSTRIAL APPLICABILITY

[0414] The present invention can be applied to a medical system for observing and treating the inside of a hollow organ, etc.

What is claimed is:

1. A medical manipulator, comprising:

a manipulator flexible portion extending in a longitudinal direction; and

two arms arranged at a distal end of the manipulator flexible portion,

wherein the two arms are arranged in a left-right direction perpendicular to the longitudinal direction,

each of the two arms includes a second bending portion, and

distal ends of second bending portions of the two arms are spaced apart in the left-right direction by bending the second bending portions.

2. The medical manipulator according to claim **1**, wherein

the second bending portion further includes a first shoulder joint provided at a distal end side and a second shoulder joint provided at a proximal end side, and

the distal ends of the second bending portions of the two arms are spaced apart in the left-right direction by bending the first shoulder joint and the second shoulder joint.

3. The medical manipulator according to claim **2**,

wherein each of the first shoulder joints of the two arms can be bent in a direction toward each other in the left-right direction, and each of the second shoulder joints of the two arms can be bent in a direction away from each other in the left-right direction.

4. The medical manipulator according to claim **2**, wherein the second bending portion includes a wire for bending the second bending portion.

5. The medical manipulator according to claim **4**, wherein the wire includes a first wire for bending the first shoulder joint and a second wire for bending the second shoulder joint.

6. The medical manipulator according to claim **1**, wherein the arm has a channel for inserting a treatment tool.

7. The medical manipulator according to claim **4**, wherein the second bending portion has a plurality of bending pieces arranged along the longitudinal direction, and

wherein each of the plurality of bending pieces has an insertion hole through which a treatment tool is inserted, a wire hole through which the wire is inserted, and a spacer protruding in the longitudinal direction.

8. The medical manipulator according to claim **7**, wherein the longitudinal length of the wire hole is longer than the longitudinal length of the spacer,

wherein the first shoulder joints of the two arms are provided such that the wire holes are positioned on the outer side in the lateral direction, and

wherein the second shoulder joints of the two arms are provided such that the wire holes are positioned on the inner side in the lateral direction.

9. The medical manipulator according to claim **1**, wherein each of the two arms includes a first bending portion through which a treatment tool can be inserted,

the first bending portion is connected to the distal end side of the second bending portion, and

the first bending portion has an artificial muscle.

10. A medical manipulator system, comprising:

a manipulator flexible portion extending in a longitudinal direction and two arms disposed at a distal end of the manipulator flexible portion;

a drive unit that drives the medical manipulator; and

a control device that controls the drive unit;

wherein the two arms are arranged in a left-right direction perpendicular to the longitudinal direction,

each of the two arms includes a second bending portion, and

distal ends of the second bending portions of the two arms are separated in the left-right direction by bending the second bending portions.

11. The medical manipulator system according to claim **10**, wherein

the second bending portion further includes a first shoulder joint provided at a distal end side and a second shoulder joint provided at a proximal end side, and

the first shoulder joint and the second shoulder joint are bent so that the distal ends of the second bending portions of the two arms are separated in the left-right direction.

12. The medical manipulator system according to claim **11**, wherein

the first shoulder joints of the two arms can be bent in a direction approaching each other in the left-right direction, and

the second shoulder joints of the two arms can be bent in a direction receding in the left-right direction.

13. The medical manipulator system according to claim **11**, further comprising an insertion manipulator through which the medical manipulator is inserted, and

wherein the insertion manipulator has a notch on its distal end surface, and the second shoulder joint fits into the notch when bent.

14. The medical manipulator system according to claim **11**,

wherein the second bending portion includes a wire for bending the second bending portion.

15. The medical manipulator system according to claim **14**,

wherein the wire includes a first wire for bending the first shoulder joint and a second wire for bending the second shoulder joint.

16. The medical manipulator system according to claim **10**, further comprising a treatment tool inserted into the medical manipulator,

wherein the arm has a channel through which the treatment tool is inserted.

17. The medical manipulator system according to claim **14**,

wherein the second bending portion includes a plurality of bending pieces arranged along the longitudinal direction, and

wherein each of the plurality of bending pieces includes an insertion hole through which the treatment tool is inserted, a wire hole through which the wire is inserted, and a spacer protruding in the longitudinal direction.

18. The medical manipulator system according to claim **17**,

wherein the longitudinal length of the wire hole is longer than the longitudinal length of the spacer,

the bending pieces are provided in the first shoulder joints of the two arms so that the wire hole is located on the outer side in the left-right direction, and

the bending pieces are provided in the second shoulder joints of the two arms so that the wire hole is located on the inner side in the left-right direction.

19. The medical manipulator system according to claim **10**,

wherein each of the two arms includes a first bending portion through which a treatment tool can be inserted,

the first bending portion is connected to the distal end side of the second bending portion, and

the first bending portion has an artificial muscle.

20. A treatment method using a medical system including a medical manipulator having two arms with a first shoulder joint and a second shoulder joint in a left-right direction, and an insertion manipulator through which the medical manipulator is inserted,

wherein each of the first shoulder joints of the two arms can be bent in a direction toward each other in the left-right direction, and each of the second shoulder joints of the two arms can be bent in a direction away from each other in the left-right direction, and

in a state in which the second shoulder joint is bent, the second shoulder joint is fitted into a notch in the insertion manipulator to perform treatment.

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